

Exploring high energy $\gamma\gamma$ interactions at the future electron-hadron collider LHeC

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ABSTRACT:

The future Large Hadron-Electron Collider (LHeC) will operate at a center-of-mass energy of 1.2 TeV, delivering an integrated electron-proton luminosity of approximately 1 ab⁻¹. With its high luminosity and, more importantly, a clean experimental environment, the LHeC will offer unique research opportunities. This paper presents a comprehensive survey of studies on high-energy photon-photon ($\gamma\gamma$) interactions at the LHeC, targeting a $\gamma\gamma$ center-of-mass energy of up to 1 TeV. The scientific potential of investigating such photon-photon processes is evaluated by calculating cross-sections and discussing signals and associated backgrounds for various $\gamma\gamma$ processes, including the exclusive production of lepton pairs, W and Z bosons, the Higgs boson, and pairs of charged supersymmetric particles.

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1 Reply to final comments by Krzysztof

1.1 EPA–LPAIR/CepGen comparison: why is LPAIR with ALLM lower than EPA?

Krzysztof’s comment: Can we explain why LPAIR with ALLM is so much lower than EPA in Table 2?

The lower CepGen/LPAIR result with ALLM could be understood as a consequence of the different treatment of the inelastic photon-emission phase space. The standalone EPA calculation uses factorized photon fluxes with explicit (Q_e^2) , (Q_p^2) , (M_N) , and $(W_{\gamma\gamma})$ cuts, whereas LPAIR evaluates the Bethe-Heitler two-photon matrix element with its own implementation of the proton-dissociative structure-function contribution. The observed difference could also be interpreted as a modeling uncertainty associated with the inelastic photon flux and proton-dissociation treatment, rather than as a contradiction between EPA and LPAIR/CepGen. This interpretation is supported by the much better agreement found in the elastic channel, where the proton electromagnetic form factors are well constrained and no inelastic structure-function modeling is involved.

A more detailed generator-level explanation, in particular regarding the exact CepGen/LPAIR implementation of the inelastic structure-function phase space, would be best confirmed with Laurent.

1.2 Anomalous tau moments: update of Δa_τ , EDM sensitivity, and SMEFT translation

Krzysztof's comment: Are the results on anomalous tau moments up to date? Is $\Delta a_\tau = 0.0042$ still valid? What about sensitivity to the tau EDM?

Additional clarification by Krzysztof: Use

$$\Delta a_\tau = 0.0005$$

for the fit, and translate the result into limits on dimension-six EFT parameters.

We replace the previous benchmark value by $\Delta a_\tau \equiv a_\tau - a_\tau^{\text{SM}} = 5.0 \times 10^{-4}$. This benchmark is implemented through the SMEFT tau dipole operator and is used to compare the Standard Model prediction with the anomalous magnetic-moment scenario in elastic photon-induced $\gamma\gamma \rightarrow \tau^+\tau^-$ production at the LHeC. Figure 1 shows the differential invariant-mass distribution $d\sigma/dM_{\tau^+\tau^-}$ for the SM and for the benchmark $\Delta a_\tau = 5.0 \times 10^{-4}$. The lower panel shows the ratio $R_{\tau\tau}(M_{\tau^+\tau^-}) = \frac{d\sigma_{\text{EFT}}/dM_{\tau^+\tau^-}}{d\sigma_{\text{SM}}/dM_{\tau^+\tau^-}}$, together with a linear fit to the ratio. The rapidity distribution of the $\tau^+\tau^-$ system is shown in Fig. 2.

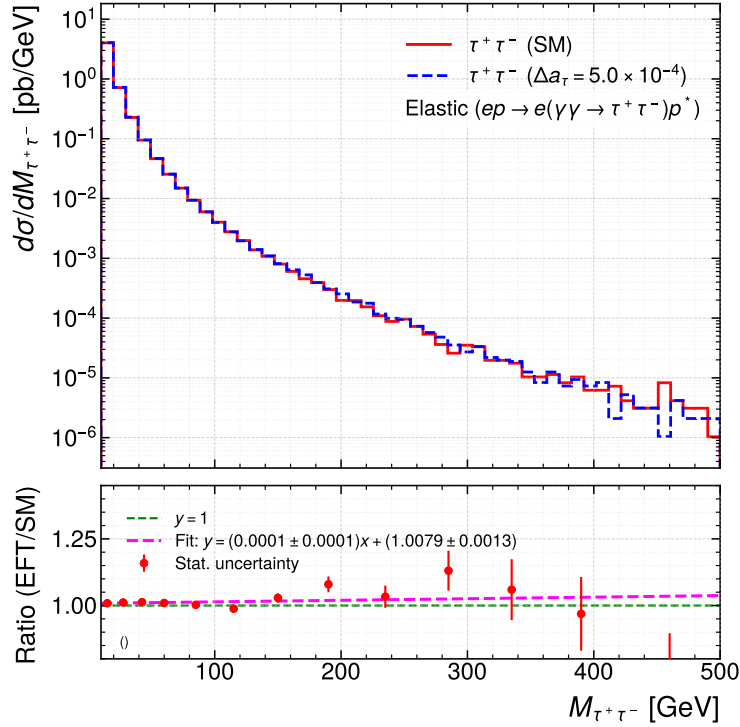


Figure 1. Differential invariant-mass distribution for elastic photon-induced $\tau^+\tau^-$ production at the LHeC. The red solid histogram shows the Standard Model prediction, while the blue dashed histogram corresponds to the SMEFT benchmark $\Delta a_\tau = 5.0 \times 10^{-4}$. The lower panel shows the ratio EFT/SM as a function of $M_{\tau^+\tau^-}$, using custom mass bins. The red points show the expected statistical uncertainty for $\mathcal{L} = 100.0 \text{ fb}^{-1}$, and the magenta dashed line shows the linear fit.

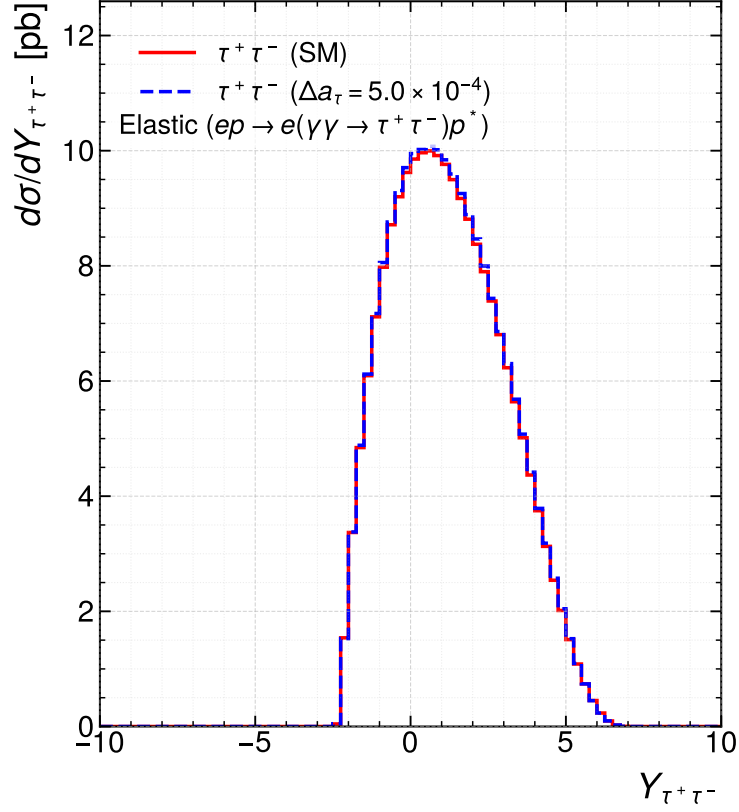


Figure 2. Rapidity distribution of the $\tau^+\tau^-$ system in elastic photon-induced production at the LHeC. The red solid histogram shows the Standard Model prediction, while the blue dashed histogram corresponds to the SMEFT benchmark $\Delta a_\tau = 5.0 \times 10^{-4}$. The distribution is shown in the range $-10 < Y_{\tau^+\tau^-} < 10$. The positive-rapidity shift reflects the asymmetric LHeC beam configuration, with the 7 TeV proton beam defining the positive z direction.

1.3 Anomalous quartic gauge couplings in $\gamma\gamma \rightarrow W^+W^-$

Krzysztof's comment: For WW , we need to quantify sensitivity to anomalous quartic gauge couplings, in a similar way to the a_τ study, by showing the ratio versus $M_{W^+W^-}$, using for example

$$\frac{f_{M2}}{\Lambda^4} = 1 \text{ TeV}^{-4}.$$

Figure 3 shows the differential distribution $d\sigma/dM_{W^+W^-}$ for the SM and for the benchmark $f_{M2}/\Lambda^4 = 1 \text{ TeV}^{-4}$. The lower panel shows the ratio $R_{WW}(M_{W^+W^-}) = \frac{d\sigma_{\text{EFT}}/dM_{W^+W^-}}{d\sigma_{\text{SM}}/dM_{W^+W^-}}$. The rapidity distribution of the W^+W^- system is shown in Fig. 4.

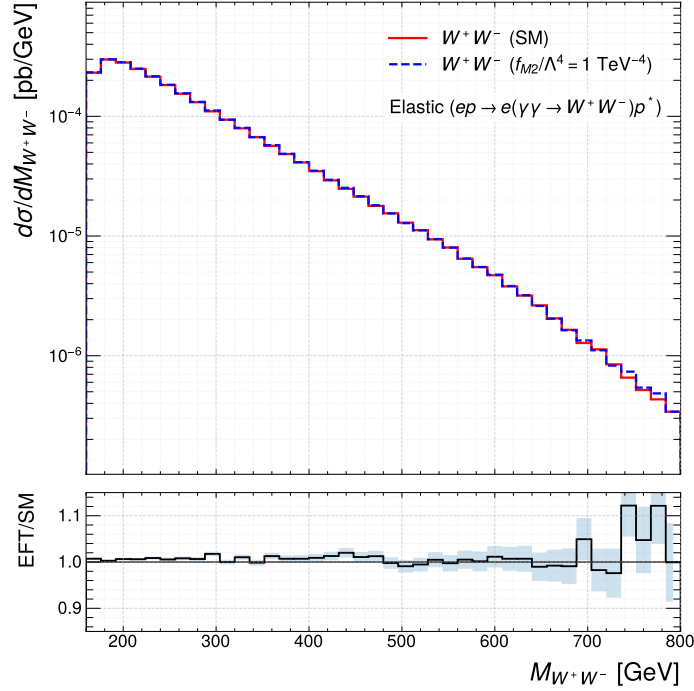


Figure 3. Differential invariant-mass distribution for elastic photon-induced W^+W^- production at the LHeC. The red solid histogram shows the Standard Model prediction, while the blue dashed histogram shows the benchmark anomalous quartic gauge-coupling scenario $f_{M2}/\Lambda^4 = 1 \text{ TeV}^{-4}$. The lower panel shows the ratio EFT/SM. The shaded band denotes the Monte Carlo statistical uncertainty on the ratio.

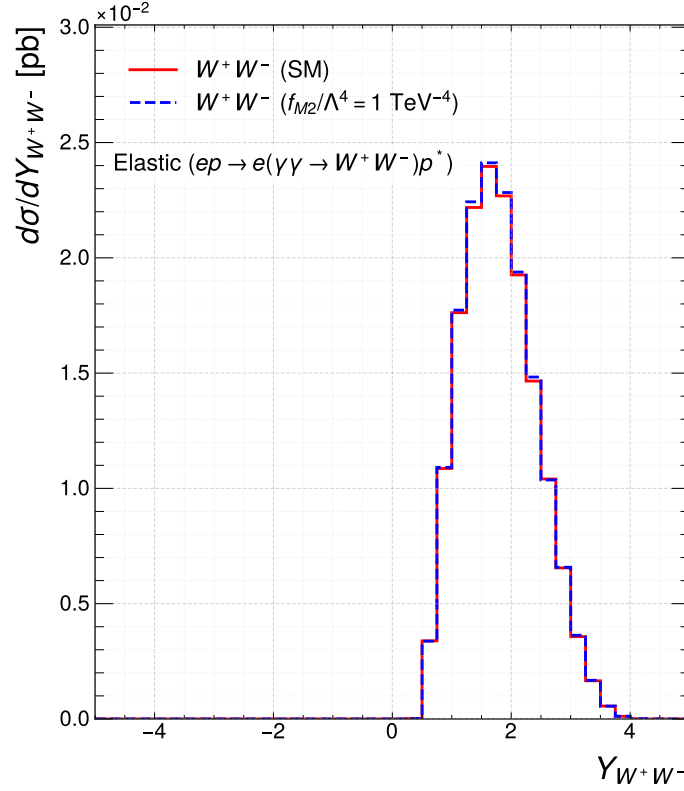


Figure 4. Rapidity distribution of the W^+W^- system in elastic photon-induced production at the LHeC. The red solid histogram shows the Standard Model prediction, while the blue dashed histogram corresponds to the benchmark $f_{M2}/\Lambda^4 = 1 \text{ TeV}^{-4}$. The positive-rapidity shift reflects the asymmetric LHeC beam configuration, with the 7 TeV proton beam defining the positive z direction.

1.4 Anomalous quartic gauge couplings in $\gamma\gamma \rightarrow ZZ$

Krzysztof's comment: For ZZ , we should quantify sensitivity to anomalous quartic gauge couplings at the total cross-section level, using for example

$$\frac{f_{M2}}{\Lambda^4} = 1 \text{ TeV}^{-4}.$$

The Standard Model contribution to $\gamma\gamma \rightarrow ZZ$ is loop-induced and therefore strongly suppressed. As a baseline, we use the EPA results obtained for the tagged elastic and inelastic topologies listed in Table 1. For an integrated luminosity of $\mathcal{L} = 1 \text{ ab}^{-1} = 10^6 \text{ pb}^{-1}$, the expected SM event yields are only of the order of a few tens of raw events before the Z -boson decay branching fractions, detector acceptance, and reconstruction efficiencies are applied.

Table 1. Baseline Standard Model cross sections for $ep \rightarrow e(\gamma\gamma \rightarrow ZZ)p^{(*)}$ at the LHeC and the corresponding raw event yields for $\mathcal{L} = 1 \text{ ab}^{-1}$. The SM contribution is loop-induced.

Topology	σ_{SM} [pb]	Raw events at 1 ab^{-1}
Tagged elastic	3.340×10^{-5}	33.4
Inelastic, $M_N < 10 \text{ GeV}$	3.305×10^{-5}	33.1
Tagged elastic + inelastic, $M_N < 10 \text{ GeV}$	6.645×10^{-5}	66.5

The AQGC contribution was generated with `MadGraph5_aMC@NLO` using the `SM_LM0123_UFO` model and the process $\gamma\gamma \rightarrow ZZ$. The UFO parameter is expressed in units of GeV^{-4} , so that

$$1 \text{ TeV}^{-4} = 10^{-12} \text{ GeV}^{-4}.$$

The generated samples correspond to the tree-level AQGC contribution only. Since the SM contribution is loop-induced, the total rate is estimated as

$$\sigma_{\text{tot}} \simeq \sigma_{\text{SM}}^{\text{loop}} + \sigma_{\text{AQGC}}.$$

The resulting AQGC cross sections and event yields are summarized in Table 2. The table uses the tagged elastic SM baseline, $\sigma_{\text{SM}}^{\text{el}} = 3.340 \times 10^{-5} \text{ pb}$, for the rate comparison.

Table 2. Tree-level AQGC contribution to $ep \rightarrow e(\gamma\gamma \rightarrow ZZ)p^{(*)}$ for different values of f_{M2}/Λ^4 . The cross sections were obtained with the MadGraph IWW photon-flux setup. The total rate is estimated as $\sigma_{\text{SM}}^{\text{el}} + \sigma_{\text{AQGC}}$, using $\sigma_{\text{SM}}^{\text{el}} = 3.340 \times 10^{-5} \text{ pb}$ as the tagged elastic SM baseline.

$\frac{f_{M2}}{\Lambda^4} [\text{TeV}^{-4}]$	σ_{AQGC} [pb]	N_{AQGC} at 1 ab^{-1}	$\sigma_{\text{SM}}^{\text{el}} + \sigma_{\text{AQGC}}$ [pb]	Enhancement
0.1	1.1716×10^{-8}	0.0117	3.3412×10^{-5}	1.0004
0.5	2.9298×10^{-7}	0.293	3.3693×10^{-5}	1.0088
1.0	1.1717×10^{-6}	1.17	3.4572×10^{-5}	1.035
5.0	2.9295×10^{-5}	29.3	6.2695×10^{-5}	1.88
10.0	1.1716×10^{-4}	117.2	1.5056×10^{-4}	4.51

To further illustrate the dependence of the neutral-boson channel on the dimension-eight aQGC coefficient, we study both the total aQGC cross section and the differential invariant-mass distribution of the ZZ system. Since the generated samples correspond to the tree-level aQGC contribution, while the SM contribution to $\gamma\gamma \rightarrow ZZ$ is loop-induced, the curves shown below should be interpreted as the EFT-induced contribution rather than the full SM+aQGC prediction. The total cross section displays the expected quadratic dependence on f_{M2}/Λ^4 , and is therefore approximately symmetric under $f_{M2} \rightarrow -f_{M2}$. This behavior confirms that the generated samples are dominated by the squared dimension-eight contribution.

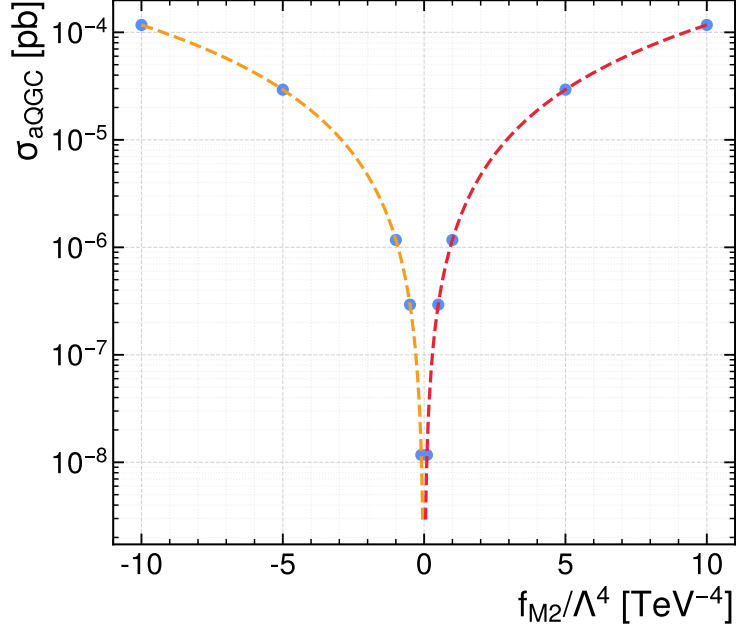


Figure 5. Total tree-level aQGC contribution to $ep \rightarrow e(\gamma\gamma \rightarrow ZZ)p^{(*)}$ as a function of the signed dimension-eight coefficient f_{M2}/Λ^4 . The cross section is shown in pb. The approximately symmetric dependence on the sign of f_{M2}/Λ^4 reflects the dominance of the squared aQGC amplitude in the generated samples. The loop-induced Standard Model contribution and the SM–aQGC interference are not included in this tree-level EFT generation.

The differential distribution $d\sigma_{\text{aQGC}}/dM_{ZZ}$ shows that the aQGC contribution is concentrated in the intermediate-to-high invariant-mass region, with a broad maximum around $M_{ZZ} \sim 600$ GeV for the LHeC beam configuration considered here. The shape is similar for different values of f_{M2}/Λ^4 , while the normalization increases rapidly with the magnitude of the coefficient. This confirms that the high- M_{ZZ} region is the most relevant part of phase space for probing the dimension-eight operator, although a full sensitivity estimate would require the inclusion of the loop-induced SM spectrum, Z -boson decay branching fractions, detector efficiencies, backgrounds, and EFT-validity considerations.

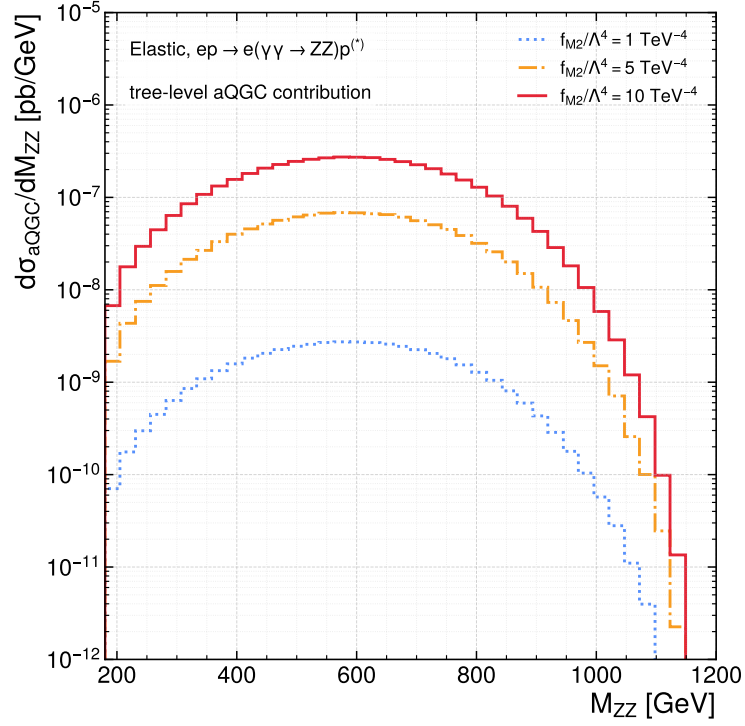


Figure 6. Differential invariant-mass distribution of the ZZ system for the tree-level aQGC contribution to $ep \rightarrow e(\gamma\gamma \rightarrow ZZ)p^{(*)}$ at the LHeC. The curves correspond to representative values $f_{M2}/\Lambda^4 = 1, 5, 10 \text{ TeV}^{-4}$. The distribution is shown in pb/GeV and peaks in the intermediate-to-high M_{ZZ} region, where dimension-eight aQGC effects are expected to be most pronounced. The normalization increases approximately as $(f_{M2}/\Lambda^4)^2$.

1.5 Compressed SUSY and higgsino-pair production: update of the most recent limits

Krzysztof's comment: We need to check and include the most recent limits on compressed SUSY.

A first survey of the most recent experimental constraints shows that compressed electroweak SUSY spectra are now significantly more constrained than in the earlier photon-induced slepton study used as motivation. The older forward-proton photon-fusion proposal showed sensitivity beyond LEP up to slepton masses of about (220 GeV) for $15 \lesssim \Delta m(\tilde{\ell}, \tilde{\chi}_1^0) \lesssim 60$ GeV, with (100 fb⁻¹) of (13 TeV) proton-proton data [1]. This remains conceptually relevant for our LHeC study because it demonstrates the usefulness of photon-induced production and kinematic closure from forward tagging. However, it is no longer sufficient as a summary of the present experimental status.

The current ATLAS and CMS searches cover a large part of the compressed electroweakino and slepton parameter space. In particular, the ATLAS Run-2 compressed electroweakino search with (139 fb⁻¹) excludes charginos in simplified models up to about (193)-(240 GeV), depending on the assumed mass splitting, and degenerate light-flavour sleptons up to about (251 GeV) [2]. A more recent ATLAS direct-slepton search in the compressed-mass corridor, using (140 fb⁻¹), excludes selectrons up to about (300 GeV) and smuons up to about (350 GeV) for $2 \text{ GeV} \lesssim \Delta m(\tilde{\ell}, \tilde{\chi}_1^0) \lesssim 100 \text{ GeV}$ [3].

Compressed higgsino scenarios are also constrained by dedicated low-momentum-track, soft-lepton, and disappearing-track searches. ATLAS excludes higgsino-like chargino masses below about (126 GeV) for $\Delta m(\tilde{\chi}_1^\pm, \tilde{\chi}_1^0) = 0.3\text{--}2 \text{ GeV}$ [4], while CMS excludes chargino masses up to about (185 GeV) for very compressed higgsino scenarios with ($\Delta m \simeq 0.55 \text{ GeV}$) [5]. CMS soft-lepton searches also exclude higgsino masses up to about (205 GeV) for ($\Delta m = 7.5 \text{ GeV}$), and about (150 GeV) for ($\Delta m = 3 \text{ GeV}$) [6].

Therefore, the low benchmark masses currently used in our exploratory LHeC plots, in particular ($m_{\tilde{\ell}} = 100 \text{ GeV}$) and ($m_{\tilde{H}} = 100 \text{ GeV}$), could be interpreted only as illustrative cross-section benchmarks. For the next step, we should keep the present plots as reference benchmarks, but add at least one higher-mass point, for example

$$m_{\tilde{\ell}} = 300 \text{ GeV} \quad \text{and} \quad m_{\tilde{H}} = 250 \text{ GeV or } 300 \text{ GeV}.$$

The final interpretation should describe the LHeC sensitivity as complementary to the LHC searches, benefiting from the clean (ep) environment, negligible pileup, and possible kinematic reconstruction with forward tagging.

The updated compressed-SUSY limits motivate considering a higher slepton-mass benchmark in addition to the illustrative $m_{\tilde{\ell}} = 100 \text{ GeV}$ case. We therefore also consider $m_{\tilde{\ell}} = 300 \text{ GeV}$, for which the pair-production threshold is $W_{\gamma\gamma}^{\text{thr}} = 2m_{\tilde{\ell}} = 600 \text{ GeV}$.

The corresponding integrated cross section as a function of the minimum two-photon invariant mass W_0 is shown in Fig. 7. A notable feature of this benchmark is that the tagged-elastic contribution vanishes in the plotted region. This is a consequence of the assumed forward-proton acceptance: the tagged elastic photon spectrum does not provide sufficient $W_{\gamma\gamma}$ reach above the 600 GeV threshold. In contrast, inelastic photon emission with larger

allowed diffractive masses and photon virtualities can still populate this high-mass region, although the resulting cross sections are very small.

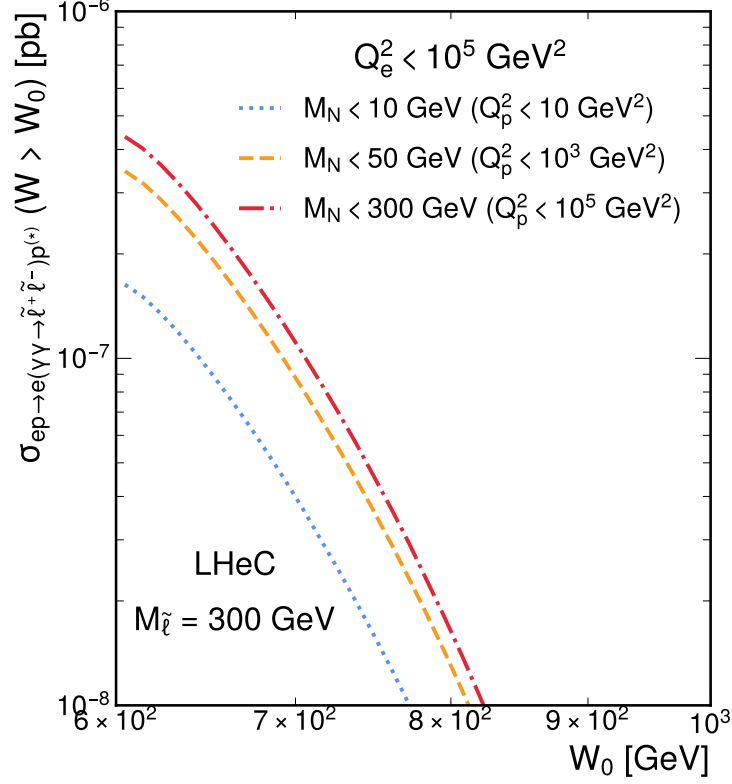


Figure 7. Integrated slepton-pair production cross section for $ep \rightarrow e(\gamma\gamma \rightarrow \tilde{\ell}^+\tilde{\ell}^-)p^{(*)}$ at the LHeC, shown as a function of the minimum two-photon invariant mass W_0 , for the benchmark mass $m_{\tilde{\ell}} = 300$ GeV. The pair-production threshold is $W_{\gamma\gamma} = 2m_{\tilde{\ell}} = 600$ GeV. Three inelastic selections are shown, corresponding to $M_N < 10$ GeV with $Q_p^2 < 10$ GeV², $M_N < 50$ GeV with $Q_p^2 < 10^3$ GeV², and $M_N < 300$ GeV with $Q_p^2 < 10^5$ GeV².

For $W_0 \simeq 600$ GeV, the integrated inelastic cross sections are of order 10^{-6} pb, corresponding to only a few raw events for an integrated luminosity of 1 ab^{-1} , as summarized in Table 3.

Table 3. Integrated cross sections and expected raw event yields for $ep \rightarrow e(\gamma\gamma \rightarrow \tilde{\ell}^+\tilde{\ell}^-)p^{(*)}$ at the LHeC for $m_{\tilde{\ell}} = 300$ GeV, evaluated at $W_0 = 600$ GeV, close to the production threshold $2m_{\tilde{\ell}} = 600$ GeV. Event yields are computed for $\mathcal{L} = 1 \text{ ab}^{-1}$, before branching fractions, detector acceptance, reconstruction efficiencies, and background rejection.

Topology	$\sigma(W > W_0)$ [pb]	Raw events at 1 ab^{-1}
Tagged elastic	0.0000	0.00
$M_N < 10 \text{ GeV}, Q_p^2 < 10 \text{ GeV}^2$	$\sim 1.61 \times 10^{-7}$	~ 0.16
$M_N < 50 \text{ GeV}, Q_p^2 < 10^3 \text{ GeV}^2$	$\sim 3.30 \times 10^{-7}$	~ 0.33
$M_N < 300 \text{ GeV}, Q_p^2 < 10^5 \text{ GeV}^2$	$\sim 4.32 \times 10^{-7}$	~ 0.43

The same high-mass benchmark can also be considered for charged-higgsino pair production. We therefore show in Fig. 8 the integrated cross section for $ep \rightarrow e(\gamma\gamma \rightarrow \tilde{H}^+\tilde{H}^-)p^{(*)}$ for $(m_{\tilde{H}} = 300 \text{ GeV})$. The corresponding production threshold is again $W_{\gamma\gamma}^{\text{thr}} = 2m_{\tilde{H}} = 600 \text{ GeV}$. As in the slepton case, the tagged-elastic contribution is not relevant in this high-mass region for the assumed forward-proton acceptance. The inelastic photon-emission topologies can still reach the threshold region, and the fermionic higgsino-pair cross section is larger than the scalar-slepton one. Nevertheless, the absolute rates remain very small, at the level of (10^{-6} pb) close to threshold.

For $(W_0 \simeq 600 \text{ GeV})$, the integrated inelastic cross sections are estimated from Fig. 8 to be of order (10^{-6} pb) , corresponding to approximately one raw event at (1 ab^{-1}) , as summarized in Table 4.

Table 4. Integrated cross sections and expected raw event yields for $ep \rightarrow e(\gamma\gamma \rightarrow \tilde{H}^+\tilde{H}^-)p^{(*)}$ at the LHeC for $m_{\tilde{H}} = 300$ GeV, estimated at $W_0 \simeq 600$ GeV, close to the production threshold $2m_{\tilde{H}} = 600$ GeV. The cross sections are read approximately from Fig. 8. Event yields are computed for $\mathcal{L} = 1 \text{ ab}^{-1} = 10^6 \text{ pb}^{-1}$, before branching fractions, detector acceptance, reconstruction efficiencies, and background rejection.

Topology	$\sigma(W > W_0)$ [pb]	Raw events at 1 ab^{-1}
Tagged elastic	0.0000	0.00
$M_N < 10 \text{ GeV}, Q_p^2 < 10 \text{ GeV}^2$	$\sim 5.5 \times 10^{-7}$	~ 0.55
$M_N < 50 \text{ GeV}, Q_p^2 < 10^3 \text{ GeV}^2$	$\sim 1.25 \times 10^{-6}$	~ 1.25
$M_N < 300 \text{ GeV}, Q_p^2 < 10^5 \text{ GeV}^2$	$\sim 1.6 \times 10^{-6}$	~ 1.6

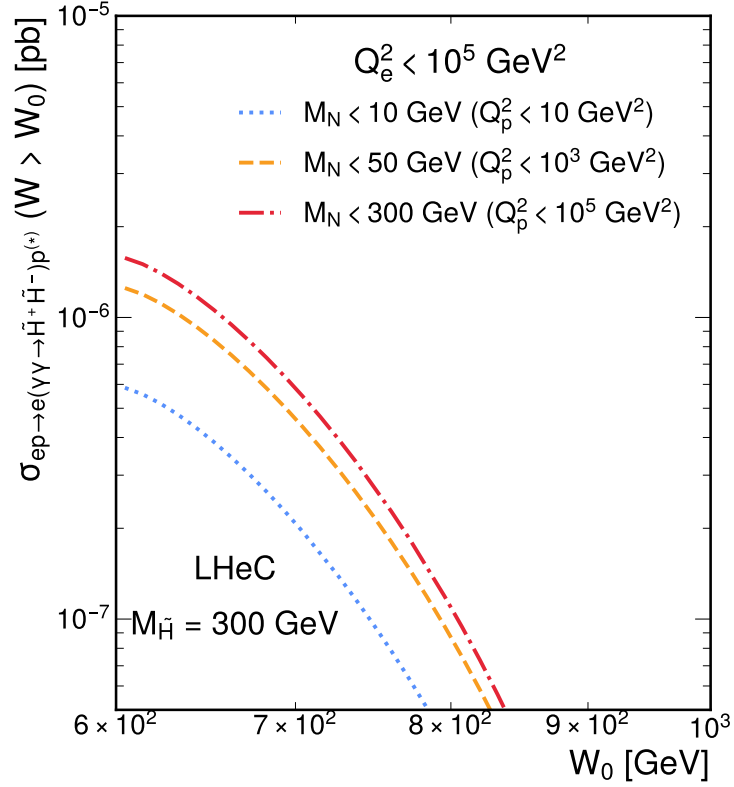


Figure 8. Integrated charged-higgsino-pair production cross section for $ep \rightarrow e(\gamma\gamma \rightarrow \tilde{H}^+\tilde{H}^-)p^{(*)}$ at the LHeC, shown as a function of the minimum two-photon invariant mass W_0 , for the benchmark mass $m_{\tilde{H}} = 300$ GeV. The pair-production threshold is $W_{\gamma\gamma} = 2m_{\tilde{H}} = 600$ GeV. Three inelastic selections are shown, corresponding to $M_N < 10$ GeV with $Q_p^2 < 10$ GeV², $M_N < 50$ GeV with $Q_p^2 < 10^3$ GeV², and $M_N < 300$ GeV with $Q_p^2 < 10^5$ GeV².

1.6 Photon-pair production: η distributions and New Physics sensitivity

Krzysztof's comment: For photon pairs, we should show η distributions for

$$W_0 = 10 \text{ GeV} \quad \text{and} \quad W_0 = 100 \text{ GeV},$$

and quantify or estimate sensitivities to New Physics.

The photon-pair final state probes Standard-Model light-by-light scattering,

$$ep \rightarrow e(\gamma\gamma \rightarrow \gamma\gamma)p^{(*)},$$

and provides a clean channel for anomalous four-photon interactions. We therefore study the inclusive photon pseudorapidity distribution, and the rapidity distribution of the diphoton system. The two thresholds $W_{\gamma\gamma} > 10 \text{ GeV}$ and $W_{\gamma\gamma} > 100 \text{ GeV}$ are used to separate the high-statistics SM region from the higher-mass region more relevant for New Physics.

The SM prediction is the loop-induced $\gamma\gamma \rightarrow \gamma\gamma$ amplitude, folded with the LHeC photon luminosities. The integrated cross sections and raw event yields are summarized in Table 5. Event yields are quoted for $\mathcal{L} = 1 \text{ ab}^{-1}$, before photon-identification efficiencies, acceptance losses, and background rejection.

Table 5. SM light-by-light cross sections and raw event yields for $ep \rightarrow e(\gamma\gamma \rightarrow \gamma\gamma)p^{(*)}$ at the LHeC.

Topology	$\sigma(W_{\gamma\gamma} > W_0) [\text{pb}]$	Raw events at 1 ab^{-1}
Tagged elastic, $W_0 = 10 \text{ GeV}$	1.294×10^{-2}	1.29×10^4
Inelastic, $W_0 = 10 \text{ GeV}$	9.404×10^{-2}	9.40×10^4
Tagged elastic, $W_0 = 100 \text{ GeV}$	5.720×10^{-5}	5.72×10^1
Inelastic, $W_0 = 100 \text{ GeV}$	1.125×10^{-4}	1.12×10^2

The resulting SM distributions are shown in Figs. 9 and 10. The diphoton rapidity is shifted toward the proton direction, as expected from the asymmetric ep beam energies. The η_γ distribution should be interpreted as a detector-acceptance diagnostic.

For the New Physics benchmark we consider the dimension-eight transverse neutral aQGC operator L_{T8} , implemented through the SM_LT8_LT9_UF0 model. In this convention the MadGraph parameter is

$$\text{FT8} = f_{T8}/\Lambda^4, \quad 1 \text{ TeV}^{-4} = 10^{-12} \text{ GeV}^{-4}.$$

Only f_{T8}/Λ^4 is varied, while all other anomalous couplings are set to zero. The generated samples correspond to the pure tree-level dimension-eight contribution,

$$\gamma\gamma \rightarrow \gamma\gamma \quad (\text{NP} = 1),$$

and therefore do not include the loop-induced SM amplitude or the SM-EFT interference. The corresponding raw yields are shown in Table 6 for a selection of f_{T8}/Λ^4 values.

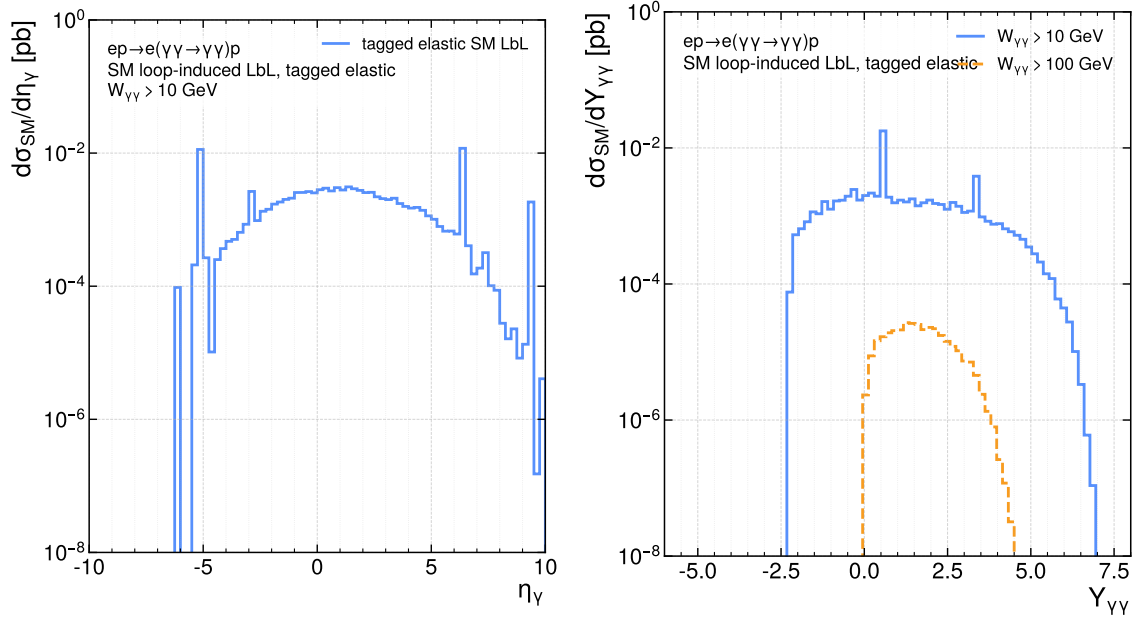


Figure 9. SM loop-induced light-by-light scattering at the LHeC in the tagged-elastic topology. Left: inclusive photon pseudorapidity distribution for $W_{\gamma\gamma} > 10$ GeV. Right: diphoton-system rapidity distributions for $W_{\gamma\gamma} > 10$ GeV and $W_{\gamma\gamma} > 100$ GeV.

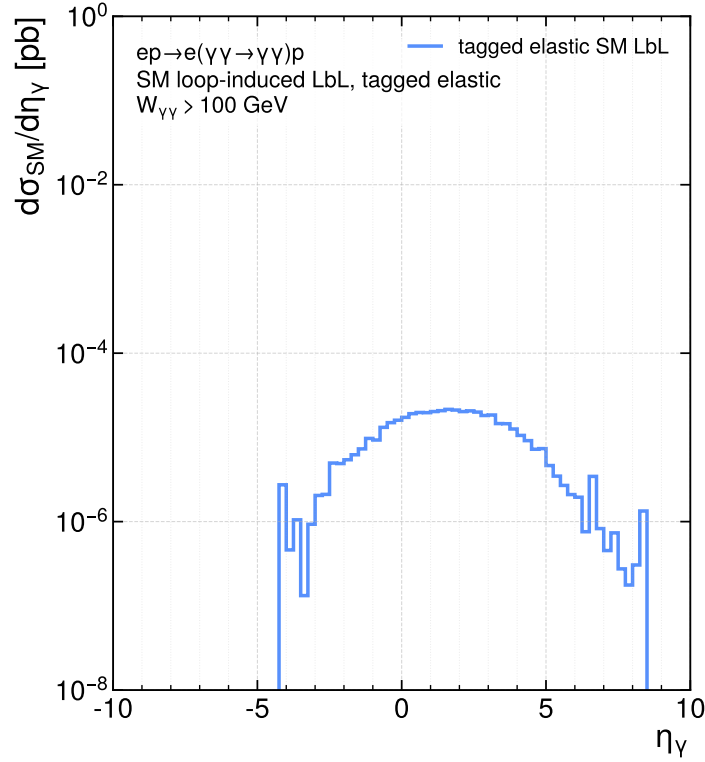


Figure 10. Inclusive photon pseudorapidity distribution for SM loop-induced light-by-light scattering in the tagged-elastic topology for $W_{\gamma\gamma} > 100$ GeV. This higher-mass selection is more directly relevant for New Physics searches, although the SM rate is much smaller than for $W_{\gamma\gamma} > 10$ GeV.

Table 6. Pure dimension-eight L_{T8} contribution to $ep \rightarrow e(\gamma\gamma \rightarrow \gamma\gamma)p^{(*)}$. The event yields correspond to $L = 1 \text{ ab}^{-1}$.

$f_{T8}/\Lambda^4 [\text{TeV}^{-4}]$	$\sigma_{\text{EFT}} [\text{pb}]$	Raw events at 1 ab^{-1}
0.5	5.69×10^{-5}	5.69×10^1
1	2.28×10^{-4}	2.28×10^2
5	5.69×10^{-3}	5.69×10^3
10	2.28×10^{-2}	2.28×10^4

The dependence of the pure dimension-eight contribution on the Wilson coefficient is shown in Fig. 11. The cross section is symmetric under $f_{T8}/\Lambda^4 \rightarrow -f_{T8}/\Lambda^4$, as expected for the generated NP = 1 samples, where only the tree-level anomalous four-photon contribution is included. In this setup the loop-induced SM light-by-light amplitude and the SM-EFT interference are absent, so the rate is dominated by the quadratic EFT term,

$$\sigma_{\text{EFT}} \propto (f_{T8}/\Lambda^4)^2.$$

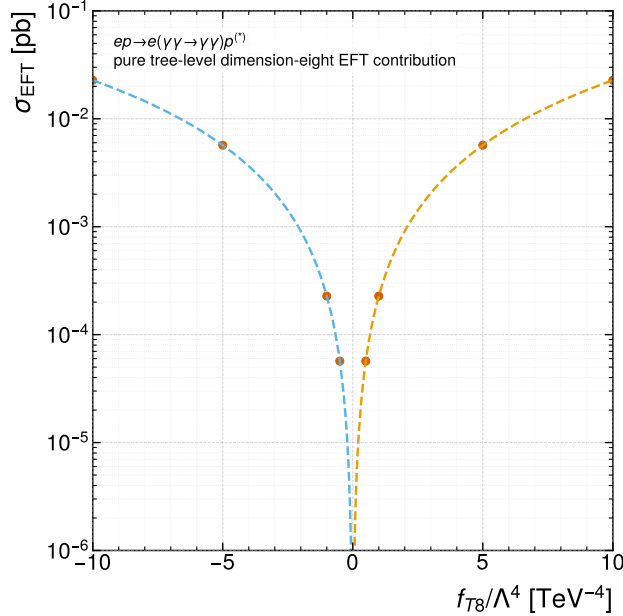


Figure 11. Pure tree-level dimension-eight EFT cross section for $ep \rightarrow e(\gamma\gamma \rightarrow \gamma\gamma)p^{(*)}$ as a function of f_{T8}/Λ^4 . The symmetry between positive and negative Wilson coefficients reflects the absence of the loop-induced SM amplitude and the SM-EFT interference in the generated NP = 1 samples.

The EFT distributions are shown in Figs. 12 and 13. As expected for a dimension-eight contact interaction, the contribution is concentrated at large $M_{\gamma\gamma}$ and sizeable photon transverse momentum, and the normalization follows the approximate f_{T8}^2 scaling. A full experimental sensitivity estimate would require the same fiducial selections applied to the SM and EFT samples, photon reconstruction efficiencies, fake-photon backgrounds, and an EFT-validity prescription such as a truncation in $W_{\gamma\gamma}$.

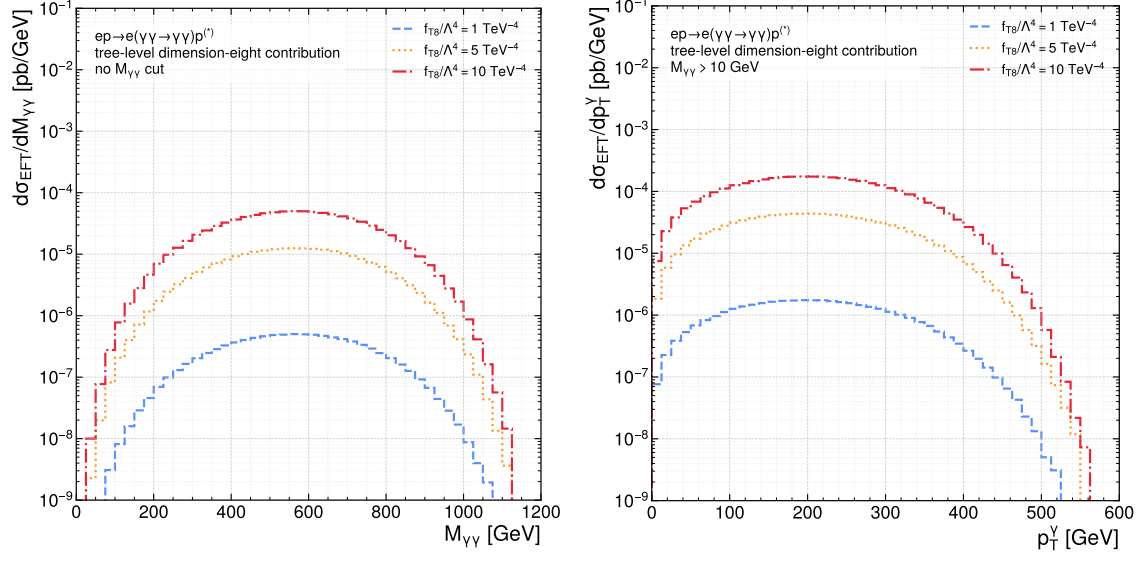


Figure 12. Pure dimension-eight L_{T8} contribution to photon-pair production at the LHeC. Left: diphoton invariant-mass distribution. Right: inclusive photon transverse-momentum distribution for $M_{\gamma\gamma} > 10$ GeV.

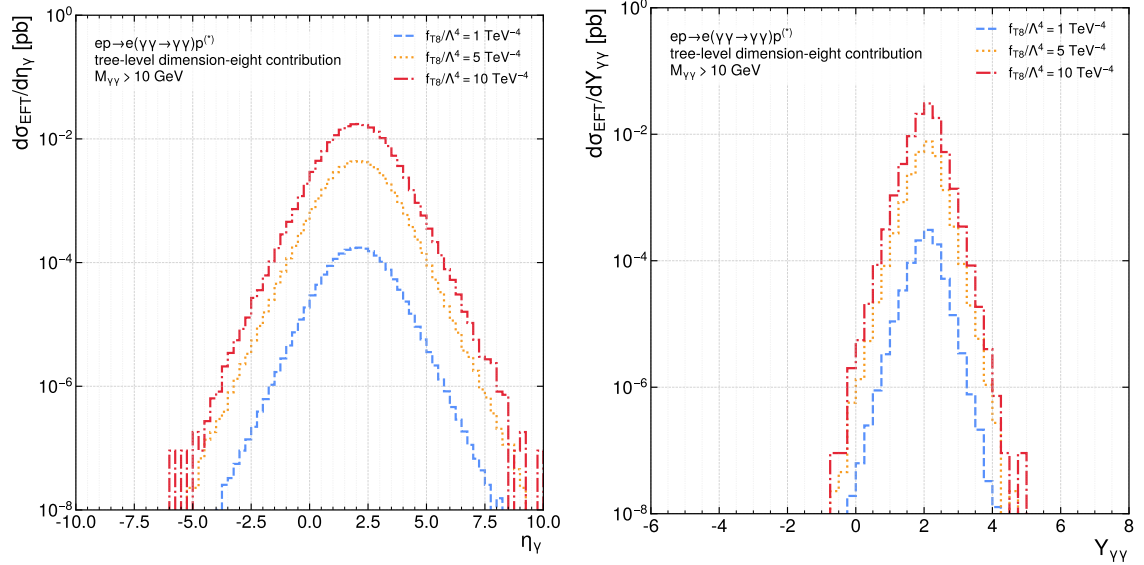


Figure 13. Pure dimension-eight L_{T8} contribution for $M_{\gamma\gamma} > 10$ GeV. Left: inclusive photon pseudorapidity distribution. Right: diphoton-system rapidity distribution. The curves correspond to $f_{T8}/\Lambda^4 = 1, 5, 10$ TeV $^{-4}$, with one Wilson coefficient switched on at a time.

2 Introduction

The proposed Large Hadron-Electron Collider (LHeC) at CERN will open fascinating fields of research [7]. Among these, the study of high-energy photon-photon ($\gamma\gamma$) interactions stands out as an exceptionally exciting and unique area of investigation. This paper aims to explore the potential and prospects of such $\gamma\gamma$ interactions at the LHeC, highlighting the novel opportunities this facility will offer for both precision measurements and the discovery of new physics phenomena.

This new collider is initially set to operate alongside the High Luminosity LHC (HL-LHC). With a center-of-mass energy of $\sqrt{s} = 1.2$ TeV for electron-proton collisions at the LHeC, and electron and proton beam energies of $E_e = 50$ GeV and $E_p = 7$ TeV, respectively, the photon-photon collision energies will often exceed the threshold for electroweak interactions and may even approach the 1 TeV scale. Moreover, the exceptionally high nominal LHeC luminosity, exceeding $10^{34} \text{ cm}^{-2}\text{s}^{-1}$, will facilitate original and stringent tests of Standard Model (SM) predictions while offering opportunities to probe novel manifestations of physics beyond the SM (BSM). The anticipated total integrated luminosity at the LHeC is approximately 1 ab^{-1} [7].

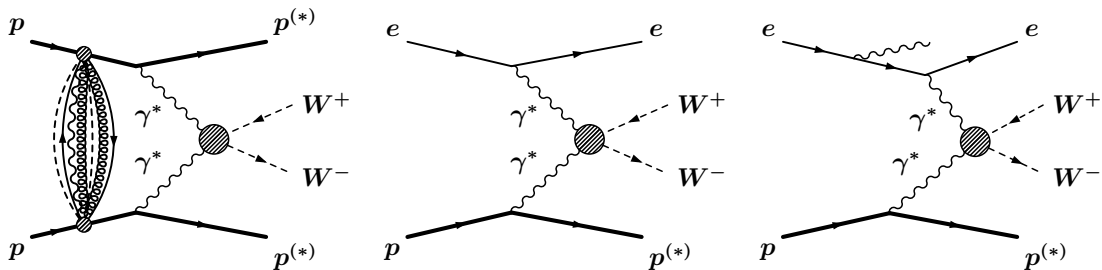


Figure 14. Feynman diagrams representing the exclusive $W^\pm$ and Z boson pair production via photon-photon fusion at the LHC (left) and future collider LHeC (middle), including with initial state radiation (right). The additional exchange between protons represents the hadronic re-scattering, absent in electron-proton collisions at the LHeC.

In large contrast to the HL-LHC, the experimental conditions at the LHeC will be highly favorable, particularly for studying exclusive production through high-energy $\gamma\gamma$ interactions. In particular, event pileup will be negligible, and certain sources of overwhelming background, such as those arising from the Drell-Yan process, will be highly suppressed. Moreover, no event suppression will occur due to hadronic re-scattering, and the associated significant theoretical and experimental uncertainties will not interfere (see the Feynman diagrams shown in Fig. (14)).

It should be noted that measurements of high-energy two-photon interactions at the LHC were originally proposed to be performed using dedicated, very forward proton detectors to cope with the substantial backgrounds in proton-proton collisions [8, 9]. However, the initial pioneering studies of $\gamma\gamma$ interactions at the LHC were conducted without the aid of such detectors. These studies were feasible, albeit limited to the exclusive production of pairs of charged leptons ($\ell^\pm$) via two-photon processes [10, 11]. Only recently were the first

measurements utilizing very forward proton detectors reported [12–18]. As discussed further below, the exceptionally "clean" experimental environment of the LHeC will significantly expand the scientific scope of high-energy $\gamma\gamma$ research at CERN.

The structure of this paper is as follows: Sec. 3 presents the theoretical framework used in this work. Sec. 4 examines the capabilities of the LHeC as a photon-photon factory for τ pairs, providing insights into τ lepton properties. Sec. 5 discusses the exclusive production of electron and muon pairs, highlighting the precision measurements possible at the LHeC. Sec. 6 focuses on the exclusive production of W boson pairs, detailing the potential for studying electroweak interactions. Sec. 7 explores the LHeC as a unique laboratory for Z boson studies. Sec. 8 investigates the production of the Higgs boson via photon-photon fusion, emphasizing the clean signatures and potential discoveries. Sec. 9 presents the two-photon exclusive production of charged supersymmetric pairs, with a particular focus on sleptons and higgsinos. Finally, Sec. 10 covers light-by-light scattering, a process that can be effectively studied at the LHeC. The paper concludes with a summary and a discussion of future prospects for high-energy $\gamma\gamma$ physics at the LHeC.

3 Theory input

The calculations of cross-sections for two-photon interactions in electron-proton collisions can be performed using the *Equivalent Photon Approximation* (EPA) framework [19, 20]. In this approach, the cross-sections are factorized in a manner similar to the partonic framework used for hadron-hadron collisions in perturbative QCD. The cross-section of a given interaction at the LHeC, proceeding via photon-photon fusion, can be accurately calculated by a convolution of the equivalent photon fluxes for electrons and protons, Φ_e and Φ_p , respectively, multiplied by the appropriate photon-photon cross-section, $\sigma_{\gamma\gamma}(W)$,

$$\sigma_{ep} = \int dy_e dy_p \Phi_e(y_e) \Phi_p(y_p) \sigma_{\gamma\gamma}(W) = \int dW S_{\gamma\gamma} \sigma_{\gamma\gamma}, \quad (3.1)$$

where $y_e = E_{\gamma(e)}/E_e$ (resp. $y_p = E_{\gamma(p)}/E_p$) refer to the respective photon-from-electron (resp. photon-from-proton) fractional energies, and the $\gamma\gamma$ center-of-mass energy is given by $W = (y_e y_p s)^{1/2}$. The photon-photon luminosity spectrum $S_{\gamma\gamma}$ is therefore equal to the flux convolution at a given value of W (see Eq. (E.5)).

Such a formula for the luminosity spectrum $S_{\gamma\gamma}$, in which the initial state interactions are supposed weak, and neglected [20], can be applied to the $\gamma\gamma$ interactions in both electron-proton and electron-nucleus collisions.

When considering the two-photon production of lepton pairs, it is essential to note certain limitations of the EPA approach. Although the EPA remains a useful framework for calculating cross-sections in many cases, there are scenarios where it fails for certain regions of the phase space assuming $W^2 \gg m_\ell^2$, particularly in processes involving electron and, to some extent, also muon and tau leptons. Indeed, in computing the cross-sections for the various $\gamma\gamma \rightarrow \ell^+ \ell^-$ subprocesses, the dependence on the virtualities photons can no longer be neglected, as discussed in Appendix E and Eq. (6.25) of Ref. [19], along with insights from Ref. [20]. As show here, the standard EPA can however be applied with smaller corrections for the region of $q^2 < W^2$. **Not sure what you mean there**

In Eq. (3.1), the photon-photon cross-section $\sigma_{\gamma\gamma}(W)$ is assumed not to be sensitive to photon virtualities. Therefore, electrons fluxes Φ_e and protons fluxes Φ_p can be integrated over the photon Q^2 ,

$$\Phi_\gamma(y) = \frac{\alpha_{e-m}}{\pi y} \int \frac{dQ^2}{Q^2} \left[(1-y) \left(1 - \frac{Q_{\min}^2}{Q^2} \right) F_E(Q^2) + \frac{y^2}{2} F_M(Q^2) \right],$$

where α_{e-m} is the fine-structure constant and F_E, F_M are the electric and magnetic form-factors, respectively [19].

For the case of electrons the trivial $F_E = F_M = 1$ form factors are assumed. Whereas for the protons, assuming *elastic* scattering and the dipole approximation, the magnetic and electric form-factors can be expressed as [8]:

$$F_M(Q^2) = G_M^2(Q^2), \quad F_E(Q^2) = \frac{4m_p^2 G_E^2 + Q^2 G_M^2}{4m_p^2 + Q^2},$$

where $G_E^2 = G_M^2/7.78 = (1 + Q^2/0.71 \text{ GeV}^2)^{-4}$ are the Sachs form factors in the dipole approximation and m_p is the proton mass. In the integral, $Q_{\max}^2$ is set according to the experimental conditions and $Q_{\min}^2 = M^2 y^2 / (1-y)$, with M the initiator particle (electron, or proton) mass.

The case of *inelastic* productions is more complex, encompassing topologies where the proton does not survive the interaction and dissociates into a diffractive state. This latter can be expressed through its diffractive invariant mass $M_N > m_p$, defining along with the Q^2 virtuality range of the emitted photon, a continuum between multiple dominant regimes. Expressed in terms of deep inelastic scattering (DIS) structure functions, inelastic form factors can be approximated as

$$F_E(Q^2) = \int dx_{\text{Bj}} \frac{F_2(x_{\text{Bj}}, Q^2)}{x_{\text{Bj}}}, \quad F_M(Q^2) = \int dx_{\text{Bj}} \frac{F_2(x_{\text{Bj}}, Q^2)}{x_{\text{Bj}}^3},$$

respectively, where $F_2(x_{\text{Bj}}, Q^2)$ is the transverse proton structure function and $M_N^2 - m_p^2 = Q^2(1/x_{\text{Bj}} - 1)$. In order to calculate the $F_E(Q^2)$ and $F_M(Q^2)$, we use the ALLM parametrization for the proton structure function, describing well the DIS data for a broad kinematical region of $10^{-6} < x_{\text{Bj}} < 0.85$ and $0 \leq Q^2 < 5000 \text{ GeV}^2$ [21, 22]. In inelastic production processes, minimal photon virtualities are generally larger than their elastic counterparts, as $Q_{\min}^2 = (M_N^2/(1-y) - m_p^2)y$ [19]. In this study we also examined other, more recent **photon** PDFs, covering a broader kinematical region of interest: namely, CT18 [23] and NNPDF4.0 [24], with $Q_{\min}^2 > Q_0^2$. Results consistent with the ALLM parametrization are obtained. **LF: do we have numbers? relative differences?** We should mention here that, in the inelastic photon emission case, tails of the photon-from-proton flux Φ_p may extend to large Q^2 values. This contrasts with the elastic case, where the photon virtuality strongly peaks at low values of Q^2 , with an upper limit around 1 GeV^2 . This is a direct consequence to the flux dependence to elastic form factors, highly suppressed at large photon virtuality values.

In Fig. 3 the elastic and inelastic luminosity spectra $S_{\gamma\gamma}$ at the LHeC, and their integrals $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma}$, are shown for selected values of photon virtualities at the electron/proton

vertex, with an additional constraint on the diffractive system mass M_N for the inelastic case. Effectively, these integrals provide the fraction of the total LHeC ep luminosity available for $\gamma\gamma$ collisions above a given minimal center-of-mass energy W_0 .

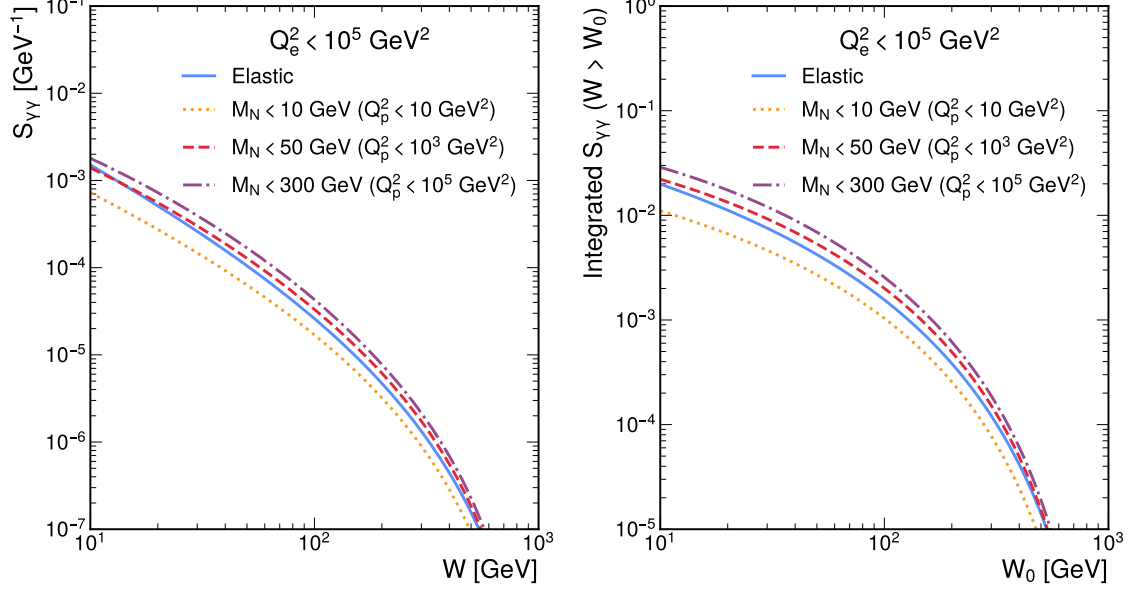


Figure 15. (Updated by exact EPA) (left) Elastic luminosity spectrum $S_{\gamma\gamma}$ at the LHeC, assuming $Q_{e,\max}^2 = 10^5 \text{ GeV}^2$ and appropriate kinematical limits for the minimal photon virtualities (see text above), as well as three inelastic luminosity spectra where the dissociative mass $M_N < 10, 50$, and 300 GeV is assumed for $Q_{p,\max}^2 = 10, 10^3$, and 10^5 GeV^2 , respectively ; (right) corresponding integrated luminosity spectra, $\int dW S_{\gamma\gamma}$, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 .

For comparison with the phased LHeC option, where the 50 GeV electron beam three-pass ERL design is replaced with a 20 GeV single-pass ERL [25], the two-photon luminosity is shown in Fig. 3 for an ep center-of-mass energy of 0.75 TeV.

It is remarkable that the total fraction of the electron-proton level luminosity, for the sum of elastic and inelastic contributions, is expected to exceed 20% for minimal two-photon masses as low as $W_0 = 10 \text{ GeV}$. This will effectively allow to convert the LHeC into, *e.g.* a precision $\tau^+\tau^-$ factory, as discussed in Sec. 4, or a "light-by-light scattering machine" (see Sec. 10). As seen in Fig. 3, the inelastic contribution dominates over the full W range, if very large dissociative masses (and large photon virtualities) are allowed. In elastic events, a significant fraction of very forward-scattered protons can be detected through near-beam detectors. This allows for improved reconstruction of event kinematics, which is particularly useful for final states containing neutrinos. However, thanks to the lack of event pileup and generally low backgrounds, detecting these protons is often not necessary to improve the purity of the majority of signal selections. Therefore, as discussed below, in many cases a large fraction of the total $\gamma\gamma$ luminosity is expected at LHeC for the study photon-photon interactions.

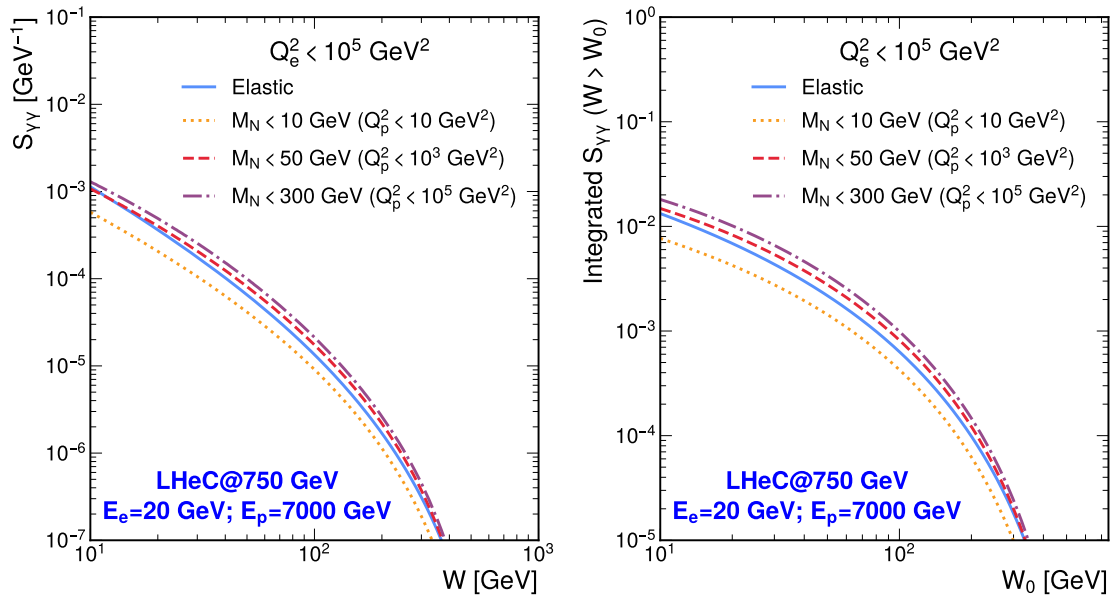


Figure 16. (Updated by exact EPA) Same as Fig. 3, for the LHeC center of mass energy of 0.75 TeV.

4 Photon-photon factory of τ pairs

The production of lepton pairs ($\ell^+\ell^-$) in photon-induced interactions draws significant interest, being a pure quantum electrodynamics (QED) process for which the SM cross-section is known within a high degree of accuracy. **LF: per-mille precision?** Such processes are marked by a particularly clean experimental signature, only requiring the detection and reconstruction of simple objects like leptons and photons, in a relatively central and well instrumented region around the interaction point. Among these, the exclusive production of τ pairs through two-photon interactions ($\gamma\gamma \rightarrow \tau^+\tau^-$) is a potentially interesting process seeking to be investigated in electron-proton collisions [7].

Their first observation in the electron-proton configuration, along with the extraction of their production rate, was reported using data recorded by the H1 [26], and ZEUS [27] detectors at HERA. Later, it has also been observed by the CMS [28] and ATLAS [29] experiments in Ultra-Peripheral Collisions (UPC) of lead nuclei. And more recently, the CMS Collaboration reported the first measurement of photon-induced production of a pair of τ leptons in proton-proton collisions at the LHC [18].

Generally speaking, the production of isolated lepton pairs ($\ell^+\ell^-$) is dominated in most of the phase space reachable experimentally by the two-photon Bethe-Heitler (2γ BH) process and can be accurately predicted within the SM [30]. Both Feynman diagrams for the elastic, and inelastic photon-photon production of τ pairs at the LHeC are shown in Fig. 17.

In terms of backgrounds, it is expected that QED-Compton type diagrams become dominant for lower invariant masses of the dilepton system, while the additional Z exchange

process is dominant in the high-mass region. As discussed later in this review, the final state of τ lepton pairs can be identified by detecting their decay products, occurring exclusively through leptonic, semi-leptonic, and fully hadronic decay modes.

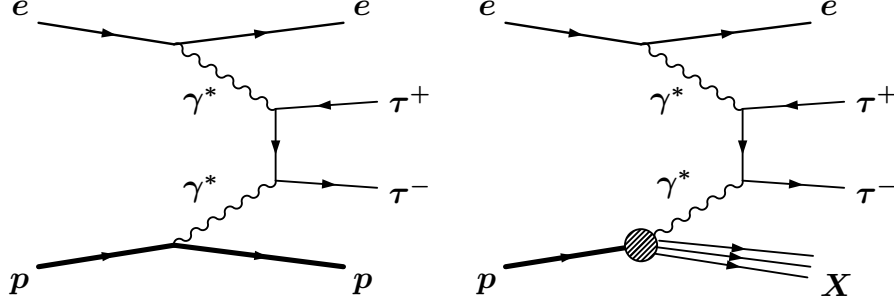


Figure 17. Schematic Feynman diagram for the elastic (left), and dissociative (right) two-photon production of τ pairs ($\gamma\gamma \rightarrow \tau^+\tau^-$) at the LHeC.

With the two-photon luminosity spectrum $S_{\gamma\gamma}$ at hand, and considering the photon-photon cross-section $\sigma(\gamma\gamma \rightarrow \tau^+\tau^-)(W)$ for tau pair production as given in Eq. (B.3), one can calculate the production cross-sections at the LHeC using the EPA approach [19, 20] as discussed earlier. The integrated τ pair production cross sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{\gamma\gamma \rightarrow \tau^+\tau^-}$, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 , both for the elastic and inelastic processes are shown in Fig. 18. The left panel displays the cross sections for photon virtualities of $Q_e^2 < 10 \text{ GeV}^2$ and $Q_p^2 < 10 \text{ GeV}^2$ for the electron and proton vertices, respectively, considering a diffractive system mass $M_N < 3 \text{ GeV}$. The right panel presents the cross sections calculated for slightly higher values of $Q_{e,p}^2 < 10^2 \text{ GeV}^2$ and $M_N < 10 \text{ GeV}$.

For the nominal LHeC luminosity of $2.3 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ [7], the total effective $\gamma\gamma$ luminosity reaches $10^{33} \text{ cm}^{-2}\text{s}^{-1}$, for $W > 10 \text{ GeV}$ (see Fig. 18). This opens yet another unique and significant research field – for studies of the two-photon production of tau lepton pairs. According to the cross-section for $\gamma\gamma \rightarrow \tau^+\tau^-$ (see Eq. B.3), convoluted with the luminosity spectrum $S_{\gamma\gamma}$ (see Fig. 18), one expects more than 10^8 τ pairs elastically produced via photon-photon fusion at $W > 10 \text{ GeV}$, and among them about two million pairs produced above 50 GeV, assuming the LHeC integrated luminosity of 1 ab^{-1} [7].

To provide further weight to the discussion above, distributions of the differential cross-sections $d\sigma/dW$ (in pb/GeV) and $d\sigma/dY_{\tau^+\tau^-}$ (in pb), calculated using the EPA approach, are presented in Fig. 19. These distributions are compared with those obtained from the CepGen Monte Carlo (MC) event generator [31] implementation of the LPAIR process [32], both for the elastic and inelastic cases, assuming $M_N < 3 \text{ GeV}$ for the diffractive system mass, and $Q_e^2 < 10 \text{ GeV}^2$ and $Q_p^2 < 10 \text{ GeV}^2$ for the photon-from-electron and photon-from-proton emissions. It is worth noting that these upper limits for the photon virtualities are imposed to ensure the numerical convergence of the EPA convolution of the two-photon fluxes, as defined in Eq. 3.1. In addition, comparisons are also made considering higher values $Q_e^2 < 10^2 \text{ GeV}^2$ and $Q_p^2 < 10^2 \text{ GeV}^2$. For the latter case, an upper limit for the diffractive system mass of $M_N < 10 \text{ GeV}$ is assumed.

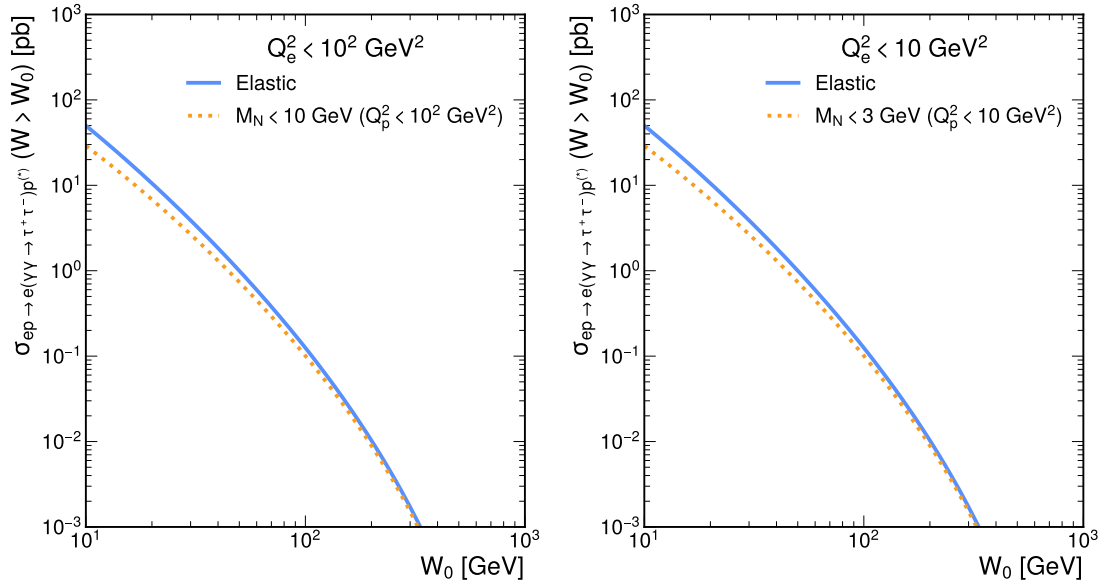


Figure 18. (Updated by exact EPA) Integrated τ pair production cross sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{\tau^+\tau^-}$, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 , through the elastic process as well as inelastic production with diffractive system mass of $M_N < 3$ GeV (left) and $M_N < 10$ GeV (right).

As depicted in Fig. 19, EPA calculations for both elastic and inelastic processes agree quite well with CEPGEN/LPAIR for the given definition of kinematics constraints shown in the plots. However, it should be noted here that the calculation of LPAIR is based on the diagrams of the two-photon BH process only, dominant in most of the phase space. In the high mass region, an additional interesting process involving Z exchange is expected to contribute significantly as well. It is currently not included in any implementations of the LPAIR process but is accounted for in GRAPE [33]. Additionally, GRAPE incorporates the effects of initial state radiation (ISR). As observed in Fig. 19, despite minor corrections mentioned earlier, it is evident that the EPA provides a good approximation for estimating the exclusive two-photon production cross-sections of tau pairs across a wide kinematic range typically accessible at LHeC energies.

Detailed comparisons between EPA, CEPGEN/LPAIR, and GRAPE calculations at the minimum two-photon invariant mass $W_0 = 10$ GeV are presented in Tables 7 and 8, for the elastic and inelastic cases respectively, for selected values of photon virtualities at the electron/proton vertex, and different diffractive system masses for the inelastic case.

In the *elastic* case, the EPA approach provides a very good estimate for exclusive $\gamma\gamma \rightarrow \tau^+\tau^-$ cross-sections at the LHeC, covering a broad kinematic range as shown in the tables. Additionally, the EPA calculations exhibit more compatibility with those of GRAPE, which includes the interesting process involving Z . Table 7 also illustrates contributions from ISR incorporated in GRAPE, leading to approximately 5-10% corrections to the total cross-section.

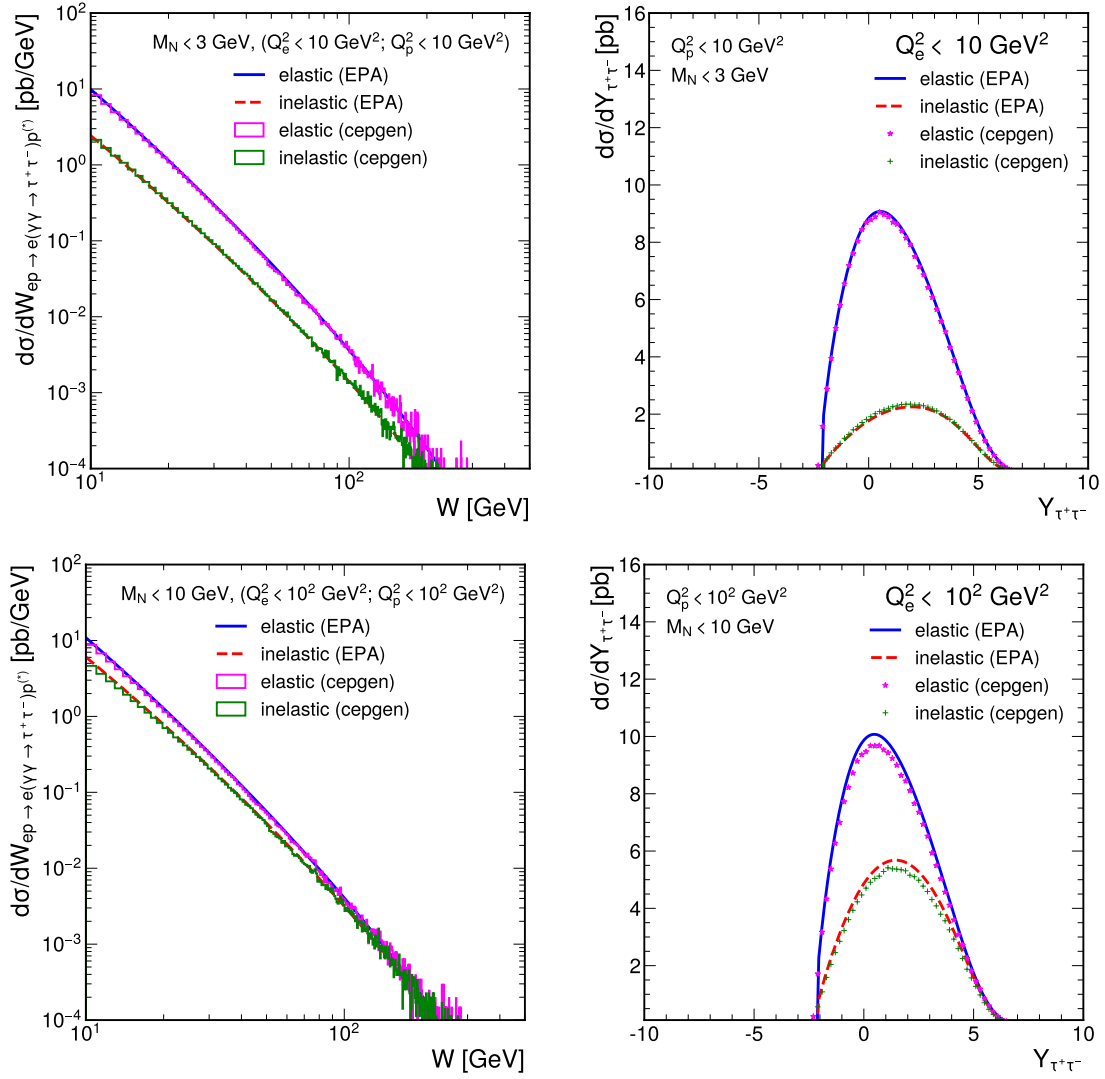


Figure 19. Central system invariant mass (left), and rapidity dependence for the $ep \rightarrow e(\gamma\gamma \rightarrow \tau^+\tau^-)p^{(*)}$ process cross section at LHeC energies, for elastic (top), and inelastic (bottom) photon emissions from the proton side. Both the EPA (lines) approach and CEPGEN/LPAIR unweighted events (histograms) are shown for comparison.

The *inelastic* $\gamma\gamma \rightarrow \tau^+\tau^-$ cross sections are presented in Table. 8 for selected values of diffractive system mass and photon virtualities at the electron/proton vertex. Values calculated using GRAPE and CEPGEN/LPAIR are also provided for comparison. As previously discussed, the EPA calculations employ the ALLM structure function [22]. However, as seen from Table. 8, it is noteworthy that using alternative structure functions, such as the hybrid LUX-like set of structure functions following the same approach as [34], also yields excellent agreement with ALLM. Nevertheless, a modern re-parametrization of ALLM is used in LUX-like: the GD11p [35] fits including HERMES, JLAB, and HERA data.

We should note here that in GRAPE, the DIS proton structure functions are also hybrid

Table 7. (Updated by exact EPA) Elastic $\tau^+\tau^-$ production cross sections (in pb) at LHeC calculated using the EPA approach, assuming different values of Q_e^2 and Q_p^2 for the electron/proton vertex and for a minimum two-photon invariant mass of $W_0 = 10$ GeV. Cross section values calculated using GRAPE and LPAIR (with dipole electromagnetic form factors) are presented for comparison.

σ^{el} (pb)	$Q_{e,p}^2 < 10 \text{ GeV}^2$	$Q_{e,p}^2 < 50 \text{ GeV}^2$	$Q_{e,p}^2 < 100 \text{ GeV}^2$
EPA	46.03	48.98	49.98
CEPGEN (LPAIR, dipole FF)	45.52	48.03	48.82
GRAPE (BH)	46.30	48.82	49.54
GRAPE (BH+ISR)	46.88	49.58	50.35
GRAPE (EW)	46.53	49.07	49.84
GRAPE (EW+ISR)	47.05	49.78	50.60

Table 8. (Updated by exact EPA) Inelastic $\tau^+\tau^-$ production cross sections (in pb) at LHeC calculated using the EPA approach, assuming different values of Q_e^2 and Q_p^2 for the electron/proton vertex and for a minimum two-photon invariant mass of $W_0 = 10$ GeV. Cross section values calculated using GRAPE and LPAIR (with Suri-Yennie proton structure functions) are presented for comparison.

σ^{inel} (pb)	$M_N < 3 \text{ GeV}$	$M_N < 10 \text{ GeV}$
	$Q_{e,p}^2 < 10 \text{ GeV}^2$	$Q_{e,p}^2 < 100 \text{ GeV}^2$
EPA (ALLM)	12.20	28.98
EPA (LUX-like)	11.63	28.30
CEPGEN (LPAIR, Suri-Yennie)	12.74	28.17
GRAPE (BH)	11.52	26.87
GRAPE (BH + ISR)	12.53	28.94
GRAPE (EW)	11.54	26.92
GRAPE (EW + ISR)	12.77	29.27

parameterizations: Fiore-Brasse *et al.* [21] for hadronic system masses lower than 2 GeV (the proton resonance region), and ALLM [22] for the remaining phase space. While the original implementation of LPAIR handled Fiore-Brasse structure functions for the resonances regime, or Suri-Yennie for the continuum region, the CEPGEN implementation of this process handles a broader collection of structure functions modelings. For the case of LPAIR, in Table. 8, we show the cross section calculated using the Suri-Yennie [36] parametrization of $F_{2,L}$ structure functions in the continuum region. The EPA calculations are found to be in good agreement with those of both GRAPE and CEPGEN/LPAIR for the kinematics shown in the table.

From an experimental point of view, several signal topologies could be considered to measure the photon-induced production of a pair of τ leptons at the LHeC. Among these, a commonly studied decay mode involves one τ lepton decaying hadronically and the other leptonically. For the production of τ leptons through photon-photon interaction, the following discussion on the main contributing backgrounds can be provided. The most significant

background arises from the production of electron or muon pairs at the LHeC, $ep \rightarrow ee^+e^-X$ and $ep \rightarrow e\mu^+\mu^-X$. In these cases, events with two leptons (ee , $\mu\mu$, or $e\mu$) may mimic events involving the production of τ pairs where both tau leptons decay leptonically. However, this background can be highly suppressed by considering the hadronic decay of at least one tau lepton. In addition, the photon-induced production of a pair of W bosons, $\gamma\gamma \rightarrow W^+W^-$, could also contribute as a background. It is worth highlighting here that unlike the LHC, where the dominant background for τ lepton production comes mainly from Drell–Yan processes involving lepton pairs, such backgrounds are entirely absent in electron-proton collisions at the LHeC.

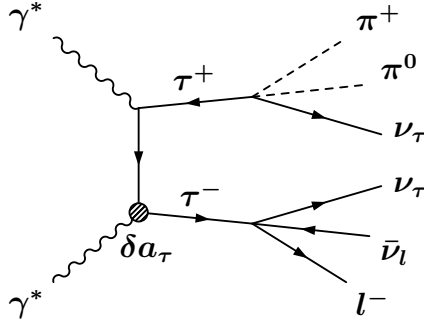


Figure 20. Feynman diagram for the two-photon production of τ pairs, with both leptonic and hadronic decays of the τ leptons (here, chosen arbitrarily). This process, through the $\tau\gamma$ couplings affecting the magnetic moment, is sensitive to BSM extensions.

4.1 Anomalous magnetic moment of τ lepton

An interesting electroweak potential of production of τ pairs via photon-photon interaction is the competitive determination of the electric and magnetic dipole moments of the τ lepton. These measurements are highly sensitive to New Physics [37–43]. Precision measurements of electromagnetic couplings, particularly the anomalous magnetic moment of the τ lepton, are powerful probes of BSM [38, 39]. Constraints on a_τ have previously been set by the DELPHI, OPAL, and L3 experiments [44–46]. The best constraint on d_τ has been reported by the Belle experiment [47], while other experiments, such as OPAL, L3, and ARGUS Collaborations, have also determined confidence intervals [45, 46, 48]. The most precise single experiment measurement of $a_{\tau,\text{exp}}$, conducted by the DELPHI collaboration [44, 49] at the Large Electron-Positron Collider (LEP), is remarkably an order of magnitude away from the theoretical central value $a_{\tau,\text{SM}}$ predicted with a precision of 10^{-5} [50]. Such poor constraints on the anomalous magnetic moment of the τ lepton highlight the potential impact of BSM [37], indicating a need for further exploration in the present experiment at LHC and future generations of high-energy, high-luminosity colliders. Most recently, the ATLAS and CMS experiments observed the $\gamma\gamma \rightarrow \tau^+\tau^-$ process in Ultra-Peripheral Collisions (UPC), which was further used to set constraints on a_τ [28, 29]. The first measurement of the photon-induced production of τ pairs in proton-proton collisions at the LHC, as well as constraints on a_τ and d_τ , was reported by the CMS Collaboration [18].

A schematic diagram representing the two-photon production of τ pairs at the LHeC is shown in Fig. 20. As illustrated, New Physics can modify the $\tau\gamma$ couplings, thereby affecting the magnetic moment. Due to the high statistics of τ pair production and favorable experimental conditions at the LHeC, as discussed above, one can consider it as a high precision probe for the a_τ coupling.

To introduce beyond SM modifications of electric and magnetic dipole moments, it is customary to use the SM effective field theory (SMEFT) approach [51, 52], assuming the scale of beyond SM physics Λ is much higher than the probe momentum transfers q in the collision. By considering the SMEFT Lagrangian, one can extract bounds on the magnetic and electric moments of the τ lepton [51].

In the specific decay channel, both leptonic decays of the τ particle (with a branching fraction of 35%) and hadronic decays of the τ particle (with an approximate branching fraction of 45%) could be considered (see Fig. 20). The τ lepton can undergo leptonic decay, where it decays into a charged lepton (ℓ), a corresponding neutrino (ν_ℓ), and a tau neutrino (ν_τ). It can also undergo hadronic decay, where it decays into a hadronic system consisting of charged pions ($\pi^\pm$), a neutral pion (π^0), and a tau neutrino (ν_τ).

In terms of background, the main sources of SM background include dielectron and dimuon production through the process $\gamma\gamma \rightarrow \ell^+\ell^-$, exclusive photo production of W boson pairs through the process $\gamma\gamma \rightarrow W^+W^-$, and diquark production through the process $\gamma\gamma \rightarrow q\bar{q}$. Considering the specific signature of the signal topology, it is possible to suppress the contributions of SM backgrounds. For example, in the signal scenario shown in Fig. 20, one can suppress the background from processes such as $\gamma\gamma \rightarrow \ell^+\ell^-$ and $\gamma\gamma \rightarrow W^+W^-$ by requiring exactly one lepton to be identified in the event. This would effectively reduce the back-to-back lepton pairs originating from these background processes.

4.2 Sensitivity of the LHeC to the anomalous magnetic moment of the τ lepton

The most general Lorentz-invariant form of the photon-lepton interaction vertex, parameterized by $ie\Gamma^\mu$, is given by:

$$\Gamma^\mu = \gamma^\mu F_1(q^2) + \frac{\sigma^{\mu\nu} q_\nu}{2m} [iF_2(q^2) + F_3(q^2)\gamma_5], \quad (4.1)$$

where F_1 , F_2 , and F_3 are form factors containing the electromagnetic structure of lepton. In particular, $F_2(0) \equiv a_\ell$ is identified as the anomalous magnetic moment of the lepton and relates to the gyromagnetic factor g_ℓ via:

$$a_\ell = \frac{g_\ell - 2}{2}, \quad (4.2)$$

while $F_3(0) = -\frac{2m_\ell}{e}d_\ell$ corresponds to the electric dipole moment. Both quantities are sensitive to new physics beyond the Standard Model.

The one-loop QED prediction for the anomalous magnetic moment is the celebrated Schwinger term:

$$a_\ell = \frac{\alpha}{2\pi} \simeq 0.00116, \quad (4.3)$$

where α is the electromagnetic fine-structure constant.

Experimental measurements of the electron anomalous magnetic moment have verified the Standard Model prediction to an accuracy of 12 significant digits. The photon-induced process $\gamma\gamma \rightarrow \tau^+\tau^-$ at the LHeC could provide a clean and promising avenue to probe the anomalous magnetic moment of the tau lepton, a_τ . Within the Standard Model, the prediction is:

$$a_\tau^{\text{SM}} = 117721(5) \times 10^{-8}. \quad (4.4)$$

The elastic photon-induced production of a τ -lepton pair is generated through MADGRAPH5_AMC@NLO v3.5. The anomalous magnetic moment a_τ and electric dipole moment d_τ of the tau lepton are parameterized using the SMEFTSIM package [53, 54]. The corresponding beyond Standard Model Lagrangian in the SM effective field theory (SMEFT) includes two dimension-six operators, leading to:

$$\mathcal{L}_{\text{BSM}} = \frac{C_{tB}}{\Lambda^2} (\bar{L}_L \sigma^{\mu\nu} \tau_R) H B_{\mu\nu} + \frac{C_{tW}}{\Lambda^2} (\bar{L}_L \sigma^{\mu\nu} \tau_R \sigma^i) H W_{\mu\nu}^i + \text{h.c.} \quad (4.5)$$

Here, Λ denotes the scale of new physics, and C_{tB} , C_{tW} are Wilson coefficients. After electroweak symmetry breaking, the effective $\tau\tau\gamma$ vertex takes the form:

$$V_{\tau\tau\gamma} = ie\gamma^\mu - \frac{v\sqrt{2}}{\Lambda^2} [\text{Re}(C_{\tau\gamma}) + i \text{Im}(C_{\tau\gamma})\gamma^5] \sigma^{\mu\nu} q_\nu, \quad (4.6)$$

where $v = 246$ GeV is the Higgs vacuum expectation value, and

$$C_{\tau\gamma} = \cos\theta_W C_{tB} - \sin\theta_W C_{tW}, \quad (4.7)$$

with θ_W the weak mixing angle.

The beyond Standard Model contributions to the anomalous magnetic moment and electric dipole moment of the tau lepton can be expressed as:

$$\delta a_\tau = \frac{2m_\tau\sqrt{2}v}{e\Lambda^2} \text{Re}(C_{\tau\gamma}), \quad (4.8)$$

$$\delta d_\tau = \frac{\sqrt{2}v}{\Lambda^2} \text{Im}(C_{\tau\gamma}). \quad (4.9)$$

In order to study the sensitivity of the LHeC to the anomalous magnetic moment of the τ lepton, signal samples are generated for a benchmark value of $a_\tau = 0.0042$, with the new physics scale fixed at $\Lambda = 1$ TeV, and incorporating a generator cut on $M_{\tau^+\tau^-} > 10$ GeV.

In the following, we investigate the impact of a benchmark value $a_\tau = 0.0042$ on relevant kinematic observables. The results are presented in Figs. 21, 22, and 23, comparing the Standard Model expectations with those including contributions from beyond Standard Model physics via anomalous τ -photon couplings and the EPA calculation.

Figure 21 shows the differential cross section as a function of the rapidity $Y_{\tau^+\tau^-}$ of the di- τ system, and in Figure 22, we display the differential cross section as a function of the invariant mass $M_{\tau^+\tau^-}$ of the $\tau^+\tau^-$ pair.

Figure 23 presents the ratio of the beyond Standard Model prediction to the SM calculation as a function of $M_{\tau^+\tau^-}$. To make the deviation more explicit, a linear fit of the form

$$ax + b \quad (4.10)$$

is performed, resulting in

$$a = (1.9 \pm 0.1) \times 10^{-3}, \quad (4.11)$$

with a reduced chi-squared value of $\chi^2/\text{dof} = 0.88$.

This analysis complements earlier studies, notably those performed by the CMS Collaboration, which have primarily focused on the anomalous magnetic moments of electrons and muons [55]. Recently, CMS has extended this program to the τ sector with a dedicated analysis of photon-induced $\tau^+\tau^-$ production in proton-proton collisions at $\sqrt{s} = 13$ TeV, using an integrated luminosity of 138 fb^{-1} . The process $\gamma\gamma \rightarrow \tau^+\tau^-$ was observed for the first time in this environment with a significance of 5.3 standard deviations.

Given this context, the LHeC, with its clean initial state, large photon fluxes, and low pile-up environment—provides a powerful and complementary platform for further exploration of the τ lepton’s electromagnetic properties. In particular, the enhanced sensitivity to high-mass $\tau^+\tau^-$ production offers an ideal opportunity to probe new physics effects at higher energy scales. LHeC is expected to offer a novel and complementary approach to constrain a_τ with improved precision.

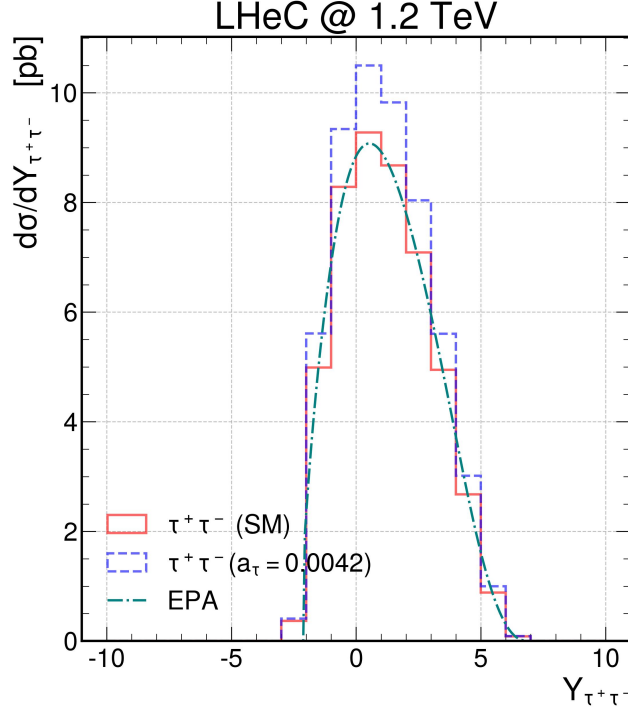


Figure 21. Differential cross section as a function of the rapidity $Y_{\tau^+\tau^-}$ for the τ lepton production at the LHeC. The red solid histogram shows the Standard Model (SM) prediction, while the blue dashed histogram includes the contribution from an anomalous electromagnetic moment of the tau lepton with $a_\tau = 0.0042$. The elastic cross-section calculated using the EPA framework is also shown for comparison.

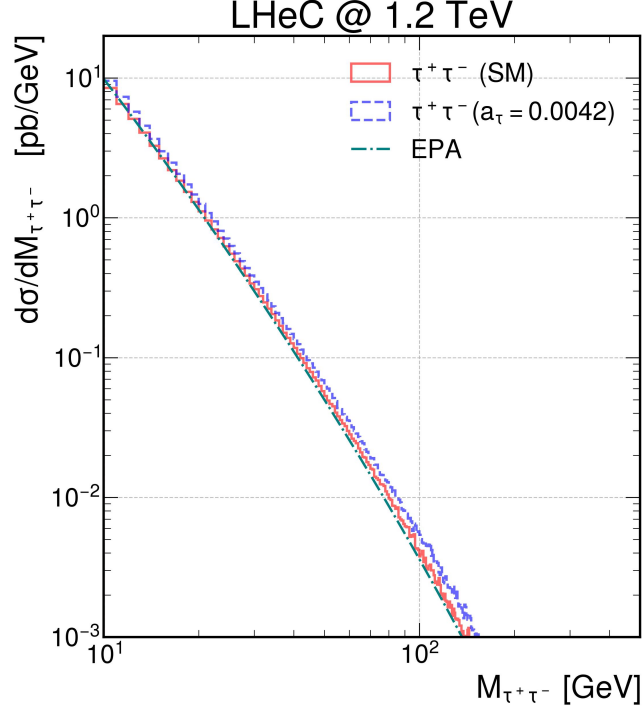


Figure 22. Differential cross section as a function of the invariant mass $M_{\tau^+\tau^-}$ of the di-tau system at the LHeC. The red solid histogram corresponds to the Standard Model (SM) prediction, while the blue dashed histogram includes the contribution from an anomalous tau electromagnetic moment with $a_\tau = 0.0042$. The green dash-dotted curve shows the prediction based on the EPA.

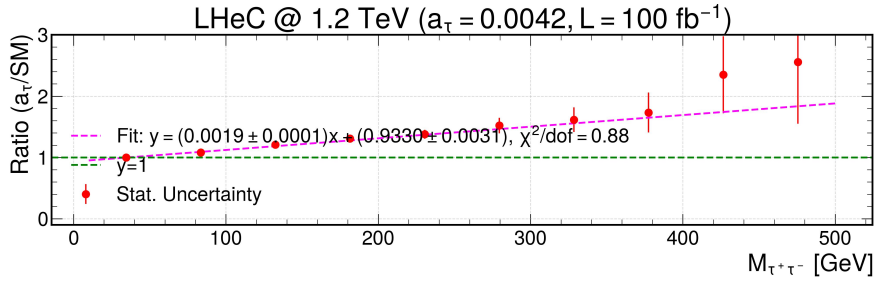


Figure 23. Ratio of the differential cross sections for the BSM scenario with an anomalous magnetic moment $a_\tau = 0.0042$ to the Standard Model (SM) prediction as a function of the invariant mass of the di-tau system $M_{\tau^+\tau^-}$ at the LHeC. The statistical uncertainties are shown as red error bars as well; and a linear fit is performed to the ratio points.

5 Exclusive production of muon and electron pairs

The cross-sections for the production of two isolated muons in electron-proton collisions were also previously measured at HERA using the H1 detector at a center-of-mass energy of $\sqrt{s} = 319$ GeV, with photon-photon BH interactions being the dominant process [56]. Beyond the HERA measurements, such processes have also been investigated in proton-proton collisions at CERN by the ATLAS and CMS collaborations. Measurements of the exclusive $\gamma\gamma \rightarrow \mu^+\mu^-$ production with the ATLAS detector have been previously performed in proton-proton collisions at $\sqrt{s} = 7$ TeV [57] and $\sqrt{s} = 13$ TeV [58]. The production of such muon pairs at $\sqrt{s} = 13$ TeV are also studied with proton-proton collision data corresponding to an integrated luminosity of 14.6 fb^{-1} [13]. Additionally, the CMS collaboration has reported the exclusive photon-photon production of muon pairs as well at the $\sqrt{s} = 7$ TeV [10].

Following the strategy used for the case of tau pair production in the previous section, one can use the EPA approach to calculate the cross-section for the two-photon production of muon pairs in both proton-proton and electron-proton collisions [20]. However, due to the small mass of the muon, it is necessary to limit ourselves to the kinematic range where the EPA is applicable. The exclusive muon pair production cross-sections at the LHeC, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 , through both elastic and inelastic production, are shown in Fig. 24. For the inelastic case, an upper limit for the diffractive system mass of $M_N < 3$ GeV is assumed. The results are shown for two selected values of photon virtualities: $Q_{e,p}^2 < 10 \text{ GeV}^2$ and $Q_{e,p}^2 < 50 \text{ GeV}^2$ for the electron/proton vertices.

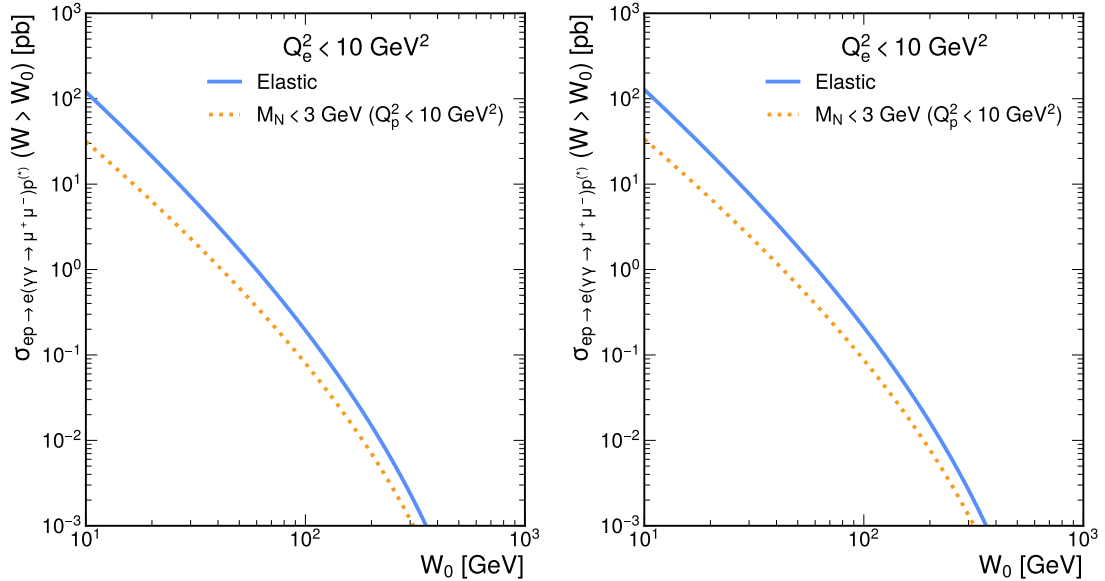


Figure 24. (Updated by exact EPA) Integrated $\mu^+\mu^-$ production cross sections via photon-photon interaction, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{\mu^+\mu^-}(W)$, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 , through the elastic process as well as inelastic production assuming diffractive system mass of $M_N < 3$ GeV.

Table 9 presents the exclusive two-photon production cross-sections of muon pairs for

Table 9. (Updated by exact EPA) Elastic $\mu^+\mu^-$ production cross-sections (in pb) at the LHeC, calculated using the EPA approach for a minimum two-photon invariant mass of $W_0 = 10$ GeV, assuming $Q_{e,p}^2 < 10$, and 50 GeV² for the electron/proton vertex. Cross-section values calculated using the GRAPE and CEPGEN/LPAIR MC event generators are also presented for comparison.

σ^{el} (pb)	$Q_{e,p}^2 < 10$ GeV ²	$Q_{e,p}^2 < 50$ GeV ²
EPA	120.12	127.73
CEPGEN (LPAIR, dipole FF)	118.31	123.93
GRAPE (BH)	121.37	127.14
GRAPE (BH+ISR)	127.61	133.87

Table 10. (Updated by exact EPA) Inelastic $\mu^+\mu^-$ production cross-sections (in pb) at the LHeC, calculated using the EPA approach for a minimum two-photon invariant mass of $W_0 = 10$ GeV, assuming $Q_{e,p}^2 < 10$, and 50 GeV² for the electron/proton vertex, and for a diffractive state mass upper limit $M_N < 3$ GeV/ c^2 . Cross-section values calculated using the GRAPE and CEPGEN/LPAIR MC event generators are also presented for comparison.

σ^{inel} (pb)	$Q_{e,p}^2 < 10$ GeV ²	$Q_{e,p}^2 < 50$ GeV ²
EPA (ALLM)	31.589	33.656
EPA (LUX-like)	30.121	32.031
GRAPE (BH)	30.857	32.135
CEPGEN (LPAIR, Suri-Yennie)	32.464	33.945
CEPGEN (LPAIR, ALLM)	27.570	28.692

selected values of photon virtualities at the electron/proton vertex, compared with those obtained using the GRAPE and CEPGEN/LPAIR MC event generators at the minimum two-photon invariant mass $W_0 = 10$ GeV. To estimate the impact of ISR on the cross-section calculations, the GRAPE results, including ISR, are also shown in Table 9. From the numbers presented in the table, it can be concluded that the EPA calculations show good compatibility with those of the GRAPE and CEPGEN/LPAIR MC event generators for the kinematics considered. The cross-section calculations for the inelastic case are presented in Table 10 as well. As previously discussed, in the case of inelastic interactions, the electromagnetic proton structure functions in GRAPE are parameterized using ALLM [22] for hadronic system invariant masses larger than 2 GeV. For the EPA approach employed in this work, we use the same proton structure functions as in GRAPE. For CEPGEN/LPAIR, different implementations of the structure functions can be used. Therefore, for the LPAIR cross-section calculations presented in Table 10, we provide the results calculated using the structure functions of Suri-Yennie and ALLM.

Considering the cross-section presented in the tables above, a large number of $\mu^+\mu^-$ pairs produced via two-photon interactions are expected at the LHeC, even for low values of Q^2 , both in the elastic process $ep \rightarrow e\mu^+\mu^-p$ and the inelastic case $ep \rightarrow e\mu^+\mu^-X$.

Experimentally speaking, in the case of the $\gamma\gamma \rightarrow \mu^+\mu^-$ signal, the following discussion regarding the main contributing backgrounds is in order. We should first note that the Drell-

Yan process, which is a major background source of two-photon production of muon pairs in proton-proton collisions, is totally absent in electron-proton collisions. This process involves the production of a virtual photon or Z boson that subsequently decays into a pair of muons. In electron-proton collisions, the QED Compton scattering ($ep \rightarrow e\gamma X \rightarrow e\mu^+\mu^-X$) can contribute to the background if the photon converts into a muon pair. Vector meson production in electron-proton scattering with subsequent decay into muons is another source of background for the two-photon production of muon pairs. Another source of background also arises from the decay of tau-leptons produced in two-photon collisions ($\gamma\gamma \rightarrow \tau^+\tau^-$).

The two-photon production of electron pairs ($\gamma\gamma \rightarrow e^+e^-$) could provide another interesting physics potential at the future collider LHeC. The production of multiple electrons in electron-proton collisions has been previously studied using the H1 detector at HERA, focusing on the restricted phase space region dominated by photon-photon collisions [59]. For such a process, considerable cross-sections are expected for such production at the LHeC as well.

For given values of photon virtualities, $Q_{e,p}^2 < 10 \text{ GeV}^2$, for the electron/proton vertices, the cross-sections for elastic production of di-electrons with invariant mass above 10 GeV are calculated using GRAPE and CEPGEN/LPAIR to be $\sigma^{\text{el}} = 237.46 \text{ (pb)}$ and $\sigma^{\text{el}} = 237.99 \text{ (pb)}$, respectively. Such considerable cross-sections indicate a very large statistics of di-electron production via photon-photon fusions at the LHeC. As discussed earlier, for the e^+e^- production, the EPA framework completely fails for certain regions of the phase space and hence does not provide a fair approximation for the total cross-section [20].

Considering the background contributions to the $\gamma\gamma \rightarrow e^+e^-$ signal, the following discussion is in order. The main experimental backgrounds involve scenarios where, in addition to a true electron, one or more fake electrons are reconstructed within the final state. A significant contribution arises from NC DIS events ($ep \rightarrow epX$), wherein, apart from the scattered electron, hadrons or emitted photons are incorrectly identified as electrons. Additionally, contributions from QED Compton scattering ($ep \rightarrow e\gamma X$) can be considered a background if the photon is incorrectly identified as an electron. The dilepton production, $\gamma\gamma \rightarrow \ell^+\ell^-$ (with $\ell = \mu, \tau$), also serves as another source of background.

Finally, the $\gamma\gamma \rightarrow \mu^+\mu^-$ and $\gamma\gamma \rightarrow e^+e^-$ pairs production discussed in this section serve as excellent calibration tools. They are instrumental for calibrating energy scales, normalizing the ep luminosity, and controlling the elastic vs. inelastic contributions to the $S_{\gamma\gamma}$.

6 Exclusive W boson pair production

The exclusive production of $W^\pm$ gauge boson pairs via photon-photon interactions ($\gamma\gamma \rightarrow W^+W^-$) at the LHeC is well-suited for studying phenomena beyond the SM, particularly the anomalous triple γWW and quartic $\gamma\gamma WW$ gauge couplings. As illustrated in the Feynman diagrams in Fig. 25, this process at leading order involves only diagrams with self-couplings of the electroweak gauge bosons, making it sensitive to anomalous gauge-boson interactions [60].

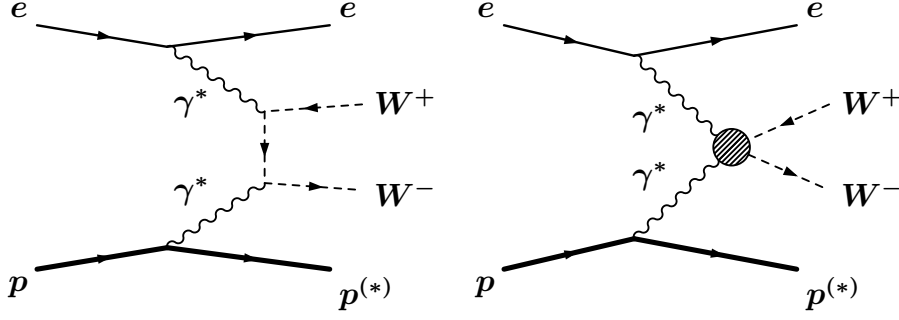


Figure 25. Leading-order Feynman diagrams contributing to the exclusive photon-induced production of W-boson pairs at the t -channel diagram (left) proceeding via the $\gamma\gamma W$ vertices, and a diagram with a quartic gauge coupling $\gamma\gamma WW$ (right).

Extensive research has been conducted on the production of W boson pairs through photon-photon interactions at the LHC. Recent experimental efforts by the ATLAS [61, 62] and CMS [15, 63, 64] Collaborations have focused on observing the photon-induced production of W-boson pairs in proton-proton collisions. The cross-section using 139 fb^{-1} of LHC proton-proton collision data taken at $\sqrt{s} = 13 \text{ TeV}$ recorded by the ATLAS experiment for the $pp \rightarrow p^{(*)}(\gamma\gamma \rightarrow W^+W^-)p^{(*)}$ process in the particular decay channel $W^+W^- \rightarrow e^\pm \nu \mu^\mp \nu$ is measured to be approximately 3.13 fb , which is in agreement with the SM theoretical predictions [61].

The future collider LHeC is expected to play an important role in investigating the two-photon production of W boson pairs as well. This can be achieved, firstly, due to the large center-of-mass energy and integrated luminosity available, and secondly, due to the clean experimental environment that the LHeC will offer.

Considering the two-photon luminosity spectrum $S_{\gamma\gamma}$ calculated above, and using the photon-photon cross-section $\sigma(\gamma\gamma \rightarrow W^+W^-)(W)$ for W pair production as given in Eq. (B.1), one can employ the EPA approach to calculate the production cross-sections at the LHeC. The exclusive W boson pair production cross sections at the LHeC, given by $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{\gamma\gamma \rightarrow W^+W^-}$, as a function of the minimal $\gamma\gamma$ center-of-mass energy W_0 , for both tagged elastic and inelastic processes, are shown in Fig. 26. We consider a kinematic range of $Q_e^2 < 10^5 \text{ GeV}^2$ for the electron vertex and $Q_p^2 < 10, 10^3$, and 10^5 GeV^2 for the proton vertex. Additionally, constraints on the diffractive system mass $M_N < 10, 50$, and 300 GeV are also applied for the inelastic case.

As illustrated in this plot, at the LHeC, even though the cross-section for two-photon exclusive W^+W^- pair production is relatively small at very high photon virtuality, it still remains significant. The cross-section values calculated using the EPA approach at a minimum two-photon invariant mass of $W_0 \sim 2M_W$ are $\sigma^{\text{el}} = 3.336 \times 10^{-2} \text{ (pb)}$ for tagged elastic ($0.01 < y_p < 0.2$) production with $Q_e^2 < 10^5 \text{ GeV}^2$ for the electron vertex and $Q_p^2 < 10 \text{ GeV}^2$ for the proton vertex, and $\sigma^{\text{inel}} = 2.931 \times 10^{-2} \text{ (pb)}$ for the inelastic case, assuming a constraint on the diffractive system mass $M_N < 10 \text{ GeV}$. The favorable experimental conditions at the LHeC, combined with negligible event pileup, make it an

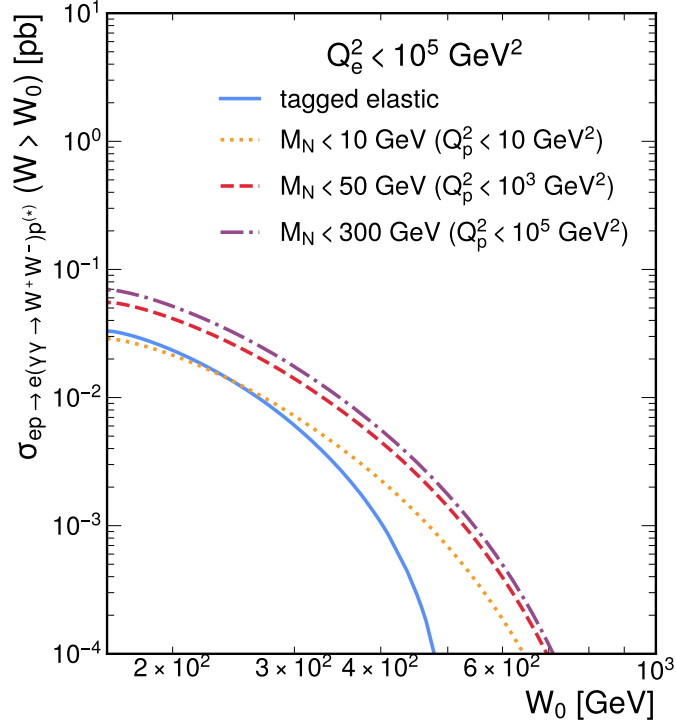


Figure 26. (Updated by exact EPA) Integrated exclusive W boson pair cross sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{W^+W^-}(W)$, as a function of the minimal $\gamma\gamma$ center-of-mass energy W_0 for both tagged elastic and inelastic production processes. Additional constraints on the diffractive system mass $M_N < 10, 50$, and 300 GeV are also considered for the inelastic case.

exceptional venue for studying the $\gamma\gamma \rightarrow W^+W^-$ process.

Several signal typologies of exclusive W boson pair production $\gamma\gamma \rightarrow W^+W^-$ could be studied at the LHeC. Therefore, different selection criteria should be used to identify signal events and to suppress relevant backgrounds. Among these typologies, the fully leptonic decays (involving muon and electron) of the W boson are particularly promising as they are considered potentially background-free. The requirement of two opposite-sign and same-flavor leptons ($ee/\mu\mu$) with sufficiently high transverse momentum can significantly suppress the backgrounds. Another interesting signal topology would consider the search for lepton pairs with different flavors, $W^+W^- \rightarrow e^\pm\nu\mu^\mp\nu$.

The dominant background at the LHeC in the context of WW pair production is typically reduced by selecting lepton pairs of different flavors, specifically $e\mu$ pairs, in order to suppress background processes such as the two-photon production of dilepton, $\gamma\gamma \rightarrow \ell^+\ell^-$. The small contribution from $\tau^+\tau^-$ processes with leptonic τ decays can still remain as one source of background. However, the contribution of this process decreases rapidly as the transverse momentum of the dilepton system increases [61]. By imposing higher requirements on the transverse momentum, this background can be further suppressed. In contrast to the LHC, the Drell–Yan (DY) production, which contributes to WW production

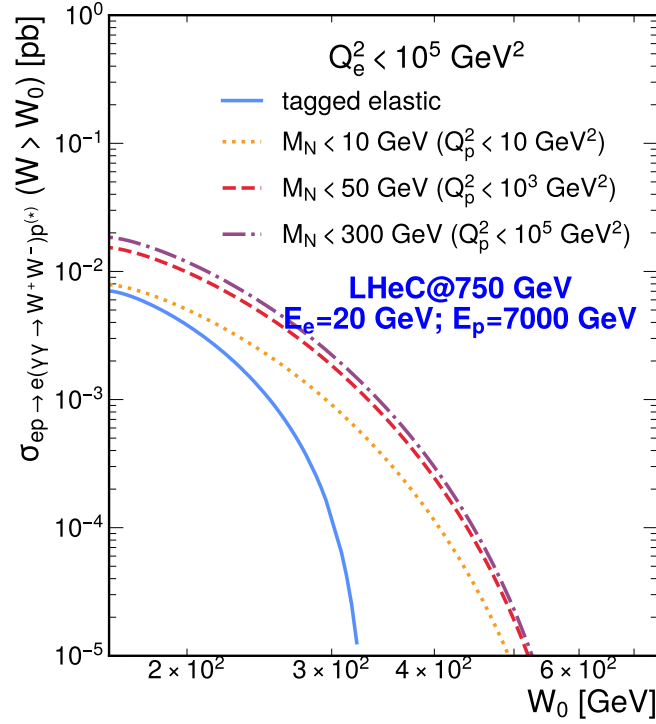


Figure 27. (Updated by exact EPA) Same as Fig. 26, for an LHeC center of mass energy of 750 GeV.

Table 11. (Updated by exact EPA) Comparison of the $\gamma\gamma \rightarrow W^+W^-$ process for the two electron beam energies of LHeC@1.2TeV, and LHeC@0.75TeV, considered in this study.

Experiment	LHeC ($E_e = 50$ GeV)	LHeC ($E_e = 20$ GeV)
σ^{el} (pb)	3.336×10^{-2}	7.044×10^{-3}
$\sigma_{M_N < 10 \text{ GeV}}^{\text{inel}}$ (pb)	2.931×10^{-2}	7.926×10^{-3}
$\sigma_{M_N < 50 \text{ GeV}}^{\text{inel}}$ (pb)	5.652×10^{-2}	1.540×10^{-2}
$\sigma_{M_N < 300 \text{ GeV}}^{\text{inel}}$ (pb)	7.081×10^{-2}	1.855×10^{-2}

via proton-proton collisions at the LHC, is completely absent at the LHeC. The background from the quark- and gluon-induced WW production ($qq(gg) \rightarrow WW$) also does not contribute.

Therefore, the distinctive fully leptonic signature of $\gamma\gamma \rightarrow W^+W^- \rightarrow \ell^+\ell^-\nu\nu$ and the experimental conditions at LHeC would offer a highly sensitive test for exploring the quartic gauge boson couplings within the SM [65]. Exclusive W boson pair production at LHeC also encompasses the exploration of anomalous triple gauge boson couplings γWW (as illustrated in Fig. 25), as well, however, the expected sensitivities are estimated to be less than the quartic couplings [66].

7 Unique Z boson laboratory

The exclusive production of ZZ pairs through two-photon interactions at the LHeC could provide an excellent platform for investigating anomalous quartic gauge couplings (AQGC), specifically those involving $\gamma\gamma ZZ$. Such anomalous couplings appear in many extensions of the SM and could directly reveal the exchange of a new, heavy boson [9, 60, 67].

From LHC experiments, a search was conducted for exclusive high-mass $\gamma\gamma \rightarrow ZZ$ production in proton-proton collisions. This search was based on data collected by the CMS and TOTEM experiments at $\sqrt{s} = 13$ TeV, corresponding to an integrated luminosity of 100 fb^{-1} [15]. No excess above the SM background prediction was observed, and upper limits were set on the $pp \rightarrow p(\gamma\gamma \rightarrow ZZ)p$ cross-sections. The results are also interpreted as new limits on dimension-6 and dimension-8 anomalous quartic gauge couplings.

It is expected that the future collider LHeC, with its large center-of-mass energy and integrated luminosity, along with a significantly cleaner experimental environment compared to the LHC, could provide enhanced sensitivity to anomalous quartic gauge couplings. Considering the EPA approach discussed so far and the calculated two-photon luminosity spectrum $S_{\gamma\gamma}$, along with the photon-photon cross-section $\sigma(\gamma\gamma \rightarrow ZZ)(W)$ for ZZ production [68], the total production cross-section at the LHeC can be obtained. The integrated ZZ pair production cross-sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{\gamma\gamma \rightarrow ZZ}$, as a function of the minimal $\gamma\gamma$ center-of-mass energy W_0 , are shown in Fig. 28. The cross-sections are presented for both tagged elastic processes and inelastic production. We consider a kinematic range of $Q_e^2 < 10^5 \text{ GeV}^2$ for the electron vertex and $Q_p^2 < 10, 10^3$, and 10^5 GeV^2 for the proton vertex. Additionally, constraints on the diffractive system mass $M_N < 10, 50$, and 300 GeV are applied for the inelastic production.

The cross-section values calculated using the EPA approach for a minimum two-photon invariant mass of $W_0 \sim 2M_Z$ are $\sigma^{\text{el}} = 3.340 \times 10^{-5} \text{ pb}$ for tagged elastic production with $Q_e^2 < 10^5 \text{ GeV}^2$ and $Q_p^2 < 10 \text{ GeV}^2$, and $\sigma^{\text{inel}} = 3.305 \times 10^{-5} \text{ pb}$ for the inelastic case ($0.01 < y_p < 0.2$), assuming a constraint on the diffractive system mass $M_N < 10 \text{ GeV}$. In the context of the LHeC and its investigation into high-energy $\gamma\gamma$ interactions, the SM cross-section for the neutral gauge boson production process $ep \rightarrow e(\gamma\gamma \rightarrow ZZ)p^*$ is expected to be notably small (see Fig. 28). This is primarily because the subprocess $\gamma^*\gamma^* \rightarrow ZZ$ is not allowed at tree level and only occurs at loop level, making it negligible. Thus, detecting even a few events in this process could provide significant evidence of anomalous quartic gauge couplings (AQGCs) and serve as an important indication of BSM effects.

Without delving into details, it is important to note that anomalous couplings can be introduced through an effective Lagrangian that models the low-energy behavior of a broad range of possible extensions to the SM. By considering a particular Lagrangian, the cross-sections for ZZ production can be expressed as a function of the associated anomalous parameters. The sensitivity to anomalous quartic gauge couplings (AQGCs) can then be explored using signatures such as the di-lepton process $\gamma\gamma \rightarrow ZZ \rightarrow \ell^+\ell^-jj$. Finally, one can compute the upper limits on the cross-section, which can be straightforwardly translated into limits on anomalous quartic couplings [9].

In Fig. 29, the distribution for the differential cross sections $d\sigma/dY_{ZZ}$ (in fb) for the Z

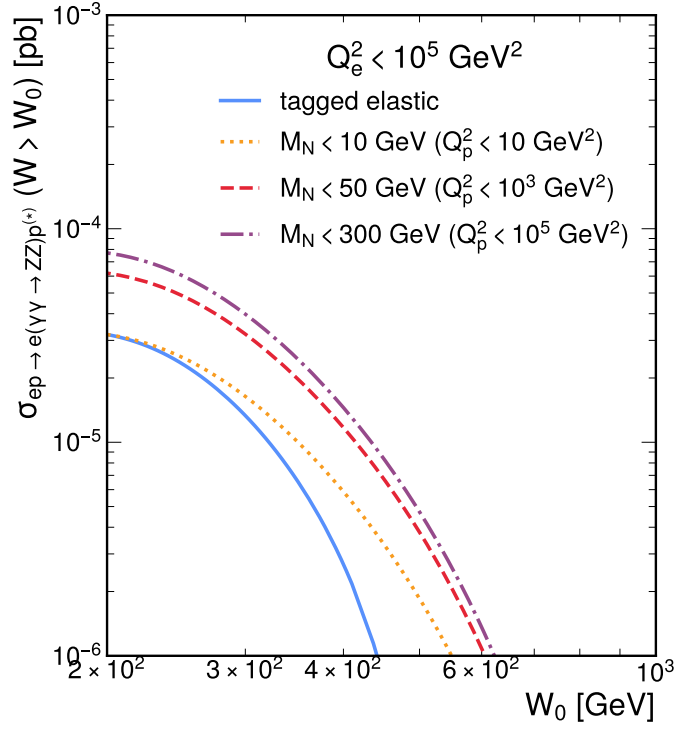


Figure 28. (Updated by exact EPA) Integrated ZZ pair production cross-sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{ZZ}(W)$, as a function of the minimal $\gamma\gamma$ center-of-mass energy W_0 , for both tagged elastic and inelastic production processes. Constraints on the diffractive system mass $M_N < 10, 50$, and 300 GeV are applied for the inelastic case.

boson pairs production, calculated using the EPA approach, is presented. The distribution is shown for both the tagged elastic and inelastic photon emission cases with $Q_e^2 < 10^5$ GeV² and $Q_p^2 < 10$ GeV². For the inelastic case, the diffractive system mass is constrained to the upper value of $M_N < 10$ GeV.

Finally, due to both leptonic and hadronic decays of the Z boson, the signal channel for the process $ep \rightarrow e(\gamma\gamma \rightarrow ZZ)p^*$ at the LHeC can result in several final states. One particularly interesting decay channel is the fully leptonic mode $\gamma\gamma \rightarrow ZZ \rightarrow \ell^+\ell^-\nu_\ell\bar{\nu}_\ell$, which features two isolated leptons and associated missing energy. Among the significant background sources, one notable background is the inclusive W boson pair production, where specific decay channels might satisfy all the identification criteria for the selected lepton flavor.

8 Production of the Higgs boson via photon-photon fusion

The future LHeC and its high-energy upgrade FCC-eh [69] have emerged as potential candidates for a ‘Higgs factory’ experiment. These colliders offer the capability to produce Higgs bosons through both neutral-current and charged-current weak boson fusion processes (NC/CC WBF). The CC WBF process, $e + p \rightarrow \nu_e + H + j$, is of great significance

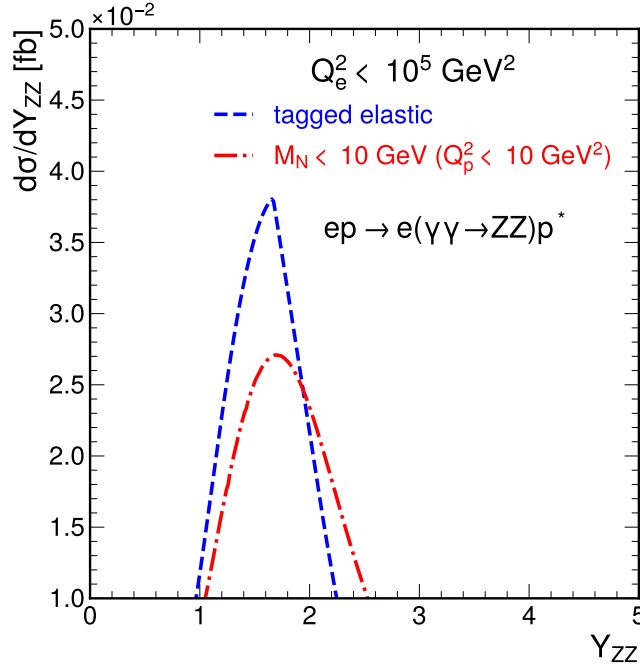


Figure 29. Rapidity dependence of the two-photon production of a Z boson pair cross section, $d\sigma/dY$ (in fb), for both tagged elastic and inelastic cases in which the diffractive system mass is constrained to the upper value of $M_N < 10$ GeV.

for the precise measurement of the Yukawa couplings [70–72]. This process, considered the leading production mode of the Higgs boson at the LHeC, exhibits a significantly larger cross-section (approximately 100 fb) compared to other production modes. The cross-section for Higgs boson production through the CC channel is approximately 5.3 times larger than that of the NC channel at the LHeC, with $E_e = 60$ GeV and $E_p = 7$ TeV. For the case of CC WBF and the decay channel $h \rightarrow b\bar{b}$, the cross-section $\sigma(ep \rightarrow H j \nu, H \rightarrow b\bar{b})$ is calculated to be around 80 fb. The presence of a tagged forward jet has been identified as a critical factor in effectively reducing background interference in studies of the $Hb\bar{b}$ and $Hc\bar{c}$ couplings. Additionally, the azimuthal angle between the neutrino and the forward jet provides a sensitive probe of the anomalous HWW coupling at the LHeC.

As discussed earlier, the LHeC, in addition to providing a clean experimental environment, will also offer high luminosity. These features collectively make the LHeC an excellent platform for studying the two-photon production of the Higgs boson. The production mechanism via photon-photon interactions opens a unique avenue to explore Higgs physics beyond the capabilities of hadron-hadron colliders like the LHC. The Feynman diagrams illustrating the Higgs photo-production process through W boson and charged fermion loops are depicted in Figure 30. In these processes, photons interact through loop diagrams involving W bosons or charged fermions to produce the Higgs boson. In the case of charged fermion loops, the dominant contributions come from the top and bottom quarks.

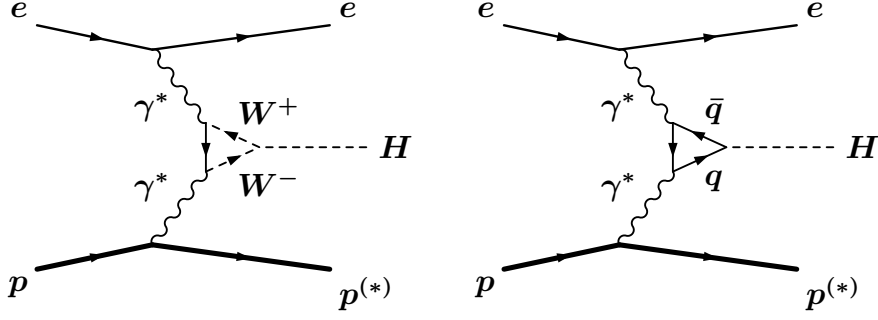


Figure 30. Feynman diagrams of the Higgs photo-production process through W (left) and charged fermion loops (right).

The total LHeC cross section for the photon-production of a Higgs boson can again be obtained through EPA, convoluting the calculated two-photon luminosity spectrum $S_{\gamma\gamma}$ with the photon-photon level production cross-section $\sigma(\gamma\gamma \rightarrow H)(W)$ (see Eq. B.4). In the elastic case for a minimum two-photon invariant mass of $W_0 \sim M_H$ it is estimated to be $\sigma^{\text{el}} = 0.141$ fb with $Q_e^2 < 10^5$ GeV² and $Q_p^2 < 10$ GeV². For the inelastic case, this cross-section intuitively depends on the diffractive system mass range considered: $\sigma^{\text{inel}} = 0.093$ fb for $M_N < 10$ GeV, $\sigma^{\text{inel}} = 0.233$ fb for $M_N < 300$ GeV. Both values assume a photon virtuality range $Q_{e,p}^2 < 10^5$ GeV². The production cross-section of the Higgs boson via photon-photon fusion at the LHeC is therefore considerably smaller than the main CC and NC WBF production channels.

It is interesting to note that its two-photon production rate is still significant and comparable at LHeC to that of FCC-ee/CEPC operating at a center-of-mass energy of 240 GeV with an integrated luminosity of approximately 5 ab^{-1} [73], with an estimated cross-section of 0.25 fb at FCC-ee/CEPC [74].

The differential rapidity cross sections distributions $d\sigma/dY_H$ for the production of the Higgs boson calculated using the EPA, are presented in Fig. 31. These are shown for an arbitrary Higgs mass $M_H = 125$ GeV, both for tagged elastic and inelastic photon emission cases with $Q_e^2 < 10^5$ GeV² and $Q_p^2 < 10$ GeV². In the inelastic case, the diffractive system mass is constrained to the upper value $M_N < 10$ GeV. In addition, the $b\bar{b}$ pairs production via photon-photon interaction is also presented for $123 < W < 127$ GeV, for both tagged elastic and inelastic cases.

Considering these results, several comments can be made. Firstly, as can be seen, the future collider LHeC could cover a more extended kinematic range in terms of Higgs boson (and decay system) rapidity. Secondly, the off-centering of the central system rapidity distributions shown above for the asymmetric electron-proton kinematics mode is well covered by the optimized acceptance of the LHeC detector geometry described e.g. in [75]. Comparing the rapidity distributions, $d\sigma/dY$ (pb), for the two-photon production of a Higgs boson and of $b\bar{b}$ pairs production via photon-photon interaction, one can conclude that

As shown in Fig. 30, various decay channels can be considered for the Higgs. In this study, the main ones are considered: $H \rightarrow b\bar{b}$, $H \rightarrow ZZ$, and $H \rightarrow W^+W^-$. For the

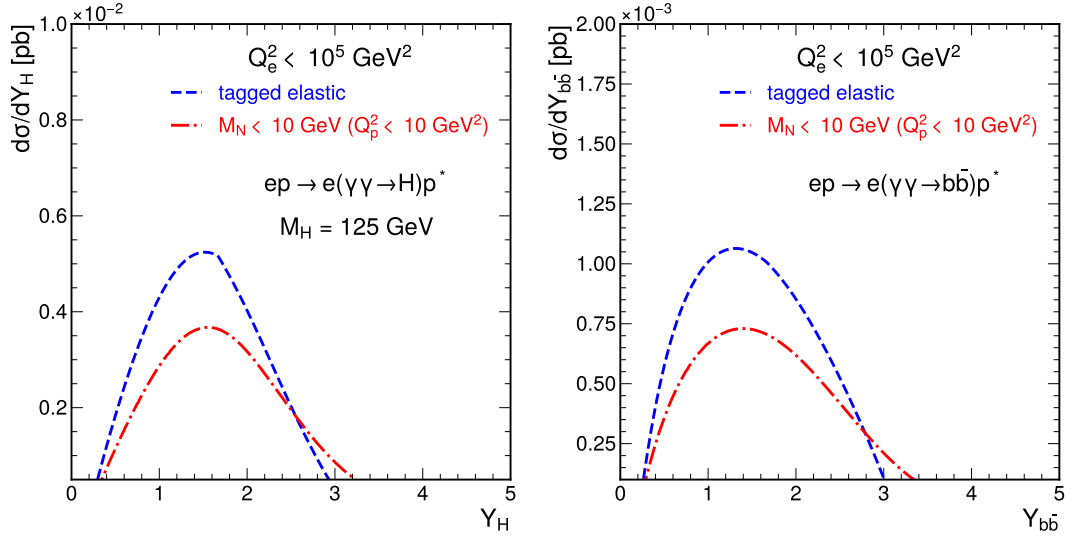


Figure 31. Differential cross section in rapidity, $d\sigma/dY$ (pb), for the two-photon production of Higgs boson ($M_H = 125$ GeV) and of $b\bar{b}$ pairs at $123 < W < 127$ GeV, for elastic and inelastic cases. For the inelastic case with $Q_p^2 < 10$ GeV², the diffractive system mass is constrained to the upper value $M_N < 10$ GeV.

$H \rightarrow b\bar{b}$ decay channel, the main background processes arise from $\gamma g \rightarrow b\bar{b}$ and $gg \rightarrow b\bar{b}$. Due to the substantial difference between gluon and photon PDFs, cross-sections for the background processes above still are orders of magnitude higher than the photo-production. Consequently, utilizing the $H \rightarrow b\bar{b}$ decay channel for this latter at the LHeC poses significant challenges. In the case of the $H \rightarrow ZZ$ channel, even when considering the decay of ZZ into two pairs of same-flavor and opposite-sign (SFOS) leptons, which could help in suppressing the QCD background, there remains a significant difference between the signal and background cross-sections. This large discrepancy makes it nearly impossible to carry out precise measurements of Higgs photo-production in this particular channel. In the $H \rightarrow W^+W^-$ channel, a possible approach is to consider the semi-leptonic decay of W bosons, where one W boson decays leptonically ($\ell\nu_\ell$) and the other decays hadronically into a final state with two jets (jj). The main background, in this case, arises from the process of $\gamma\gamma \rightarrow W^+W^-$ with semi-leptonic decays of the W bosons. Additionally, the NC WBF process will also contribute as a background to the analysis of the Higgs boson production in the $H \rightarrow W^+W^-$ channel. It has been shown in the past literature that, even in the most favorable semi-leptonic W^+W^- channel, which has an acceptable signal-to-background ratio, it remains challenging to identify the photo-production of the Higgs boson due to an irreducible background from the photo-production of W^+W^- (see e.g. [73]).

9 Two-photon exclusive production of supersymmetric pairs

Next in line is the production of non-strongly interacting supersymmetric (SUSY) particles, and more specifically a pair of sleptons ($\tilde{\ell}$), or higgsinos ($\tilde{H}$). At the LHeC, the photon-photon collisions could also hold considerable interest. The anticipated high luminosity

available at the LHeC, combined with its clean experimental environment compared to hadron colliders, provides a unique opportunity to study these processes with improved statistical significance and reduced background noise. In the following sections, we present and discuss these two processes in details.

The Feynman diagram representing the exclusive pair production of sleptons at the LHeC, $ep \rightarrow e(\gamma\gamma \rightarrow \tilde{\ell}\tilde{\ell})p^*$, is shown in Fig. 32. In this process, the sleptons decay into their Standard Model partners ℓ and the lightest neutralino $\tilde{\chi}_1^0$, resulting in a very clean signature of the final state: two leptons (ee , $e\mu$, or $\mu\mu$) of opposite charge detected in the central region, accompanied by missing momentum resulting from the undetectable neutralinos. Additionally, the final state electron/proton, scattered at an almost zero-degree angle, may be observed through dedicated very forward detectors (VFDs).

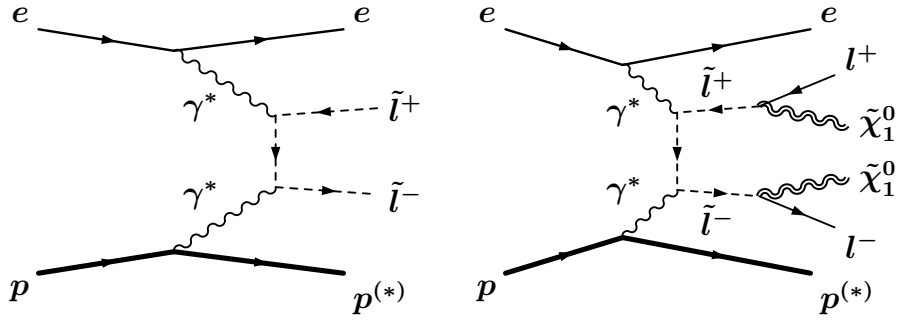


Figure 32. (Left) exclusive pair production of scalar leptons (sleptons) $\tilde{\ell}$ at the LHeC, (right) decay of sleptons into their SM partners ℓ and the lightest neutralino dark matter $\tilde{\chi}_1^0$.

In terms of experimental observables, measuring the outgoing proton kinematics allows to determine the colliding photon momenta and the missing momentum to be expected from the interaction. According to the various assumptions on very forward proton detectors acceptance at the LHeC, lightest SUSY particles such as sleptons, may be observed with considerable resolution.

Again, the EPA allows to compute total LHeC ep -level cross-sections, given the two-photon luminosity spectrum $S_{\gamma\gamma}$ and the two-photon level cross-section for slepton pair production $\sigma_{\gamma\gamma \rightarrow \tilde{\ell}^+\tilde{\ell}^-}(W)$ (see Eq. B.5). The integrated slepton pair production cross-sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{\tilde{\ell}^+\tilde{\ell}^-}(W)$, as a function of the minimal $\gamma\gamma$ center-of-mass energy W_0 , are shown in Fig. 33 both for the tagged elastic, and inelastic production processes. We consider a kinematic range of $Q_e^2 < 10^5 \text{ GeV}^2$ (resp. $Q_p^2 < 10, 10^3$, and 10^5 GeV^2) for the photon-from-electron (resp. photon-from-proton) emission. For inelastic production, additional constraints on the diffractive system mass $M_N < 10, 50$, and 300 GeV are also applied. In this study, we assume a slepton mass $m_{\tilde{\ell}} = 100 \text{ GeV}$ to compute integrated production cross-sections.

For a photon-from-electron virtuality range $Q_e^2 < 10^5 \text{ GeV}^2$, cross-section values calculated for a minimum two-photon invariant mass of $W_0 \sim 2M_{\tilde{\ell}}$ are $\sigma^{\text{el}} = 0.197 \text{ fb}$ for tagged elastic production ($0.01 < y_p < 0.2$) with $Q_p^2 < 10 \text{ GeV}^2$. In the inelastic case, total

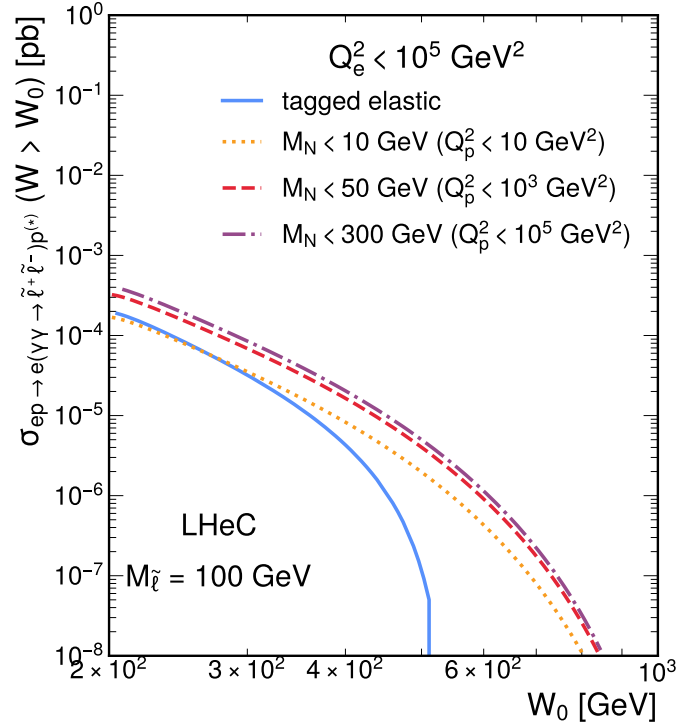


Figure 33. (Updated by exact EPA) Integrated slepton ($m_{\tilde{\ell}} = 100$ GeV) pair production cross-sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{\tilde{\ell}^+ \tilde{\ell}^-}$, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 , for both tagged elastic processes and inelastic production. For the latter, additional constraints on the diffractive system mass $M_N < 10, 50$, and 300 GeV are applied.

cross-sections again depend on the diffractive system mass range: for $M_N < 10$ GeV, and $M_N < 300$ GeV, $\sigma^{\text{inel}} = 0.170$ fb and 0.409 fb respectively, assuming $Q_p^2 < 10^5$ GeV 2 .

The cross-section does not exhibit logarithmic enhancement (see Eq. B.5), indicating that the production rate for sleptons is expected to be low for larger values of $m_{\tilde{\ell}}$. Experimental searches for supersymmetric particles could impose limits on their masses [20]. In Ref. [1], a novel search strategy is proposed, aiming to utilize the LHC as a photon collider to improve sensitivity to the production of scalar leptons ($\tilde{\ell}$) in a compressed mass scenario, which involves masses ranging from 15 to 60 GeV above that of the neutralino dark matter ($\tilde{\chi}_1^0$). The primary outcomes of the study reveal increased sensitivity in comparison to LEP, expanding the detection capabilities to slepton masses up to 220 GeV for $15 \lesssim \Delta m(\tilde{\ell}, \tilde{\chi}_1^0) \lesssim 60$ GeV, considering 100 fb^{-1} of 13 TeV proton collisions.

Despite the anticipated high $\gamma\gamma$ luminosity at the LHeC, the production cross-section of sleptons via photon-photon fusion is relatively low (see Fig. 33). This is partially compensated not only by the clean experimental conditions but also by the simple topology of the initial state and the potentially clean environment of the final state. Therefore, the LHeC would be a perfect candidate for novel and complementary searches for sleptons using photon-photon collisions.

Finally, in Fig. 34, the distributions for the differential cross sections $d\sigma/dY_{\tilde{\ell}^+\tilde{\ell}^-}$ (in fb) for the slepton pair production, calculated using the EPA, are presented. The distributions are shown for arbitrary slepton masses of $M_{\tilde{\ell}} = 100$ and 200 GeV, both for tagged elastic and inelastic photon emission cases with $Q_e^2 < 10^5 \text{ GeV}^2$ and $Q_p^2 < 10 \text{ GeV}^2$. For the inelastic case, the diffractive system mass is constrained to the upper value $M_N < 10 \text{ GeV}$.

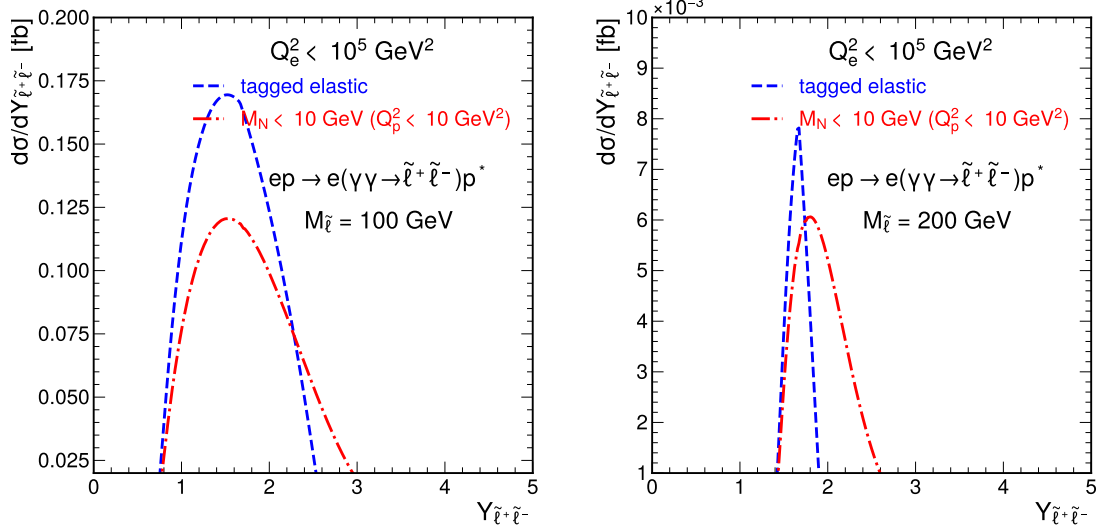


Figure 34. Differential cross section $d\sigma/dY_{\tilde{\ell}^+\tilde{\ell}^-}$ (in fb) as a function of the slepton rapidity $Y_{\tilde{\ell}^+\tilde{\ell}^-}$ for two given values of $M_{\tilde{\ell}} = 100$ (left), and 200 GeV (right). Distributions are shown for elastic, and inelastic cases. For the latter, an upper limit $M_N < 10 \text{ GeV}$ is applied on the diffractive system mass.

Considering the signal topology $\gamma\gamma \rightarrow \tilde{\ell}^+\tilde{\ell}^-$, where the sleptons decay to their SM partners ℓ and the lightest neutralino dark matter $\tilde{\chi}_1^0$, the dominant expected background primarily arises from the exclusive two-photon production of W boson pairs, followed by their leptonic decay modes ($\gamma\gamma \rightarrow W^+W^-$). This background process closely mimics the signal due to the presence of multiple leptons in the final state. Additionally, the production of lepton pairs ($\gamma\gamma \rightarrow \ell^+\ell^-$) via photon-photon interactions also contributes as a significant background. Given the similarities in final state particles, these backgrounds pose challenges in distinguishing the signal from the background in experimental analyses.

Several considerations regarding the discrimination of signals from backgrounds merit discussion. Given the substantial increase in the cross sections for forward-backward scattering, it becomes feasible to implement cuts on the rapidity $|y|$ and the transverse momentum p_T of the final state particles to enhance signal sensitivity. These cuts can help suppress the contributions from these background processes. Additionally, the invariant mass M of the produced slepton pairs should satisfy the condition $M > 2m_{\tilde{\ell}}$, ensuring that only events consistent with the production of heavy sleptons are selected [76]. This mass threshold

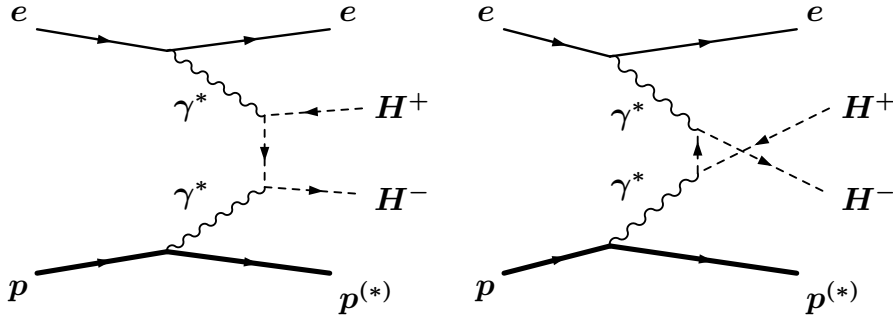


Figure 35. Feynman diagrams for the t and u -channel two-photon production of higgsinos at the LHeC.

helps to further discriminate between the signal and backgrounds, particularly in scenarios where backgrounds predominantly result in lower invariant mass distributions.

Unlike in signal events where the transverse momenta of the final state W boson or ℓ differ due to the escaping invisible neutralino, in $\gamma\gamma \rightarrow \ell^+\ell^-$ and $\gamma\gamma \rightarrow W^+W^-$ events, the transverse momenta of the W bosons or ℓ are typically balanced. This characteristic provides a key discriminator between signal and background. By applying a cut to selectively reject events where the transverse momenta of the W or ℓ particles are balanced within a few GeV, one can substantially reduce the backgrounds while having a minimal impact on the signal. As discussed earlier, as the slepton mass $m(\tilde{\ell})$ increases, the $\gamma\gamma \rightarrow \tilde{\ell}^+\tilde{\ell}^-$ production cross-section decreases, leading to a statistical limitation in the search at future colliders like the LHeC. Despite this reduction in cross-section, signals corresponding to larger $m(\tilde{\ell})$ could become more distinguishable from the $\gamma\gamma \rightarrow W^+W^-$ background due to a clearer separation from the W boson mass [1]. This mass difference aids in isolating the signal, making it easier to differentiate between the slepton production events and the background. Additionally, the $\gamma\gamma \rightarrow \ell^+\ell^-$ background could be further suppressed by imposing a requirement for large missing energy, which is characteristic of signal events where the neutralino escapes detection. This approach effectively enhances the signal-to-background ratio, particularly in scenarios with heavier sleptons, thereby improving the prospects for discovery at the LHeC.

In the following, we present and discuss the production of higgsinos through photon-photon interactions at the LHeC. Higgsinos, superpartners of the Higgs bosons in supersymmetric models, can be produced in pairs as shown in the Feynman diagrams in Fig. (35). The potential of the LHeC to explore this production channel, combined with the clean experimental conditions provided by the electron-proton collisions, positions it as a powerful tool for the search for new BSM physics. In addition, the photon-photon processes are generally straightforward, with production and decay mechanisms that tend to be less complex than those in hadronic collisions at LHC. This simplicity translates to a cleaner experimental environment at LHeC, particularly advantageous when searching for decay products of SUSY particles like higgsinos. Indeed, leptonic final states [77], expected in such processes, further contribute to the cleanliness of the event topologies.

Considering the two-photon luminosity spectrum $S_{\gamma\gamma}$, and considering that higgsinos couple to photons similarly to standard fermions with spin- $\frac{1}{2}$ [20, 76] in which allows to use the $\gamma\gamma$ cross-section for the charged fermion pair production (see Eq. B.3), the production cross-section at the LHeC can be computed using the EPA approach as discussed above. The total cross-section, as a function of the minimal photon-photon center-of-mass energy (W_0), is illustrated in Fig. 36 for a higgsino mass of $M_{\tilde{H}} = 100$ GeV. These integrated cross-section values assume a kinematic range of $Q_e^2 < 10^5$ GeV² for the electron vertex, and $Q_p^2 < 10, 10^3$ and 10^5 GeV² for the proton vertex. For the inelastic case, additional constraints are applied on the diffractive system mass of $M_N < 10, 50$ and 300 GeV.

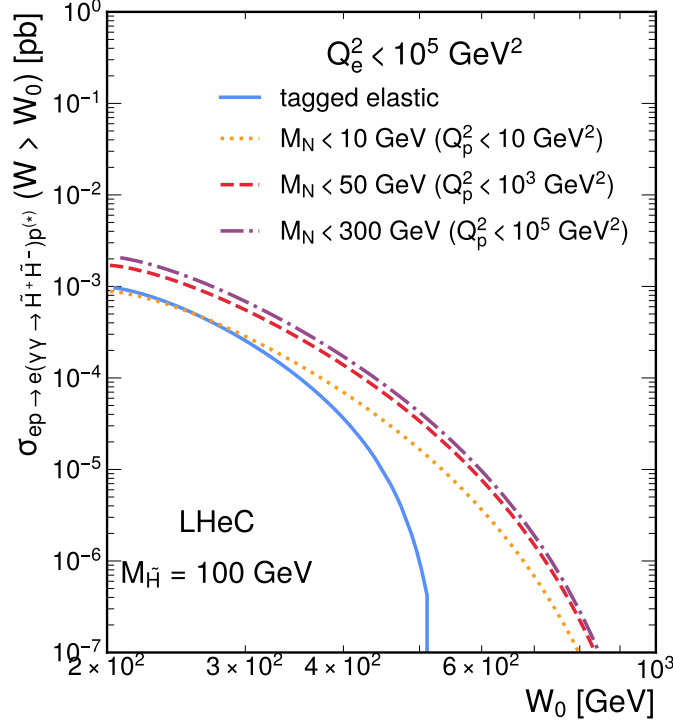


Figure 36. (Updated by exact EPA) Integrated higgsinos production cross sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{\tilde{H}^+\tilde{H}^-}(W)$, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 , through the tagged elastic and inelastic production. For the latter, additional constraints are applied on the diffractive system mass of $M_N < 10, 50$ and 300 GeV. The integrated production cross-sections are calculated for a given higgsino mass value of $M_{\tilde{H}} = 100$ GeV.

The cross-section values calculated using the EPA approach for a minimum two-photon invariant mass of $W_0 \sim 2M_{\tilde{H}}$ are $\sigma^{\text{el}} = 9.836 \times 10^{-4}$ pb for tagged elastic ($0.01 < y_p < 0.2$) production with $Q_e^2 < 10^5$ GeV² and $Q_p^2 < 10$ GeV²; and $\sigma^{\text{inel}} = 8.873 \times 10^{-4}$ pb for the inelastic case, assuming a constraint on the diffractive system mass $M_N < 10$ GeV. Furthermore, when considering higher photon virtualities $Q_{e,p}^2 < 10^5$ GeV² and a diffractive system mass of $M_N < 300$ GeV, the calculated cross-section are estimated to be $\sigma^{\text{inel}} = 2.133 \times 10^{-3}$ pb for the inelastic case.

Table 12. (Updated by exact EPA) Cross-sections at the generator level for a minimum two-photon invariant mass of $W_0 = 200$ and 400 GeV for the $\gamma\gamma \rightarrow \tilde{H}^+\tilde{H}^-$ process using EPA approach, GRAPE and CEPGEN (with the LPAIR process) event generators. Values are presented for two higgsino mass scenarios, $M_{\tilde{H}} = 100$ and 200 GeV, assuming a constraint on the diffractive system mass $M_N < 10$ GeV.

Generator	$Q_{e,p}^2 < 10^5 \text{ GeV}^2$			
	$W_0 = 200 \text{ GeV}$		$W_0 = 400 \text{ GeV}$	
	σ^{el} (pb)	σ^{inel} (pb)	σ^{el} (pb)	σ^{inel} (pb)
EPA	1.322×10^{-3}	1.165×10^{-3}	3.383×10^{-5}	2.896×10^{-5}
LPAIR (CEPGEN)	1.270×10^{-3}	1.182×10^{-3}	3.258×10^{-5}	3.091×10^{-5}
GRAPE (BH)	1.298×10^{-3}	1.127×10^{-3}	3.343×10^{-5}	2.899×10^{-5}
GRAPE (EW)	1.375×10^{-3}	1.444×10^{-3}	3.756×10^{-5}	3.153×10^{-5}
GRAPE (BH+ISR)	1.335×10^{-3}	1.284×10^{-3}	3.469×10^{-5}	3.210×10^{-5}

Given the low magnitude of these cross-sections within the range of kinematics probed, achieving significant statistical accuracy requires very large integrated luminosity, which is projected to be achievable at the LHeC. The combination of high-resolution detectors capable of accurately reconstructing events, the high invariant mass of the two-photon system, and the precisely defined initial states enables the clean identification of events. Moreover, the distinct signatures of these exclusive processes, characterized by low backgrounds, could potentially allow for precise and accurate measurements to be carried out at the LHeC, even with the challenging cross-sections involved. This unique environment thus enhances the discovery potential for New Physics, such as the production of higgsinos.

To give further weight to the discussion above, in the following, we present the cross-section values calculated using the EPA approach at given values of minimum two-photon invariant mass $W_0 = 200$ and 400 GeV and compare them with the generator-level cross-sections calculated with those of and CEPGEN (with the LPAIR process) [31, 32] and GRAPE [33] event generators. The integrated production cross-sections are calculated for the higgsino mass of $M_{\tilde{H}} = 100$ and 200 GeV. As detailed in Table 12, for the specified kinematic conditions, the cross-section values obtained using the EPA framework show good agreement with those from the GRAPE and CEPGEN/LPAIR generators. As mentioned earlier, the LPAIR calculation is based on the diagrams of the two-photon BH process, which dominates a large fraction of the available phase space. In contrast, GRAPE not only includes the additional contributions arising from Z^0 exchange in the high mass region but also accounts for the effects of ISR. The cross-sections calculated using GRAPE in the presence of ISR are also shown in the table.

Highlighting the comparisons made earlier, we present the distributions for the differential cross sections $d\sigma/dW$ and $d\sigma/dY_{\tilde{H}^+\tilde{H}^-}$ calculated using the EPA framework in Fig. 37. These results are compared with those obtained from the CepGen event generator implementation of the LPAIR Monte Carlo. The differential cross sections illustrated in this figure are calculated for a representative higgsino mass of $M_{\tilde{H}} = 100$ GeV, considering both elastic and inelastic production modes. For the inelastic case, the diffractive system mass is constrained to an upper limit of $M_N < 10$ GeV. To ensure numerical stability in the EPA

convolution of photon fluxes, upper limits of $Q_{e,p}^2 < 10^3 \text{ GeV}^2$ for the photon virtualities are also applied.

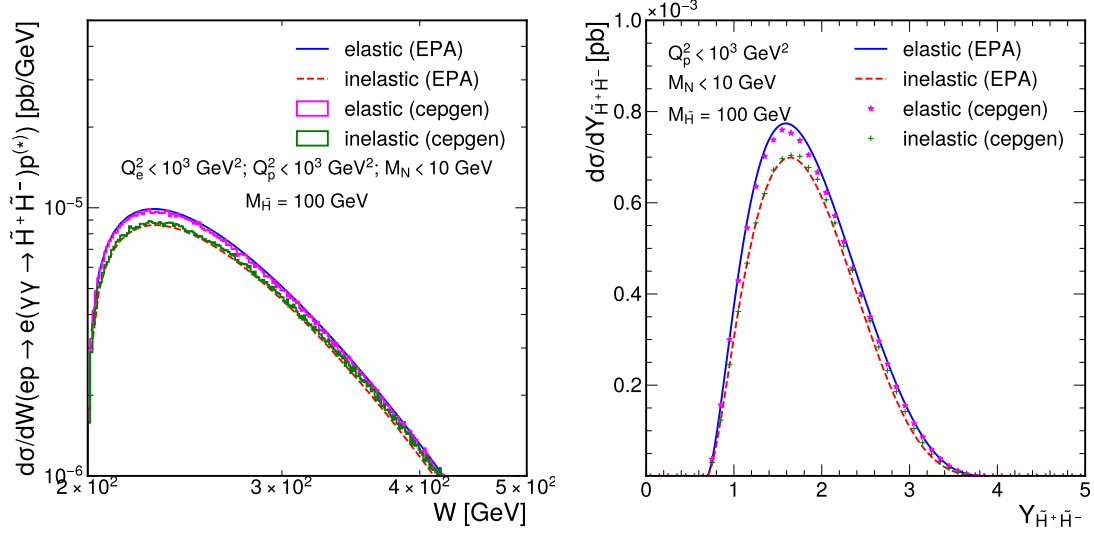


Figure 37. Differential cross sections for higgsino pair production: (left) $d\sigma/dW$ as a function of the photon-photon center-of-mass energy W ; (right) $d\sigma/dY_{\tilde{H}^+\tilde{H}^-}$ as a function of the higgsinos rapidity $Y_{\tilde{H}^+\tilde{H}^-}$. For the inelastic case, a photon virtualities upper limits $Q_{e,p}^2 < 10^3 \text{ GeV}^2$ is applied.

As shown in Fig. 37, the EPA calculations for both elastic and inelastic production modes agree well with the results obtained using the LPAIR generator, assuming equivalent kinematic constraints. This agreement demonstrates that despite the minor corrections mentioned earlier, the EPA provides a reliable approximation for estimating high-mass exclusive two-photon production cross-sections at the LHeC across a broad kinematic range.

10 Light-by-light scattering

The interaction of two photons is a purely quantum mechanical process that occurs at leading-order (LO) in the fine structure constant, involving a loop-level interaction described by box diagrams, as shown in Fig. 38. In this latter, all charged particles—including charged fermions (leptons and quarks) and bosons ($W^\pm$)—contribute to the scattering amplitude. Light-by-light scattering (LbL) has a very small cross-section for optical photons due to the higher-order nature of the interaction [78, 79]. However, at higher photon energies, such as those accessible at the LHC and future colliders, the cross-section becomes significant enough for experimental observation.

Recently, LbL was observed at the LHC in ultra-peripheral heavy-ion collisions by the CMS [80] and ATLAS [81, 82] Collaborations, at a photon-photon center-of-mass energy of approximately $W_{\gamma\gamma} \approx 5 \text{ GeV}$. Heavy-ions provide an ideal environment for studying LbL,

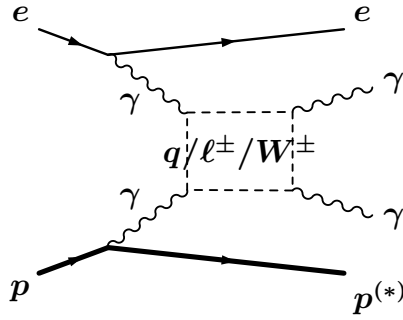


Figure 38. Feynman diagram for the (in)elastic LbL scattering ($\gamma\gamma \rightarrow \gamma\gamma$) with electron-proton initial, and final state.

as initial photon fluxes are enhanced by a factor proportional to Z^4 with respect to single nucleon cases, making these collisions particularly suitable for observing $\gamma\gamma$ scattering.

More recently, it was also sought for in proton-proton collisions, along with the detection of scattered protons in CMS-TOTEM [14, 17]. The detection of forward protons with a kinematics compatible with the two-photon system provides insights into anomalous production processes. In addition, through the backgrounds from inclusive sources they allow to reduce, this technique improves the precision of the measurements.

With several prior theoretical and experimental studies, the feasibility of measuring cross-sections for LbL at future facilities, such as HL-LHC, FCC and LHeC was emphasized, along with its sensitivity to new physics beyond the Standard Model, such as anomalous quartic gauge couplings (aQGCs), or the exchange of scalar, or tensor resonances. Both approaches can be probed through deviations from the predicted Standard Model cross section [78, 83–85]. The measurement of light-by-light scattering at such high photon-photon energies could provide a unique opportunity to probe physics at energy scales that are inaccessible to other types of interactions.

In Fig. 39, the cross-section of the light-by-light scattering process is shown as a function of the photon-photon center-of-mass energy W . Considering the EPA approach discussed so far, along with the calculated two-photon luminosity spectrum $S_{\gamma\gamma}$, and the photon-photon cross-section $\sigma(\gamma\gamma \rightarrow \gamma\gamma)(W)$ for the light-by-light scattering process, the total production cross-section at the LHeC can be obtained. The integrated cross-sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{\gamma\gamma \rightarrow \gamma\gamma}(W)$, as a function of the minimum $\gamma\gamma$ center-of-mass energy W_0 , are shown in Fig. 39. All values are presented for both tagged elastic and inelastic production processes. We consider a kinematic range of $Q_e^2 < 10^5 \text{ GeV}^2$ for the electron vertex and $Q_p^2 < 10, 10^3, \text{ and } 10^5 \text{ GeV}^2$ for the proton vertex. Additionally, constraints on the diffractive system mass $M_N < 10, 50, \text{ and } 300 \text{ GeV}$ are applied for inelastic production.

The cross-section values calculated using the EPA approach at a minimum two-photon invariant mass of $W_0 = 10 \text{ GeV}$ are $\sigma^{\text{el}} = 1.077 \times 10^{-2} \text{ (pb)}$ for tagged elastic ($0.01 < y_p < 0.2$) production with $Q_e^2 < 10^5 \text{ GeV}^2$ for the electron vertex and $Q_p^2 < 10 \text{ GeV}^2$ for the proton vertex, and $\sigma^{\text{inel}} = 1.914 \times 10^{-2} \text{ (pb)}$ for the inelastic case, assuming a constraint on the diffractive system mass $M_N < 10 \text{ GeV}$. These numbers indicate that the light-by-light

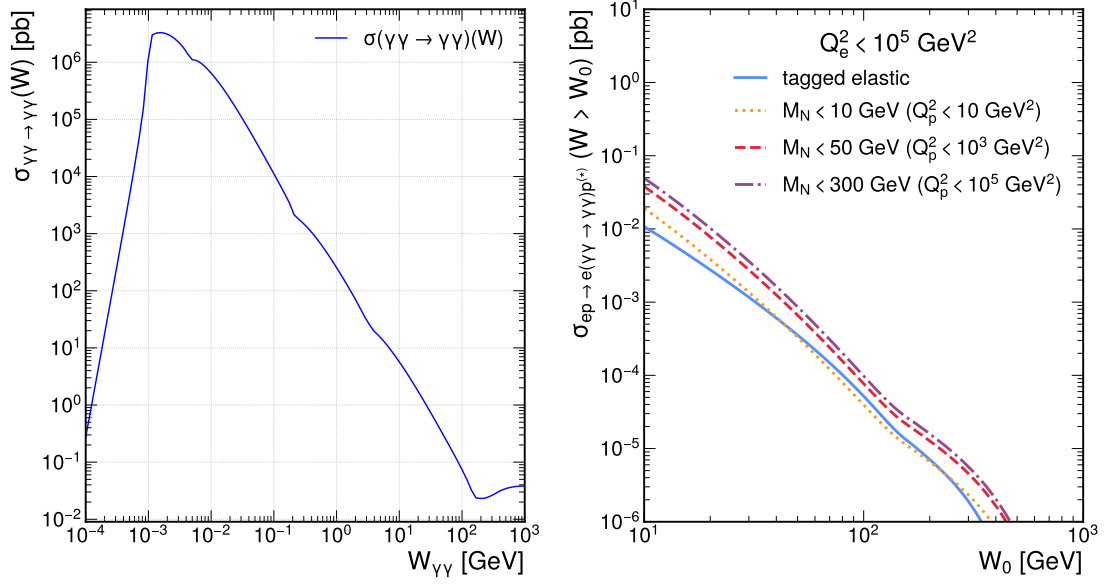


Figure 39. (Updated by exact EPA) (left) Cross section for the light-by-light scattering process $\sigma(\gamma\gamma \rightarrow \gamma\gamma)$ as a function of the photon-photon center of mass energy W ; (right) integrated cross sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{\gamma\gamma \rightarrow \gamma\gamma}(W)$, as a function of the minimal $\gamma\gamma$ center-of-mass energy W_0 for both tagged elastic and inelastic production processes. Additional constraints on the diffractive system mass $M_N < 10, 50$, and 300 GeV are also considered for the inelastic case.

scattering, $\gamma\gamma \rightarrow \gamma\gamma$, can be measured in electromagnetic interactions of electron-proton collision at the LHeC with considerable statics, thanks to the large photon fluxes available at very high-energy collisions.

11 Two-photon exclusive production of top quark-antiquark pairs

The calculated cross-sections for the exclusive production of top quark-antiquark pairs based on the EPA, is shown in Fig. 40 for both tagged elastic and inelastic processes.

The cross-section values calculated using the EPA approach for a minimum two-photon invariant mass of $W_0 \sim 2M_{\text{top}}$ are $\sigma^{\text{el}} = 1.889 \times 10^{-5}$ pb for tagged elastic production with $Q_e^2 < 10^5$ GeV² and $Q_p^2 < 10$ GeV²; and $\sigma^{\text{inel}} = 3.048 \times 10^{-5}$ pb for the inelastic case, assuming a constraint on the diffractive system mass $M_N < 10$ GeV. Furthermore, when considering higher photon virtualities $Q_{e,p}^2 < 10^5$ GeV² and a diffractive system mass of $M_N < 300$ GeV, the calculated cross-section are estimated to be $\sigma^{\text{inel}} = 7.471 \times 10^{-5}$ pb for the inelastic case.

These results show that, despite the relatively small cross-section values, the clean experimental environment of the LHeC and the high photon-photon luminosity make the study of exclusive top quark pair production a valuable probe for both SM and beyond SM

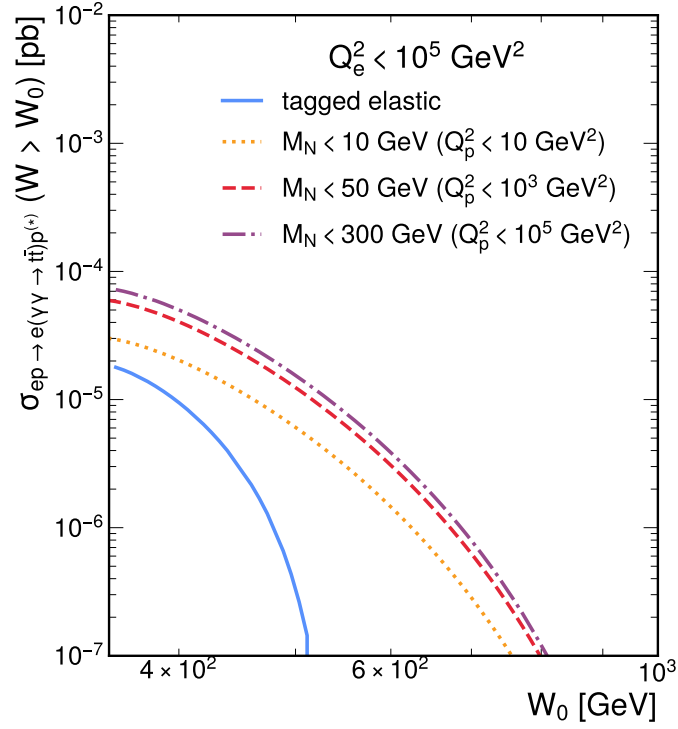


Figure 40. (Updated by exact EPA) Integrated exclusive top quark pair production cross sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{t\bar{t}}(W)$, as a function of the minimal $\gamma\gamma$ center-of-mass energy W_0 for both tagged elastic and inelastic production processes. Additional constraints on the diffractive system mass $M_N < 10, 50$, and 300 GeV are also considered for the inelastic case.

physics.

<https://arxiv.labs.arxiv.org/html/2210.13884> <https://inspirehep.net/literature/2170538>
Search for central exclusive production of top quark pairs with the CMS and TOTEM experiments

<https://arxiv.labs.arxiv.org/html/2007.04565> <https://inspirehep.net/literature/1806036>
Top quark pair production in the exclusive processes at the LHC

12 High energy $\gamma\gamma$ interactions at FCC-eh

The proposed Future Circular Collider with hadron-electron interactions (FCC-he) will use an electron beam with an energy of around 60 GeV combined with a proton beam of up to 50 TeV, resulting in a center-of-mass energy of approximately 3.5 TeV.

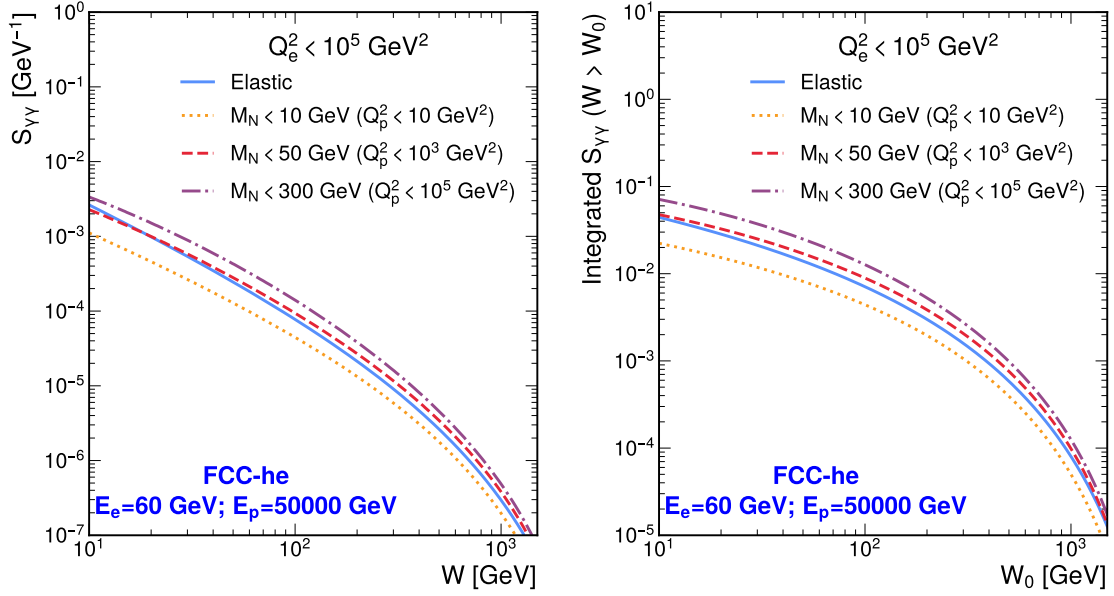


Figure 41. (Updated by exact EPA)(left) Elastic luminosity spectrum $S_{\gamma\gamma}$ at the FCC-he, assuming $Q_e^2 < 10^5 \text{ GeV}^2$ and the appropriate kinematical limits for the minimal photon virtualities (see inline), as well as three inelastic luminosity spectra where the maximum dissociative mass $M_N < 10, 50$, and 300 GeV is assumed for $Q_p^2 < 10, 10^3$, and 10^5 GeV^2 , respectively; (right) corresponding integrated luminosity spectra, $\int dW S_{\gamma\gamma}$, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 .

Table 13. (Updated by exact EPA) Summary of total cross section (in pb) for the two-photon exclusive productions of W^+W^- , ZZ , $t\bar{t}$, $\gamma\gamma$, $\tilde{H}^+\tilde{H}^-$, $\tilde{\ell}^+\tilde{\ell}^-$ and Higgs boson at FCC-he using the EPA approach. Results are shown for a minimum two-photon invariant mass of $W_0 \sim 2M_X$ with $X = W, Z, \text{top}, \tilde{H}, \tilde{\ell}$. For the single Higgs boson production process, we consider $W_0 \sim M_{\text{Higgs}}$. For the case of LbL, the cross-section value at $W_0 = 10 \text{ GeV}$ is reported.

$\sigma(ep \rightarrow e(X)p^{(*)})$	σ^{el}	$\sigma_{M_N < 10 \text{ GeV}}^{\text{inel}}$	$\sigma_{M_N < 50 \text{ GeV}}^{\text{inel}}$	$\sigma_{M_N < 300 \text{ GeV}}^{\text{inel}}$
W^+W^-	3.205×10^{-1}	2.083×10^{-1}	4.114×10^{-1}	5.652×10^{-1}
ZZ	4.976×10^{-4}	3.286×10^{-4}	6.427×10^{-4}	8.687×10^{-4}
$t\bar{t}$	8.333×10^{-4}	5.613×10^{-4}	1.085×10^{-3}	1.439×10^{-3}
$\gamma\gamma$ (LbL)	6.932×10^{-2}	3.116×10^{-2}	6.614×10^{-2}	9.929×10^{-2}
$\tilde{H}^+\tilde{H}^-$	9.595×10^{-3}	6.263×10^{-3}	1.234×10^{-2}	1.698×10^{-2}
$\tilde{\ell}^+\tilde{\ell}^-$	1.630×10^{-3}	1.055×10^{-3}	2.091×10^{-3}	2.898×10^{-3}
H	4.643×10^{-4}	2.751×10^{-4}	5.721×10^{-4}	8.707×10^{-4}

13 Summary and Conclusions

For the first time an exploratory survey of production cross sections for the exclusive production of lepton pairs, W and Z boson pairs, the Higgs boson, and pairs of charged supersymmetric particles via photon-photon interactions was performed for the future LHeC

and FCC-he, the two major proposals for electron-hadron colliders at CERN.

With their high luminosity and clean experimental environment, these latter are particularly well-suited for these investigations, revealing excellent prospects for advancing our understanding of these processes. The combination of these factors, along with advanced high-resolution detectors in a reasonable range of rapidities, creates optimal conditions for detailed studies of $\gamma\gamma$ processes.

The results presented in this paper highlight the potential of electron-proton colliders to explore a broad range of photon-photon interactions, including the exclusive production of various particles with large statistics.

This study suggests that the LHeC can play a significant role in probing both the Standard Model and potential New Physics scenarios with high precision.

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A The standard photon luminosity spectrum $S_{\gamma\gamma}$

The luminosity spectrum $S_{\gamma\gamma}$ as given in Eq. 3.1 is defined by the convolution of photon fluxes from the electron and proton:

$$S_{\gamma\gamma} = \frac{2}{W} \int_{W^2/s}^1 dy_e \Phi_e(y_e) y_p \Phi_p(y_p), \quad (\text{A.1})$$

where $y_p = \frac{W^2}{y_e s}$. Consequently, the luminosity spectrum can also be expressed as:

$$S_{\gamma\gamma} = \frac{2W}{s} \int_{W^2/s}^1 \frac{dy_e}{y_e} \Phi_e(y_e) \Phi_p\left(\frac{W^2}{y_e s}\right). \quad (\text{A.2})$$

Considering the differential cross-section $\frac{d\sigma_{ep}}{dW}$, which is given by

$$\frac{d\sigma_{ep}}{dW} = S_{\gamma\gamma} \sigma_{\gamma\gamma}(W), \quad (\text{A.3})$$

one can then calculate the differential cross-section $\frac{d\sigma_{ep}}{dY}$ as follows:

$$\frac{d\sigma_{ep}}{dY} = \frac{2}{s} \int_{W_0}^{\sqrt{s}} dW \sigma_{\gamma\gamma}(W) W \Phi_e(y_e) \Phi_p(y_p) = \frac{1}{s} \int_{s_0}^s dW^2 \sigma_{\gamma\gamma}(W) \Phi_e(y_e) \Phi_p(y_p), \quad (\text{A.4})$$

where $Y = \frac{1}{2} \ln\left(\frac{E_p y_p}{E_e y_e}\right)$ is the rapidity of the $\gamma\gamma$ system, with $y_p = \frac{W e^Y}{2E_p}$ and $y_e = \frac{W e^{-Y}}{2E_e}$. During the integration, it is crucial to ensure that the photon fluxes are set to zero when $y_p > 1$ or $y_e > 1$.

B The $\gamma\gamma$ level cross-sections $\sigma_{\gamma\gamma}(W)$

- The $\gamma\gamma$ cross-section for the production of W boson pairs near the threshold, below the $\gamma\gamma$ center-of-mass energy $W = 300$ GeV [68], is given by

$$\sigma(\gamma\gamma \rightarrow W^+ W^-)(W) \simeq \frac{19\pi(\hbar c)^2 \alpha^2}{2M_W^2} \sqrt{1 - \frac{4M_W^2}{W^2}}, \quad (\text{B.1})$$

where $M_W = 80.4$ GeV is the mass of the W boson, and $\alpha = \frac{1}{137}$ is the fine-structure constant. Above $W = 300$ GeV, the cross-section is assumed to be constant and equal to

$$\sigma(\gamma\gamma \rightarrow W^+ W^-) \approx \frac{8\pi(\hbar c)^2 \alpha^2}{M_W^2}. \quad (\text{B.2})$$

- The $\gamma\gamma$ cross-section for the production of a charged fermion-antifermion pair is given by [86]:

$$\sigma(\gamma\gamma \rightarrow f^+ f^-)(W) = \frac{4\pi(\hbar c)^2 \alpha^2}{W^2} \beta \left(\frac{3 - \beta^4}{2\beta} \ln\left(\frac{1 + \beta}{1 - \beta}\right) - 2 + \beta^2 \right), \quad (\text{B.3})$$

where $\beta = \sqrt{1 - \frac{4m^2}{W^2}}$, and m is the fermion mass.

- The Higgs boson production cross-section in photon-photon collisions, $\sigma(\gamma\gamma \rightarrow H)$, can be calculated using the narrow resonance approximation, which is given by [74, 87, 88]:

$$\sigma(\gamma\gamma \rightarrow H)(W) \simeq 8\pi^2 \frac{\Gamma_{\gamma\gamma}(\hbar c)^2}{M_H} \frac{1}{\pi} \frac{M_H \Gamma}{(M_H^2 - W^2)^2 + M_H^2 \Gamma^2} \simeq 4\pi^2 \frac{\Gamma_{\gamma\gamma}(\hbar c)^2}{M_H^2} \delta(W - M_H), \quad (\text{B.4})$$

where $M_H = 125$ GeV is the Higgs boson mass, $\Gamma_{\gamma\gamma}$ is the two-photon decay width, and Γ is the total decay width of the Higgs boson. These quantities are given by $\Gamma_{\gamma\gamma} \simeq 2.27 \times 10^{-3} \Gamma$ and $\Gamma \simeq 4.07$ MeV [89].

- The cross-section for slepton pair production via two-photon fusion ($\gamma\gamma \rightarrow \tilde{\ell}^+ \tilde{\ell}^-$) is given by [76, 86]:

$$\sigma(\gamma\gamma \rightarrow \tilde{\ell}^+ \tilde{\ell}^-)(W) = \frac{2\pi(\hbar c)^2 \alpha^2}{W^2} \beta \left[2 - \beta^2 - \frac{1 - \beta^4}{2\beta} \ln \left(\frac{1 + \beta}{1 - \beta} \right) \right], \quad (\text{B.5})$$

where $\beta = \sqrt{1 - \frac{4m_{\tilde{\ell}}^2}{W^2}}$, with $m_{\tilde{\ell}}$ being the mass of the slepton.

Similarly, the production of charged Higgs boson pairs in photon-photon collisions ($\gamma\gamma \rightarrow H^+ H^-$) occurs at the same rate as slepton pair production, assuming equal masses for the sleptons and charged Higgs bosons.

C Kinematics correction: The exact elastic case $W^2 = f(y_e, y_p, Q_e^2, Q_p^2) = -Q_e^2 + y_e y_p s$

- The photon flux for electrons, $\Phi_e(y_e)$, can be written as:

$$\Phi_e(y_e) = \frac{\alpha}{\pi y_e} \int dQ^2 \frac{1}{Q^2} \left[(1 - y_e) \left(1 - \frac{Q_{\min}^2}{Q^2} \right) F_E(Q^2) + \frac{y_e^2}{2} F_M(Q^2) \right] \quad (\text{C.1})$$

Where α is the fine-structure constant and $F_E(Q^2)$ and $F_M(Q^2)$ are the electric and magnetic form factors, respectively.

For electrons:

$$F_E = F_M = 1$$

- The photon flux for protons, $\Phi_p(y_p)$, can be written as:

$$\Phi_p(y_p) = \frac{\alpha}{\pi y_p} \int dQ^2 \frac{1}{Q^2} \left[(1 - y_p) \left(1 - \frac{Q_{\min}^2}{Q^2} \right) F_E(Q^2) + \frac{y_p^2}{2} F_M(Q^2) \right] \quad (\text{C.2})$$

Where $F_E(Q^2)$ and $F_M(Q^2)$ are given by the dipole approximation:

$$F_E = \frac{4M_p^2 G_E^2 + Q^2 G_M^2}{4M_p^2 + Q^2}, \quad F_M = G_M^2$$

With:

$$G_E = G_M / 7.78 = \left(1 + \frac{Q^2}{0.71 \text{ GeV}^2} \right)^{-4}$$

The correct formula for the photon-photon center-of-mass energy squared W^2 in the case of elastic interaction can be written as,

$$W^2 = -Q_e^2 + y_e y_p s \quad (\text{C.3})$$

$$W = \sqrt{-Q_e^2 + y_e y_p s} \quad (\text{C.4})$$

$$y_p^* = \frac{W^2 + Q_e^2}{y_e s} \quad (\text{C.5})$$

$$\frac{\partial W}{\partial y_p} = \frac{\partial}{\partial y_p} \left(\sqrt{-Q_e^2 + y_e y_p s} \right) \quad (\text{C.6})$$

$$\frac{\partial W}{\partial y_p} = \frac{1}{2} \cdot (-Q_e^2 + y_e y_p s)^{-1/2} \cdot (y_e s) \quad (\text{C.7})$$

$$\frac{\partial W}{\partial y_p} = \frac{y_e s}{2W} \quad (\text{C.8})$$

Process (elastic)	EPA ($W^2 = y_e y_p s$)	GRAPE (BH)	lpair	EPA ($W^2 = -Q_e^2 + y_e y_p s$)	MG(EPA:iww)
$(Q_e^2/Q_p^2 = 1/1 \text{ GeV}^2)$	41.15	41.35	40.69	41.01	43.92
$(Q_e^2/Q_p^2 = 10/10 \text{ GeV}^2)$	46.32	46.30	45.49	46.01	49.41
$(Q_e^2/Q_p^2 = 50/50 \text{ GeV}^2)$	49.74	48.82	48.82	48.95	53.11
$(Q_e^2/Q_p^2 = 100/100 \text{ GeV}^2)$	51.22	49.54	49.56	49.96	54.68
$(Q_e^2/Q_p^2 = 10^3/10^3 \text{ GeV}^2)$	56.11	50.41	49.71	51.70	59.89
$(Q_e^2/Q_p^2 = 10^4/10^4 \text{ GeV}^2)$	60.77	50.47	49.73	51.94	65.09
$(Q_e^2/Q_p^2 = 10^5/10^5 \text{ GeV}^2)$	65.65	50.48	49.67	51.94	70.37

Table 14. The elastic $\tau^+\tau^-$ production cross sections (in pb) at LHeC calculated using the EPA approach at $W_0 = 10 \text{ GeV}$, assuming different values of $Q_{e,max}^2$ and $Q_{p,max}^2$ for electron/proton vertex. Cross-sections using the lpair, GRAPE and **MadGraph5_aMC@NLO (with EPA [iww])** MC event generators are also presented for comparison.

Process (elastic)	EPA (Old)	GRAPE (EW)	GRAPE (BH)	EPA(Corrected)
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_e^2/Q_p^2 = 10/10 \text{ GeV}^2)$	9.153×10^{-4}	9.157×10^{-4}	9.155×10^{-4}	9.148×10^{-4}
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_e^2/Q_p^2 = 10^2/10^2 \text{ GeV}^2)$	1.037×10^{-4}	1.038×10^{-3}	1.037×10^{-3}	1.037×10^{-3}
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_e^2/Q_p^2 = 10^3/10^3 \text{ GeV}^2)$	1.159×10^{-3}	1.163×10^{-3}	1.157×10^{-3}	1.157×10^{-3}
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_e^2/Q_p^2 = 10^4/10^4 \text{ GeV}^2)$	1.281×10^{-3}	1.295×10^{-3}	1.259×10^{-3}	1.266×10^{-3}
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_e^2/Q_p^2 = 10^5/10^5 \text{ GeV}^2)$	1.396×10^{-3}	1.375×10^{-3}	1.298×10^{-3}	1.322×10^{-3}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_e^2/Q_p^2 = 10/10 \text{ GeV}^2)$	2.238×10^{-5}	2.240×10^{-5}	2.239×10^{-5}	2.236×10^{-5}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_e^2/Q_p^2 = 10^2/10^2 \text{ GeV}^2)$	2.556×10^{-5}	2.562×10^{-5}	2.557×10^{-5}	2.256×10^{-5}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_e^2/Q_p^2 = 10^3/10^3 \text{ GeV}^2)$	2.874×10^{-5}	2.910×10^{-5}	2.871×10^{-5}	2.871×10^{-5}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_e^2/Q_p^2 = 10^4/10^4 \text{ GeV}^2)$	3.192×10^{-5}	3.356×10^{-5}	3.169×10^{-5}	3.173×10^{-5}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_e^2/Q_p^2 = 10^5/10^5 \text{ GeV}^2)$	3.494×10^{-5}	3.756×10^{-5}	3.346×10^{-5}	3.383×10^{-5}

Table 15. The **elastic** higgsinos production cross sections (in pb) at LHeC calculated using the EPA approach at $W_0 = 2 m_{\tilde{H}}$, assuming different values of $Q_{e,max}^2$ and $Q_{p,max}^2$ for electron/proton vertex. Cross-sections calculated using the GRAPE MC event generator are also presented for comparison.

The Jacobian is given by the absolute value of the partial derivative:

$$J = \left| \frac{\partial W}{\partial y_p} \right| = \left| \frac{y_e s}{2W} \right| = \left| \frac{y_e s}{2\sqrt{-Q_e^2 + y_e y_p s}} \right| \quad (\text{C.9})$$

The final expression for the elastic $S_{\gamma\gamma}(W)$ considering the property of the Jacobian is given by,

$$S_{\gamma\gamma}(W) = \int_{y_{e,\min}}^{y_{e,\max}} \int_{Q_{e,\min}^2}^{Q_{e,\max}^2} \Phi_e(y_e, Q_e^2) \frac{\Phi_p(y_p^*)}{\left| \frac{\partial W}{\partial y_p} \right|_{y_p=y_p^*}} dQ_e^2 dy_e \quad (\text{C.10})$$

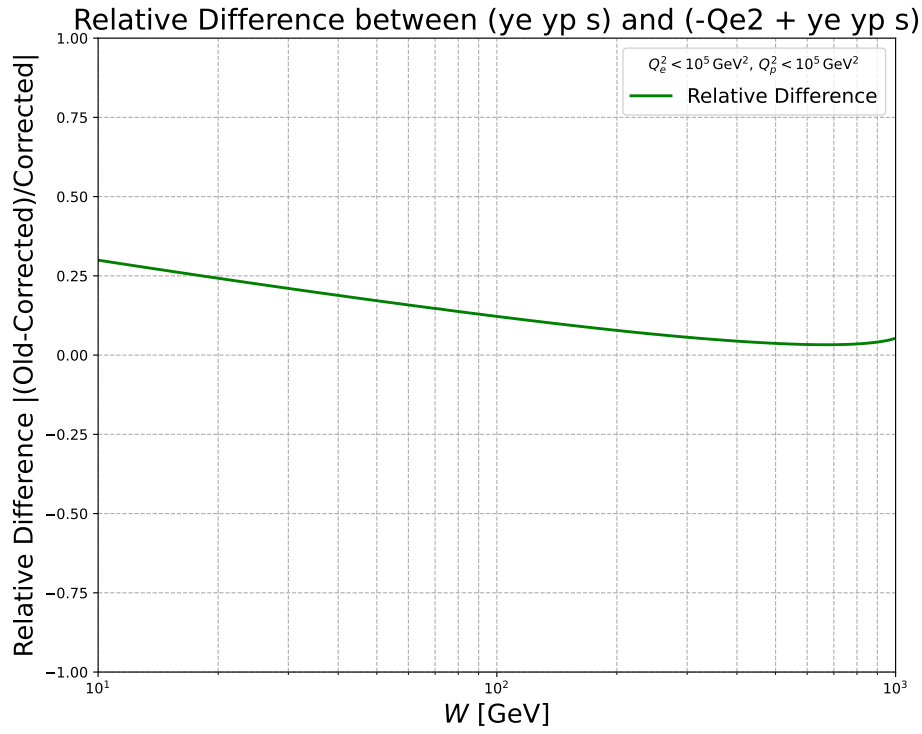
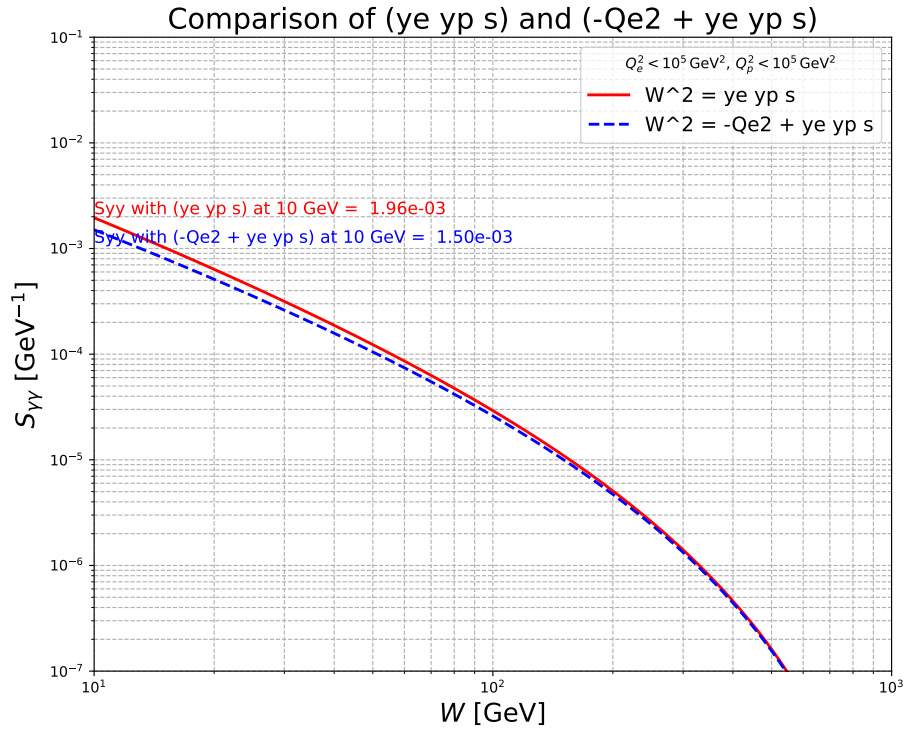


Figure 42. Comparison of elastic photon-photon luminosity spectrum and the relative difference calculated using exact $W^2 = -Q_e^2 + y_e y_p s$ and the approximate $W^2 = y_e y_p s$.

D Kinematics correction: The exact inelastic case : $W^2 = f(y_e, y_p, Q_e^2, Q_p^2)$

The correct formula for the photon-photon center-of-mass energy squared W^2 in the case of inelastic interaction can be written as,

$$W^2 = -Q_e^2 - Q_p^2 + y_e y_p s + Q_e^2(Q_p^2 + M_N^2 - M_p^2)/s \quad (\text{D.1})$$

$$W^2 = -Q_e^2 - Q_p^2 + y_e y_p 4E_e E_p + Q_e^2(Q_p^2 + M_N^2 - M_p^2)/(4E_e E_p) \quad (\text{D.2})$$

$$\Phi_e(y_e) = \frac{\alpha}{\pi y_e} \int dQ^2 \frac{1}{Q^2} \left[(1 - y_e) \left(1 - \frac{Q_{\min}^2}{Q^2} \right) + \frac{y_e^2}{2} \right] \quad (\text{D.3})$$

$$\Phi_p(y_p) = \frac{\alpha}{\pi y_p} \int dQ^2 \frac{1}{Q^2} \left[(1 - y_p) \left(1 - \frac{Q_{\min}^2}{Q^2} \right) F_E(Q^2) + \frac{y_p^2}{2} F_M(Q^2) \right] \quad (\text{D.4})$$

$$F_E(Q^2) = \int dx \frac{F_2(x, Q^2)}{x} \quad (\text{D.5})$$

$$F_M(Q^2) = \int dx \frac{F_2(x, Q^2)}{x^3} \quad (\text{D.6})$$

$$x = \frac{Q^2}{M_N^2 - M_p^2 + Q^2} \quad (\text{D.7})$$

$$\frac{dx}{dM_N} = \frac{2M_N Q^2}{(M_N^2 - M_p^2 + Q^2)^2} \quad (\text{D.8})$$

$$F_E(Q^2) = \int_{M_N^{\min}}^{M_N^{\max}} dM_N F_2 \left(\frac{Q^2}{M_N^2 - M_p^2 + Q^2}, Q^2 \right) \cdot \frac{2M_N}{M_N^2 - M_p^2 + Q^2} \quad (\text{D.9})$$

$$F_M(Q^2) = \int_{M_N^{\min}}^{M_N^{\max}} dM_N F_2 \left(\frac{Q^2}{M_N^2 - M_p^2 + Q^2}, Q^2 \right) \cdot \frac{2M_N(M_N^2 - M_p^2 + Q^2)}{Q^2 \times Q^2} \quad (\text{D.10})$$

$$F_E(Q^2, M_N) = F_2 \left(\frac{Q^2}{M_N^2 - M_p^2 + Q^2}, Q^2 \right) \cdot \frac{2M_N}{M_N^2 - M_p^2 + Q^2} \quad (\text{D.11})$$

$$F_M(Q^2, M_N) = F_2 \left(\frac{Q^2}{M_N^2 - M_p^2 + Q^2}, Q^2 \right) \cdot \frac{2M_N(M_N^2 - M_p^2 + Q^2)}{Q^2 \times Q^2} \quad (\text{D.12})$$

$$\Phi_e(y_e, Q_e^2) = \frac{\alpha}{\pi y_e} \frac{1}{Q_e^2} \left[(1 - y_e) \left(1 - \frac{Q_{\min}^2}{Q_e^2} \right) + \frac{y_e^2}{2} \right] \quad (\text{D.13})$$

$$\Phi_p(y_p, Q_p^2, M_N) = \frac{\alpha}{\pi y_p} \frac{1}{Q_p^2} \left[(1 - y_p) \left(1 - \frac{Q_{\min}^2}{Q_p^2} \right) F_E(Q_p^2, M_N) + \frac{y_p^2}{2} F_M(Q_p^2, M_N) \right] \quad (\text{D.14})$$

The general expression for the photon luminosity spectrum $S_{\gamma\gamma}(W)$,

$$S_{\gamma\gamma}(W) = \int_{y_{e,\min}}^{y_{e,\max}} \int_{Q_{e,\min}^2}^{Q_{e,\max}^2} \int_{Q_{p,\min}^2}^{Q_{p,\max}^2} \int_{M_N^{\min}}^{M_N^{\max}} \int_{y_{p,\min}}^{y_{p,\max}} \Phi_e(y_e, Q_e^2) \Phi_p(y_p, Q_p^2, M_N) dy_p dM_N dQ_p^2 dQ_e^2 dy_e \quad (\text{D.15})$$

$$W = \sqrt{-Q_e^2 - Q_p^2 + y_e y_p 4E_e E_p + \frac{Q_e^2(Q_p^2 + M_N^2 - M_p^2)}{(4E_e E_p)}} \quad (\text{D.16})$$

$$J_{y_p} = \frac{\partial W}{\partial y_p} = \frac{2y_e E_e E_p}{W} \quad (\text{D.17})$$

$$y_p^* = \frac{W^2 + Q_e^2 + Q_p^2 - \frac{Q_e^2(Q_p^2 + M_N^2 - M_p^2)}{4E_e E_p}}{4y_e E_e E_p} \quad (\text{D.18})$$

Final expression for the photon luminosity spectrum $S_{\gamma\gamma}(W)$,

$$S_{\gamma\gamma}(W) = \int_{y_{e,\min}}^{y_{e,\max}} \int_{Q_{e,\min}^2}^{Q_{e,\max}^2} \int_{M_N^{\min}}^{M_N^{\max}} \int_{Q_{p,\min}^2}^{Q_{p,\max}^2} \Phi_e(y_e, Q_e^2) \Phi_p(y_p^*, Q_p^2, M_N) \left[\frac{1}{\left| \frac{\partial W}{\partial y_p} \right|_{y_p=y_p^*}} \right] dQ_p^2 dM_N dQ_e^2 dy_e \quad (\text{D.19})$$

$$M_N^{\min} = M_p + M_{\pi^0} \quad (\text{D.20})$$

$$Q_{e,\min}^2 = \frac{m_e^2 y_e^2}{1 - y_e}, \quad \text{for } 0 < y_e < 1 \quad (\text{D.21})$$

$$Q_{p,\min}^2 = \left(\frac{M_N^2}{1 - y_p} - M_p^2 \right) y_p, \quad \text{for } 0 < y_p < 1 \quad (\text{D.22})$$

Process (inelastic)	EPA ($W^2 = y_e y_p s$)	GRAPE (BH)	lpair/cepgen	EPA (Based on Eq. (D.1))
$(Q_e^2/Q_p^2 = 10/10 \text{ GeV}^2); M_N < 10 \text{ GeV}$	24.07	22.49	23.37	23.72
$(Q_e^2/Q_p^2 = 50/50 \text{ GeV}^2); M_N < 10 \text{ GeV}$	29.15	26.34	27.47	28.19
$(Q_e^2/Q_p^2 = 10^2/10^2 \text{ GeV}^2); M_N < 10 \text{ GeV}$	30.25	26.87	28.11	28.97
$(Q_e^2/Q_p^2 = 10^3/10^3 \text{ GeV}^2); M_N < 10 \text{ GeV}$	33.09	27.36	28.76	30.07
$(Q_e^2/Q_p^2 = 10^4/10^4 \text{ GeV}^2); M_N < 10 \text{ GeV}$	35.89	27.41	28.77	30.27
$(Q_e^2/Q_p^2 = 10^5/10^5 \text{ GeV}^2); M_N < 10 \text{ GeV}$	38.69	27.41	28.74	30.27
$(Q_e^2/Q_p^2 = 10^5/10 \text{ GeV}^2); M_N < 10 \text{ GeV}$	33.94	24.57	25.59	26.12

Table 16. Inelastic $\tau^+\tau^-$ production cross sections (in pb) at LHeC calculated using the EPA approach at $W_0 = 10 \text{ GeV}$ assuming a constraint on the diffractive system mass $M_N < 10 \text{ GeV}$. Cross-sections using the lpair and GRAPE MC event generators are also presented for comparison. For the exact inelastic $S_{\gamma\gamma}(W)$ calculation in Eq. D.19, we use Monte Carlo/VEGAS integrations.

Process (inelastic)	EPA (luxlike)	GRAPE (BH)	lpair (Suri-Yennie/Brasse)	EPA (ALLM)
$(Q_e^2/Q_p^2 = 10/10 \text{ GeV}^2); M_N < 50 \text{ GeV}$	30.77	30.99	28.01/44.94	30.99
$(Q_e^2/Q_p^2 = 50/50 \text{ GeV}^2); M_N < 50 \text{ GeV}$	44.32	41.95	37.35/48.91	42.19
$(Q_e^2/Q_p^2 = 10^2/10^2 \text{ GeV}^2); M_N < 50 \text{ GeV}$	48.32	44.62	39.62/49.48	48.08
$(Q_e^2/Q_p^2 = 10^3/10^3 \text{ GeV}^2); M_N < 50 \text{ GeV}$	53.26	45.38	42.19/50.49	52.96
$(Q_e^2/Q_p^2 = 10^4/10^4 \text{ GeV}^2); M_N < 50 \text{ GeV}$	53.67	45.44	42.37/50.29	53.34
$(Q_e^2/Q_p^2 = 10^5/10^5 \text{ GeV}^2); M_N < 50 \text{ GeV}$	53.66	45.45	42.32/50.49	53.39

Table 17. Same as Table.16, for $M_N < 50 \text{ GeV}$. The inelastic results in this table are presented for two different sets of structure functions; LUX-like and ALLM. Both EPA results are based on the corrected W^2 formula in Eq. (D.1). For the case of lpair/cepgen, the results calculated using two structure functions of Suri-Yennie and Fiore-Brasse are presented for comparison.

Process (inelastic)	EPA (Yennie)	GRAPE (BH)	lpair (Yennie/Brasse)	EPA (ALLM)
$(Q_e^2/Q_p^2 = 10/10 \text{ GeV}^2); M_N < 300 \text{ GeV}$	31.67	33.37	28.44/45.20	32.29
$(Q_e^2/Q_p^2 = 50/50 \text{ GeV}^2); M_N < 300 \text{ GeV}$	45.32	47.84	38.69/48.96	49.91
$(Q_e^2/Q_p^2 = 10^2/10^2 \text{ GeV}^2); M_N < 300 \text{ GeV}$	50.33	52.22	41.75/49.73	56.66
$(Q_e^2/Q_p^2 = 10^3/10^3 \text{ GeV}^2); M_N < 300 \text{ GeV}$	59.25	53.05	45.32/50.59	67.90
$(Q_e^2/Q_p^2 = 10^4/10^4 \text{ GeV}^2); M_N < 300 \text{ GeV}$	60.72	53.12	45.80/50.63	69.25
$(Q_e^2/Q_p^2 = 10^5/10^5 \text{ GeV}^2); M_N < 300 \text{ GeV}$	60.85	53.13	45.60/50.45	69.37

Table 18. Same as Table.16, for $M_N < 300 \text{ GeV}$. The inelastic results in this table are presented for two different sets of structure functions; Suri-Yennie and ALLM. Both EPA results are based on the corrected W^2 formula in Eq. (D.1). For the case of lpair/cepgen, the results calculated using two structure functions of Suri-Yennie, and Fiore-Brasse are presented for comparison.

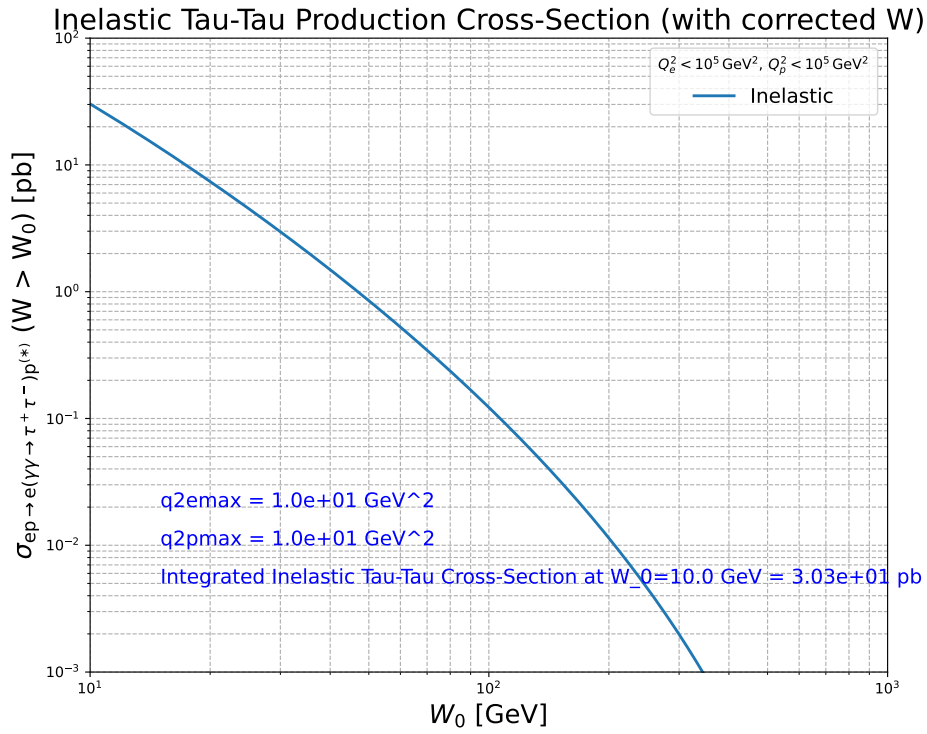
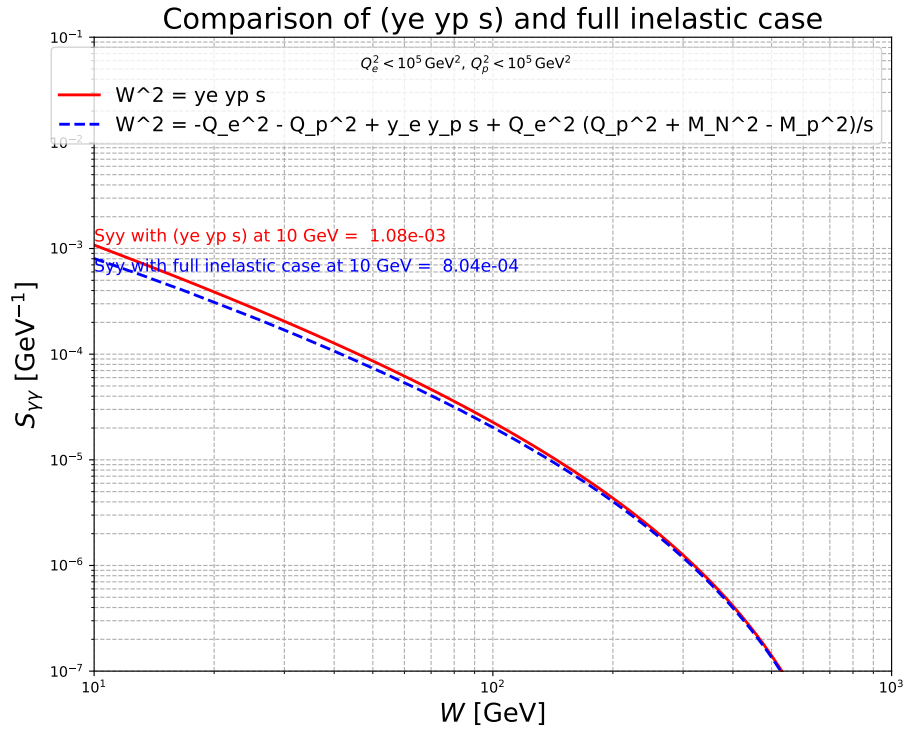


Figure 43. (top) Comparison of inelastic photon-photon luminosity spectrum calculated using $W^2 = -Q_e^2 - Q_p^2 + y_e y_p s + Q_e^2 (Q_p^2 + M_N^2 - M_p^2)/s$ and $W^2 = y_e y_p s$; (down) The corrected inelastic $\tau^+ \tau^-$ production cross-section.

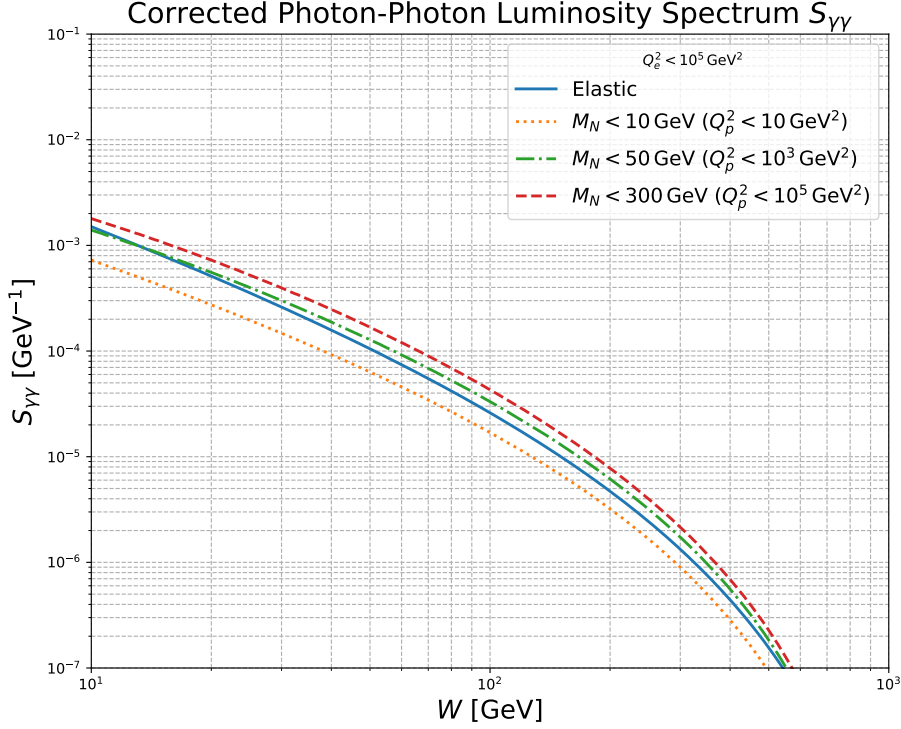


Figure 44. The **corrected** elastic luminosity spectrum $S_{\gamma\gamma}$ at the LHeC, assuming $Q_{e,\max}^2 = 10^5 \text{ GeV}^2$ and the appropriate kinematical limits for the minimal photon virtualities (see text above), as well as three **corrected** inelastic luminosity spectra where the maximum dissociative mass M_N of 10, 50, and 300 GeV is assumed for $Q_{p,\max}^2 = 10, 10^3$, and 10^5 GeV^2 , respectively.

Process (inelastic)	EPA (Old)	GRAPE (BH)	lpair (Yennie)	EPA (Corrected)
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_{e,\max}^2/Q_{p,\max}^2 = 10/10 \text{ GeV}^2)$	6.113×10^{-4}	6.034×10^{-4}	6.226×10^{-4}	6.138×10^{-4}
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_{e,\max}^2/Q_{p,\max}^2 = 10^2/10^2 \text{ GeV}^2)$	9.038×10^{-4}	8.951×10^{-4}	9.356×10^{-4}	9.080×10^{-4}
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_{e,\max}^2/Q_{p,\max}^2 = 10^3/10^3 \text{ GeV}^2)$	1.015×10^{-3}	1.004×10^{-3}	1.050×10^{-3}	1.019×10^{-3}
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_{e,\max}^2/Q_{p,\max}^2 = 10^4/10^4 \text{ GeV}^2)$	1.122×10^{-3}	1.093×10^{-3}	1.142×10^{-3}	1.115×10^{-3}
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_{e,\max}^2/Q_{p,\max}^2 = 10^5/10^5 \text{ GeV}^2)$	1.229×10^{-3}	1.127×10^{-3}	1.182×10^{-3}	1.165×10^{-3}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_{e,\max}^2/Q_{p,\max}^2 = 10/10 \text{ GeV}^2)$	1.291×10^{-5}	1.313×10^{-5}	1.400×10^{-5}	1.298×10^{-5}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_{e,\max}^2/Q_{p,\max}^2 = 10^2/10^2 \text{ GeV}^2)$	2.149×10^{-5}	2.187×10^{-5}	2.320×10^{-5}	2.159×10^{-5}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_{e,\max}^2/Q_{p,\max}^2 = 10^3/10^3 \text{ GeV}^2)$	2.449×10^{-5}	2.492×10^{-5}	2.654×10^{-5}	2.548×10^{-5}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_{e,\max}^2/Q_{p,\max}^2 = 10^4/10^4 \text{ GeV}^2)$	2.721×10^{-5}	2.749×10^{-5}	2.930×10^{-5}	2.717×10^{-5}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_{e,\max}^2/Q_{p,\max}^2 = 10^5/10^5 \text{ GeV}^2)$	2.993×10^{-5}	2.899×10^{-5}	3.091×10^{-5}	2.896×10^{-5}

Table 19. Same as Table.16, for the inelastic higgsinos production cross sections (in pb), **assuming a constraint on the diffractive system mass $M_N < 10 \text{ GeV}$.**

Process (inelastic)	GRAPE (EW)	GRAPE (BH)	lpair (Yennie)	EPA (Corrected)
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_{e,\text{max}}^2/Q_{p,\text{max}}^2 = 10/10 \text{ GeV}^2)$	6.037×10^{-4}	6.034×10^{-4}	6.226×10^{-4}	6.138×10^{-4}
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_{e,\text{max}}^2/Q_{p,\text{max}}^2 = 10^2/10^2 \text{ GeV}^2)$	8.955×10^{-3}	8.951×10^{-3}	9.356×10^{-3}	9.080×10^{-3}
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_{e,\text{max}}^2/Q_{p,\text{max}}^2 = 10^3/10^3 \text{ GeV}^2)$	1.007×10^{-3}	1.004×10^{-3}	1.050×10^{-3}	1.019×10^{-3}
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_{e,\text{max}}^2/Q_{p,\text{max}}^2 = 10^4/10^4 \text{ GeV}^2)$	1.113×10^{-3}	1.093×10^{-3}	1.142×10^{-3}	1.115×10^{-3}
$m_{\tilde{H}} = 100 \text{ GeV}, (Q_{e,\text{max}}^2/Q_{p,\text{max}}^2 = 10^5/10^5 \text{ GeV}^2)$	1.444×10^{-3}	1.127×10^{-3}	1.182×10^{-3}	1.165×10^{-3}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_{e,\text{max}}^2/Q_{p,\text{max}}^2 = 10/10 \text{ GeV}^2)$	1.331×10^{-5}	1.331×10^{-5}	1.400×10^{-5}	1.298×10^{-5}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_{e,\text{max}}^2/Q_{p,\text{max}}^2 = 10^2/10^2 \text{ GeV}^2)$	2.189×10^{-5}	2.187×10^{-5}	2.320×10^{-5}	2.159×10^{-5}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_{e,\text{max}}^2/Q_{p,\text{max}}^2 = 10^3/10^3 \text{ GeV}^2)$	2.514×10^{-5}	2.492×10^{-5}	2.654×10^{-5}	2.548×10^{-5}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_{e,\text{max}}^2/Q_{p,\text{max}}^2 = 10^4/10^4 \text{ GeV}^2)$	2.868×10^{-5}	2.749×10^{-5}	2.930×10^{-5}	2.717×10^{-5}
$m_{\tilde{H}} = 200 \text{ GeV}, (Q_{e,\text{max}}^2/Q_{p,\text{max}}^2 = 10^5/10^5 \text{ GeV}^2)$	3.153×10^{-5}	2.899×10^{-5}	3.091×10^{-5}	2.896×10^{-5}

Table 20. Same as Table.19, GRAPE (EW) has been included to compare with GRAPE (BH), assuming a constraint on the diffractive system mass $M_N < 10 \text{ GeV}$.

Process (inelastic)	GRAPE (EW)	GRAPE (BH)	lpair (Yennie)	EPA (Corrected)
$m_{\tilde{H}} = 100, (Q_e^2/Q_p^2 = 10/10 \text{ GeV}^2)$	6.059×10^{-4}	6.056×10^{-4}	6.222×10^{-4}	6.155×10^{-4}
$m_{\tilde{H}} = 100, (Q_e^2/Q_p^2 = 10^2/10^2 \text{ GeV}^2)$	1.063×10^{-3}	1.062×10^{-3}	1.077×10^{-3}	1.078×10^{-3}
$m_{\tilde{H}} = 100, (Q_e^2/Q_p^2 = 10^3/10^3 \text{ GeV}^2)$	1.483×10^{-3}	1.479×10^{-3}	1.546×10^{-3}	1.496×10^{-3}
$m_{\tilde{H}} = 100, (Q_e^2/Q_p^2 = 10^4/10^4 \text{ GeV}^2)$	1.662×10^{-3}	1.643×10^{-3}	1.747×10^{-3}	1.672×10^{-3}
$m_{\tilde{H}} = 100, (Q_e^2/Q_p^2 = 10^5/10^5 \text{ GeV}^2)$	2.009×10^{-3}	1.693×10^{-3}	1.802×10^{-3}	1.745×10^{-3}
$m_{\tilde{H}} = 200, (Q_e^2/Q_p^2 = 10/10 \text{ GeV}^2)$	1.331×10^{-5}	1.331×10^{-5}	1.399×10^{-5}	1.295×10^{-5}
$m_{\tilde{H}} = 200, (Q_e^2/Q_p^2 = 10^2/10^2 \text{ GeV}^2)$	2.359×10^{-5}	2.357×10^{-5}	2.484×10^{-5}	2.330×10^{-5}
$m_{\tilde{H}} = 200, (Q_e^2/Q_p^2 = 10^3/10^3 \text{ GeV}^2)$	3.393×10^{-5}	3.371×10^{-5}	3.714×10^{-5}	3.340×10^{-5}
$m_{\tilde{H}} = 200, (Q_e^2/Q_p^2 = 10^4/10^4 \text{ GeV}^2)$	3.994×10^{-5}	3.875×10^{-5}	4.386×10^{-5}	3.850×10^{-5}
$m_{\tilde{H}} = 200, (Q_e^2/Q_p^2 = 10^5/10^5 \text{ GeV}^2)$	4.340×10^{-5}	4.087×10^{-5}	4.602×10^{-5}	4.098×10^{-5}

Table 21. Inelastic higgsinos production cross sections (in pb) assuming a constraint on the diffractive system mass $M_N < 50 \text{ GeV}$.

Process (inelastic)	GRAPE (EW)	GRAPE (BH)	lpair (Yennie)	EPA (Corrected)
$m_{\tilde{H}} = 100, (Q_e^2/Q_p^2 = 10/10 \text{ GeV}^2)$	6.059×10^{-4}	6.057×10^{-4}	6.217×10^{-4}	6.153×10^{-4}
$m_{\tilde{H}} = 100, (Q_e^2/Q_p^2 = 10^2/10^2 \text{ GeV}^2)$	1.064×10^{-3}	1.062×10^{-3}	1.077×10^{-3}	1.078×10^{-3}
$m_{\tilde{H}} = 100, (Q_e^2/Q_p^2 = 10^3/10^3 \text{ GeV}^2)$	1.546×10^{-3}	1.543×10^{-3}	1.594×10^{-3}	1.556×10^{-3}
$m_{\tilde{H}} = 100, (Q_e^2/Q_p^2 = 10^4/10^4 \text{ GeV}^2)$	1.991×10^{-3}	1.976×10^{-3}	2.098×10^{-3}	1.973×10^{-3}
$m_{\tilde{H}} = 100, (Q_e^2/Q_p^2 = 10^5/10^5 \text{ GeV}^2)$	2.415×10^{-3}	2.103×10^{-3}	2.272×10^{-3}	2.132×10^{-3}
$m_{\tilde{H}} = 200, (Q_e^2/Q_p^2 = 10/10 \text{ GeV}^2)$	1.331×10^{-5}	1.331×10^{-5}	1.400×10^{-5}	1.296×10^{-5}
$m_{\tilde{H}} = 200, (Q_e^2/Q_p^2 = 10^2/10^2 \text{ GeV}^2)$	2.360×10^{-5}	3.357×10^{-5}	2.479×10^{-5}	2.330×10^{-5}
$m_{\tilde{H}} = 200, (Q_e^2/Q_p^2 = 10^3/10^3 \text{ GeV}^2)$	3.413×10^{-5}	3.390×10^{-5}	3.729×10^{-5}	3.357×10^{-5}
$m_{\tilde{H}} = 200, (Q_e^2/Q_p^2 = 10^4/10^4 \text{ GeV}^2)$	4.495×10^{-5}	4.375×10^{-5}	5.068×10^{-5}	4.308×10^{-5}
$m_{\tilde{H}} = 200, (Q_e^2/Q_p^2 = 10^5/10^5 \text{ GeV}^2)$	5.149×10^{-5}	4.896×10^{-5}	5.836×10^{-5}	4.872×10^{-5}

Table 22. Inelastic higgsinos production cross sections (in pb) assuming a constraint on the diffractive system mass $M_N < 300 \text{ GeV}$.

E Correction to the standard EPA framework - The Correction Factor

The EPA is a well-established method for estimating photon fluxes in high-energy collisions involving charged particles. However, for collisions involving photons from electrons and protons at the LHeC, the EPA becomes less accurate for the case of dilepton productions ($\ell = \mu, \tau$) when the photon virtuality, denoted by Q^2 , increases. Specifically, the standard EPA assumes that the photon-photon cross-section, $\sigma_{\gamma\gamma}(W)$, is independent of photon virtualities, an assumption that fails when Q^2 becomes comparable to or exceeds the invariant mass squared of the photon-photon system, W^2 . In this section, we discuss the correction needed to ensure the applicability of the EPA in such cases by introducing a suppression factor.

E.1 Introduction of the correction factor

To address the limitations of the EPA at high photon virtualities, a correction factor is introduced to dampen the contribution from large Q^2 values, ensuring that the EPA remains applicable. The correction factor used in this analysis is of an exponential form:

$$S(Q^2, W) = \exp\left(-\frac{Q^2}{cW^2}\right), \quad (\text{E.1})$$

where:

- Q^2 is the photon virtuality.
- W is the center-of-mass energy of the photon-photon system.
- c is a parameter that can be optimized through comparison with experimental data.

The motivation for introducing the suppression factor is to reduce the contribution from high Q^2 photon emissions. At large virtualities, the standard EPA tends to overestimate photon-photon interactions because it was initially developed under the assumption of soft photon emissions (i.e., low virtuality). This suppression factor helps bring the model closer to physical reality, particularly at the energy scales probed by the LHeC.

E.2 Modified photon flux

The modified photon flux for the case of dilepton production via photo-photon interaction, considering the correction factor can be expressed as:

$$f_{\text{modified}}(y, Q^2, W) = f_{\text{EPA}}(y, Q^2) \times S(Q^2, W), \quad (\text{E.2})$$

where $f_{\text{EPA}}(y, Q^2)$ represents the flux calculated using the original EPA formulation, and $S(Q^2, W)$ is the correction factor as defined above. This modification helps maintain the validity of EPA predictions even when photon virtualities are significant.

E.3 Photon flux with correction factor

The photon fluxes for electron or proton, are corrected by including the correction factor in the EPA calculation.

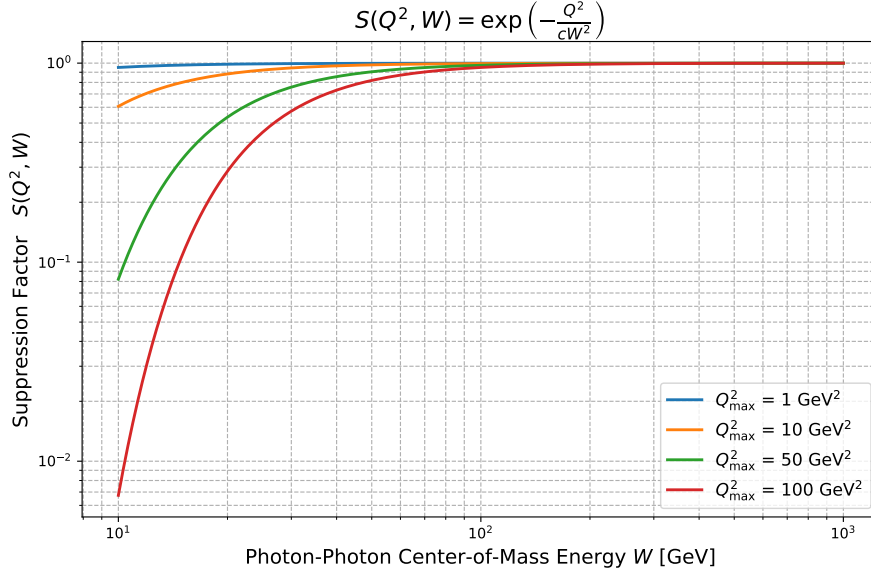


Figure 45. The correction factor $S(Q^2, W)$ as a function of photon-photon center of mass energy W , and for some selected values of $Q_{\max}^2$.

E.3.1 Photon flux for electrons

For photons emitted by electrons, the corrected flux incorporating the correction factor is given by,

$$\Phi_e(y, W) = \frac{\alpha}{\pi y} \int_{Q_{\min}^2}^{Q_{\max}^2} \frac{dQ^2}{Q^2} \left[(1-y) \left(1 - \frac{Q_{\min}^2}{Q^2} \right) + \frac{y^2}{2} \right] \times S(Q^2, W), \quad (\text{E.3})$$

where y is the energy fraction carried by the photon, $Q_{\min}^2$ and $Q_{\max}^2$ denote the minimum and maximum photon virtualities, respectively, and $S(Q^2, W)$ is the correction factor discussed above.

E.3.2 Photon flux for protons

Similarly, the photon flux for protons is modified by incorporating the correction factor as follows,

$$\Phi_p(y, W) = \frac{\alpha}{\pi y} \int_{Q_{\min}^2}^{Q_{\max}^2} \frac{dQ^2}{Q^2} \left[(1-y) \left(1 - \frac{Q_{\min}^2}{Q^2} \right) F_E(Q^2) + \frac{y^2}{2} F_M(Q^2) \right] \times S(Q^2, W), \quad (\text{E.4})$$

where $F_E(Q^2)$ and $F_M(Q^2)$ are the electric and magnetic form factors, respectively. The correction factor $S(Q^2, W)$ ensures that contributions from large virtualities are dampened effectively.

E.4 Parameter optimization (c)

The choice of the parameter c is crucial to accurately represent the reduction in photon flux due to high virtualities. The parameter c must be optimized based on experimental data to ensure that the correction factor accurately accounts for deviations observed at high Q^2 . In this work, the optimization c parameter is performed by Comparing the calculated photon-photon cross-sections with GRPAE MC calculations for the $\tau^+\tau^-$ production. Another possible approach to determine c is to perform a fit of the model to experimental data from similar energy scales, adjusting c to minimize the deviation between theoretical predictions and data. Sensitivity studies can also be conducted to evaluate the impact of different values of c on the predicted cross-sections.

The standard EPA is typically applicable when photon virtualities are small, $Q^2 \ll W^2$, such that the cross-section of the photon-induced process does not depend significantly on Q^2 . For electron-proton collisions like those at the LHeC, the EPA works well for the dilepton production for photon virtualities up to 100 GeV^2 . Beyond this range, corrections such as the correction factor become necessary to maintain accuracy. However, as shown in the paper, the standard EPA works well in producing massive objects such as higgsinos, even for the large photon virtualities, i.e. 10^6 GeV^2 .

E.5 The luminosity spectrum

The corrected luminosity spectrum $S_{\gamma\gamma}$ can be calculated by the convolution of photon fluxes from the electron and proton given above,

$$S_{\gamma\gamma}(W) = \frac{2}{W} \int_{W^2/s}^1 dy_e \Phi_e(y_e) y_p \Phi_p(y_p), \quad (\text{E.5})$$

$$\text{where } y_p = \frac{W^2}{y_e s}.$$

E.6 Results considering the corrected luminosity spectrum

In Table 25, the $\tau^+\tau^-$ production cross sections (in pb) at the LHeC are presented, calculated using the standard EPA approach for various values of $Q_{e,\text{max}}^2$ and $Q_{p,\text{max}}^2$, corresponding to the electron and proton vertices, respectively. Cross sections computed using the LPAIR and GRAPE Monte Carlo event generators are also included for comparison. Additionally, results obtained using the corrected EPA, incorporating the correction factor for high photon virtualities, are presented. Table 24 shows the production cross sections (in pb) for Higgsinos, calculated similarly under the same EPA corrections.

E.7 Some comparisons plots with and without correction factors

Process (elastic)	EPA	EPA (Corrected by Eq. (E.2))	GRAPE (BH)	lpair
$\tau^+\tau^-$ ($Q_{e,\max}^2/Q_{p,\max}^2 = 1/1$ GeV ²)	41.15	40.92	41.35	40.69
$\tau^+\tau^-$ ($Q_{e,\max}^2/Q_{p,\max}^2 = 10/10$ GeV ²)	46.32	45.81	46.30	45.49
$\tau^+\tau^-$ ($Q_{e,\max}^2/Q_{p,\max}^2 = 50/50$ GeV ²)	49.74	48.34	48.82	48.82
$\tau^+\tau^-$ ($Q_{e,\max}^2/Q_{p,\max}^2 = 100/100$ GeV ²)	51.22	49.05	49.54	49.56
$\tau^+\tau^-$ ($Q_{e,\max}^2/Q_{p,\max}^2 = 1000/1000$ GeV ²)	56.11	50.01	50.41	49.71
$\tau^+\tau^-$ ($Q_{e,\max}^2/Q_{p,\max}^2 = 10^4/10^4$ GeV ²)	60.77	50.15	50.47	49.73
$\tau^+\tau^-$ ($Q_{e,\max}^2/Q_{p,\max}^2 = 10^5/10^5$ GeV ²)	65.65	50.16	50.48	49.67

Table 23. $\tau^+\tau^-$ production cross sections (in pb) at LHeC calculated using the EPA approach at $W_0 = 10$ GeV, assuming different values of $Q_{e,\max}^2$ and $Q_{p,\max}^2$ for electron/proton vertex. Cross-sections using the lpair and GRAPE MC event generators are also presented for comparison. The results for the **corrected EPA assuming suppression factor have been shown.**

Process (elastic)	EPA	GRAPE (BH)	EPA (Corrected by Eq. (E.2))
$m_{\tilde{H}} = 100$ GeV, ($Q_{e,\max}^2/Q_{p,\max}^2 = 10/10$ GeV ²)	9.153×10^{-4}	9.155×10^{-4}	9.143×10^{-4}
$m_{\tilde{H}} = 100$ GeV, ($Q_{e,\max}^2/Q_{p,\max}^2 = 100/100$ GeV ²)	1.037×10^{-4}	1.037×10^{-3}	1.036×10^{-3}
$m_{\tilde{H}} = 100$ GeV, ($Q_{e,\max}^2/Q_{p,\max}^2 = 10^3/10^3$ GeV ²)	1.159×10^{-3}	1.156×10^{-3}	1.155×10^{-3}
$m_{\tilde{H}} = 100$ GeV, ($Q_{e,\max}^2/Q_{p,\max}^2 = 10^4/10^4$ GeV ²)	1.281×10^{-3}	1.259×10^{-3}	1.259×10^{-3}
$m_{\tilde{H}} = 200$ GeV, ($Q_{e,\max}^2/Q_{p,\max}^2 = 10/10$ GeV ²)	2.238×10^{-5}	2.239×10^{-5}	2.234×10^{-5}
$m_{\tilde{H}} = 200$ GeV, ($Q_{e,\max}^2/Q_{p,\max}^2 = 100/100$ GeV ²)	2.556×10^{-5}	2.553×10^{-5}	2.552×10^{-5}
$m_{\tilde{H}} = 200$ GeV, ($Q_{e,\max}^2/Q_{p,\max}^2 = 10^3/10^3$ GeV ²)	2.874×10^{-5}	2.870×10^{-5}	2.868×10^{-5}
$m_{\tilde{H}} = 200$ GeV, ($Q_{e,\max}^2/Q_{p,\max}^2 = 10^4/10^4$ GeV ²)	3.192×10^{-5}	3.168×10^{-5}	3.167×10^{-5}

Table 24. Higgsinos production cross sections (in pb) at LHeC calculated using the EPA approach $W_0 = 2m_{\tilde{H}}$, assuming different values of $Q_{e,\max}^2$ and $Q_{p,\max}^2$ for electron/proton vertex. Cross-sections using the GRAPE MC event generator are also presented for comparison. The results for the **corrected EPA assuming suppression factor have been shown.**

Process (elastic)	EPA	EPA (Corrected by Eq.(E.2))	GRAPE (BH)	lpair
$\mu^+\mu^-$ ($Q_{e,\max}^2/Q_{p,\max}^2 = 1/1$ GeV ²)	107.05	106.83	109.11	106.10
$\mu^+\mu^-$ ($Q_{e,\max}^2/Q_{p,\max}^2 = 10/10$ GeV ²)	120.42	119.47	121.37	118.14
$\mu^+\mu^-$ ($Q_{e,\max}^2/Q_{p,\max}^2 = 50/50$ GeV ²)	129.29	125.97	127.14	123.89
$\mu^+\mu^-$ ($Q_{e,\max}^2/Q_{p,\max}^2 = 100/100$ GeV ²)	133.11	127.76	128.61	125.02
$\mu^+\mu^-$ ($Q_{e,\max}^2/Q_{p,\max}^2 = 10^3/10^3$ GeV ²)	145.79	130.07	130.28	126.22
$\mu^+\mu^-$ ($Q_{e,\max}^2/Q_{p,\max}^2 = 10^4/10^4$ GeV ²)	158.48	130.39	130.39	126.57

Table 25. $\mu^+\mu^-$ production cross sections (in pb) at LHeC calculated using the EPA approach $W_0 = 10$ GeV, assuming different values of $Q_{e,\max}^2$ and $Q_{p,\max}^2$ for electron/proton vertex. Cross-sections using the lpair and GRAPE MC event generators are also presented for comparison. The results for the **corrected EPA assuming suppression factor have been shown.**

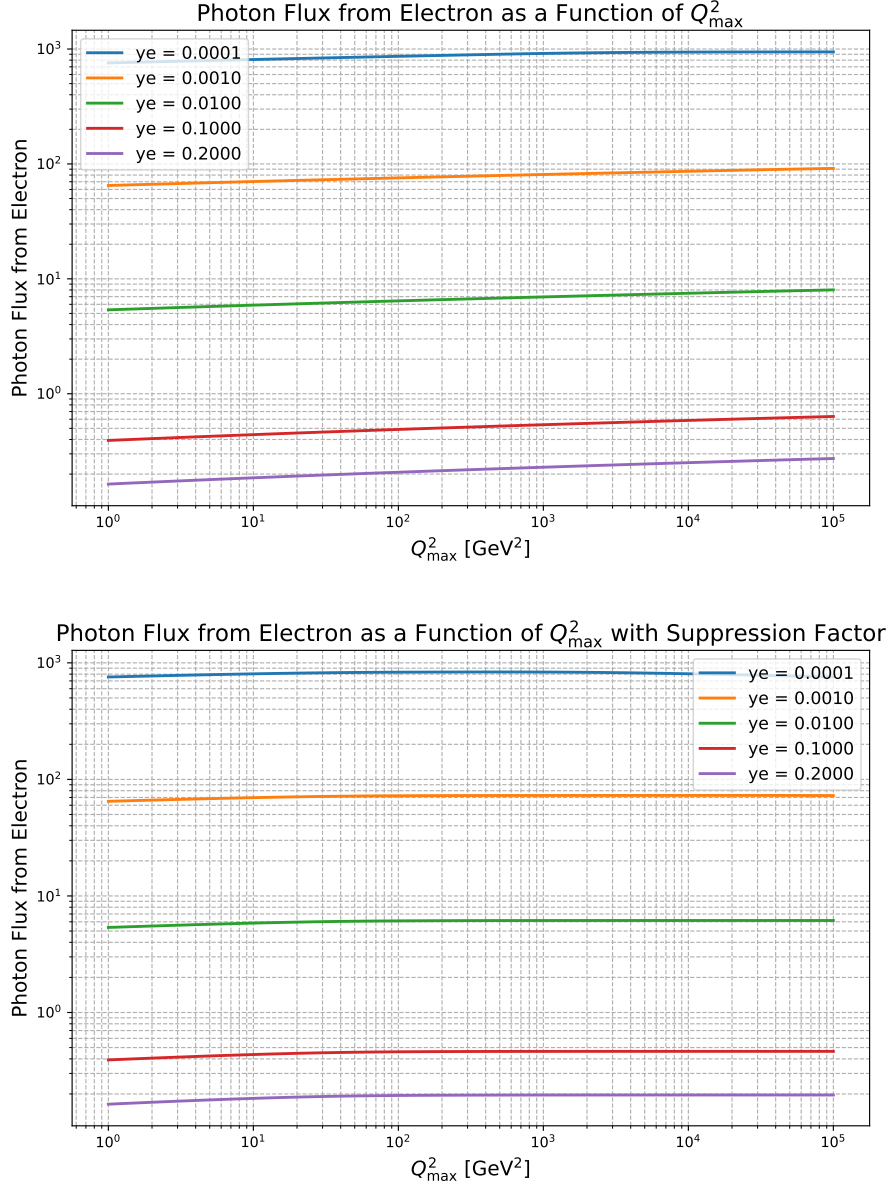


Figure 46. Photon flux for electrons (Eq. E.3) as a function of $Q_{\max}^2$ for different values of y_e . Flux is calculated with and without correction factor $S(Q^2, W)$ for the $W = 10$ GeV.

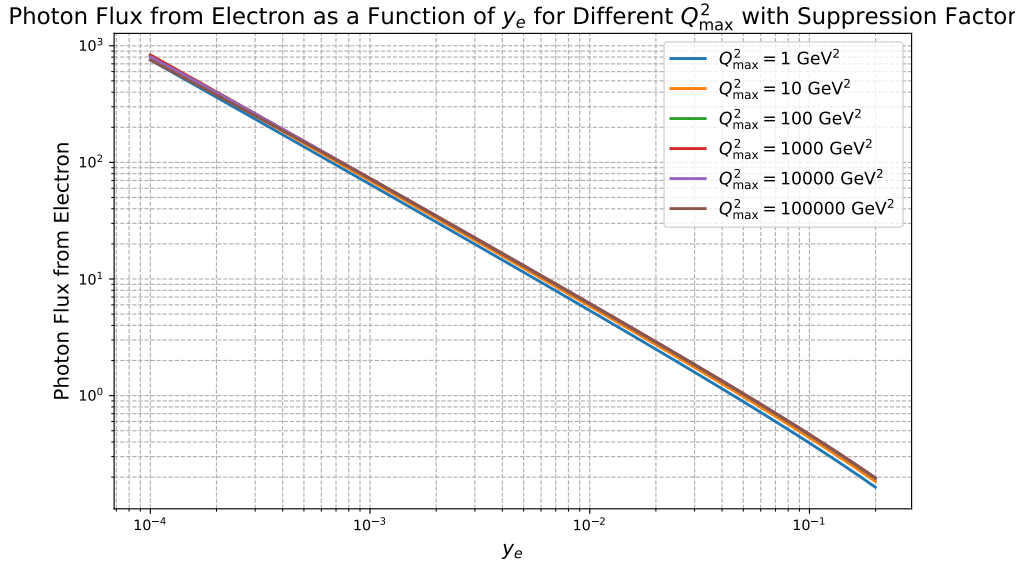
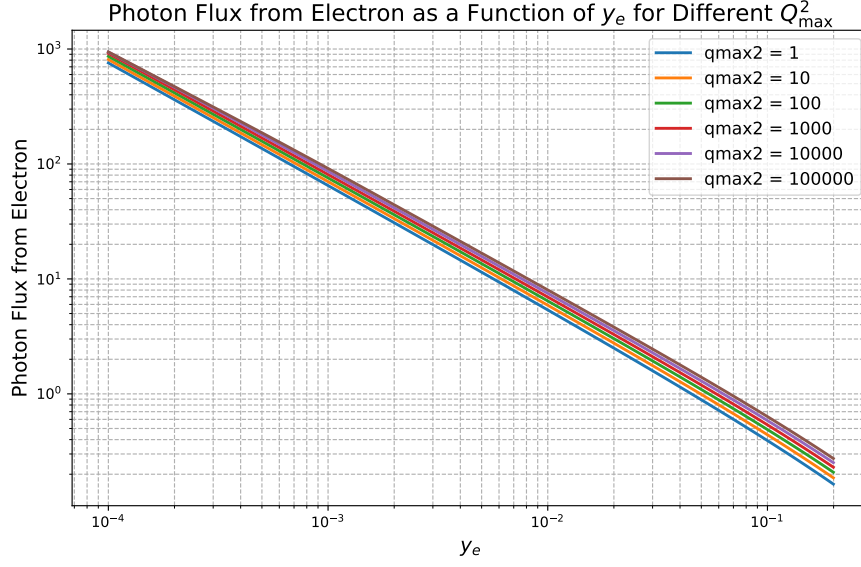


Figure 47. Photon flux for electrons (Eq. E.3) as a function of y_e for different values of $Q_{\max}^2$. Flux is calculated with and without correction factor $S(Q^2, W)$ for the $W = 10$ GeV.

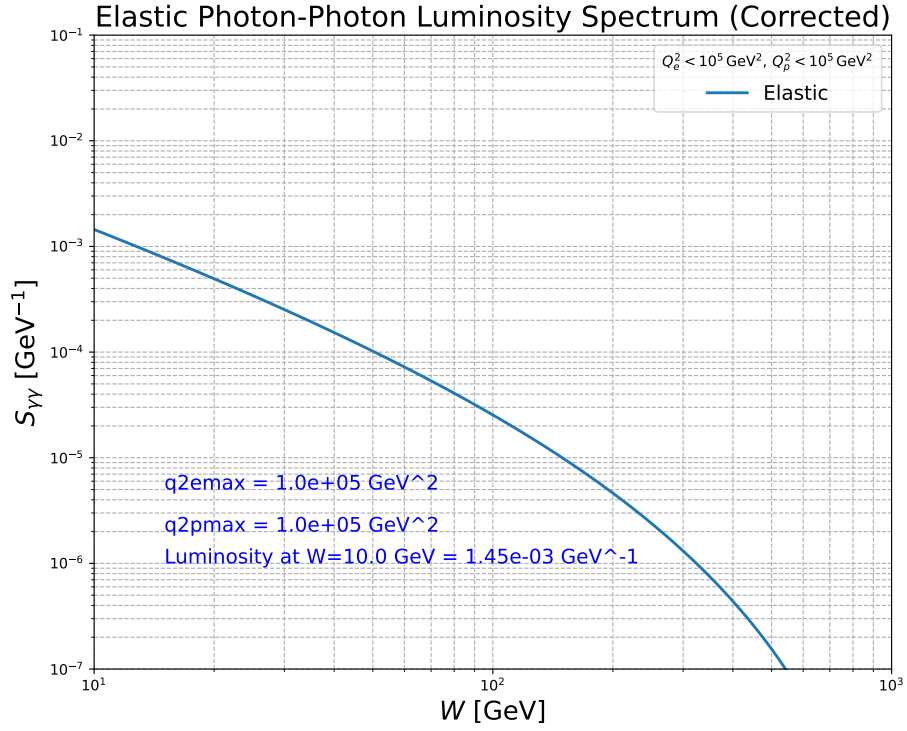
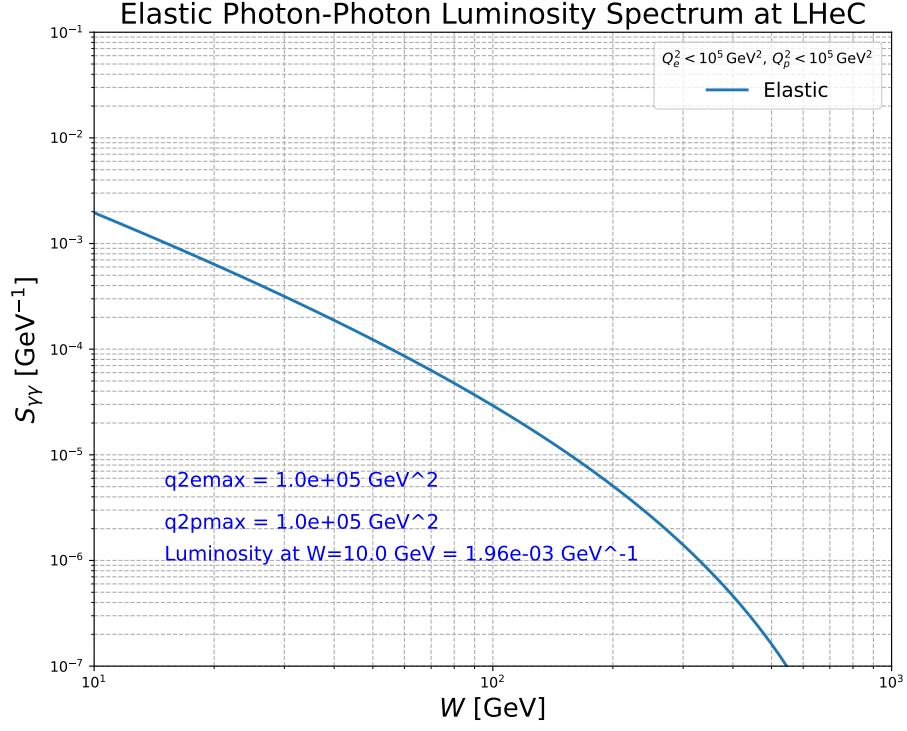


Figure 48. The elastic luminosity spectrum $S_{\gamma\gamma}$ at the LHeC as a function of W , assuming $Q_{\text{max}}^2 = 10^5 \text{ GeV}^2$ for both electron and proton vertex. The result with the correction factor is also shown for comparison.

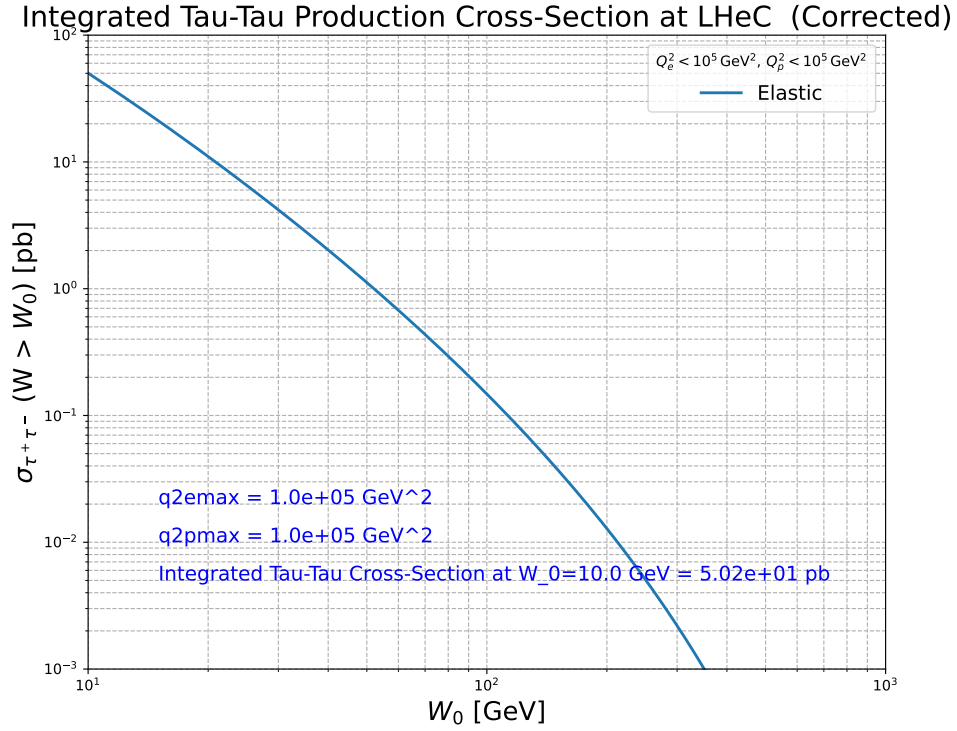
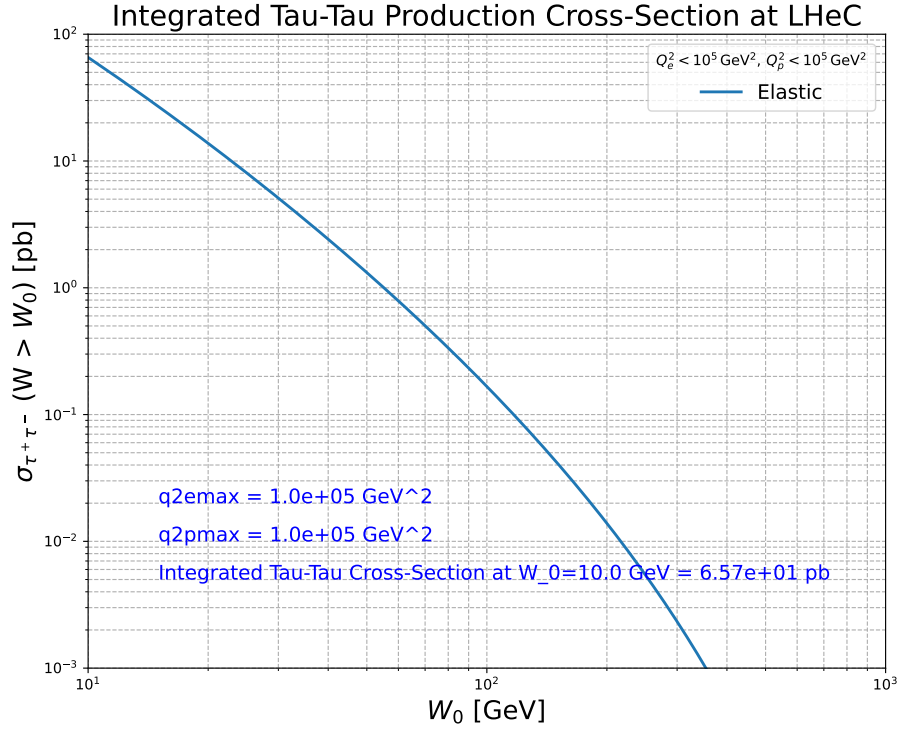


Figure 49. Integrated τ pair production cross sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{\tau^+\tau^-}$, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 , through the elastic process, assuming $Q_{\text{max}}^2 = 10^5 \text{ GeV}^2$ for both electron and proton vertex. The result with the correction factor is also shown for comparison.

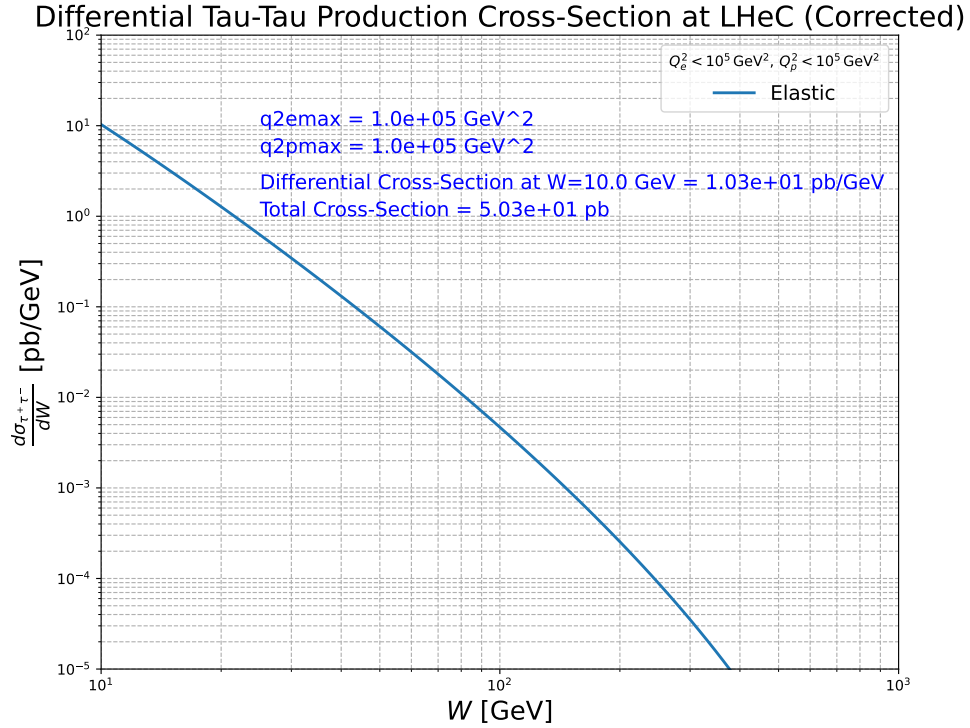
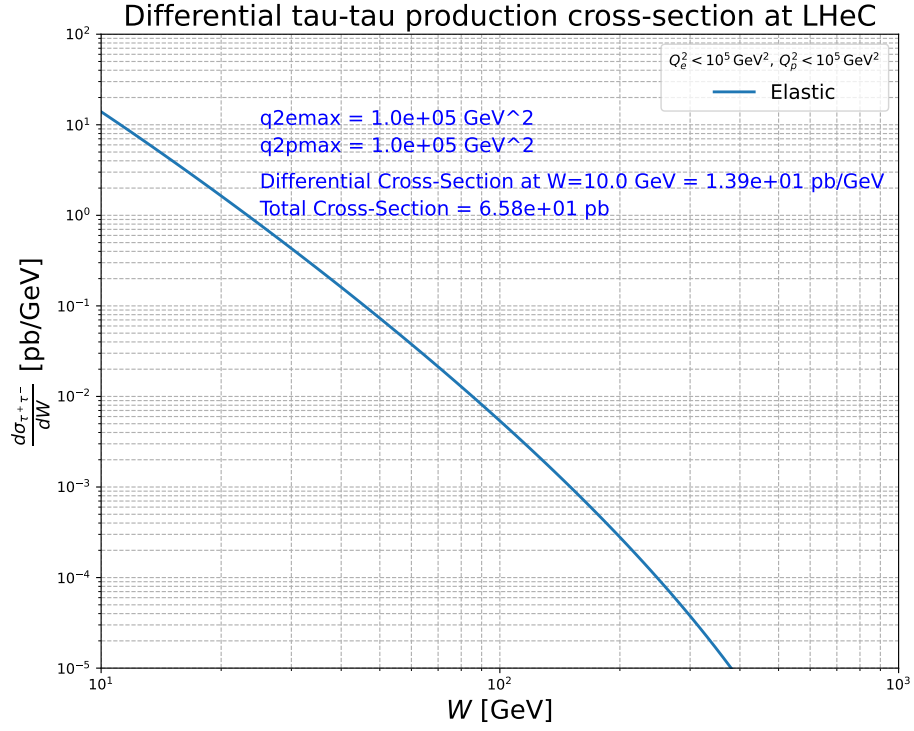


Figure 50. The differential cross section $d\sigma/dW$ in pb/GeV as a function of the photon-photon center-of-mass energy W , through the elastic process, assuming $Q_{\text{max}}^2 = 10^5 \text{ GeV}^2$ for both electron and proton vertex. The result with the correction factor is also shown for comparison.

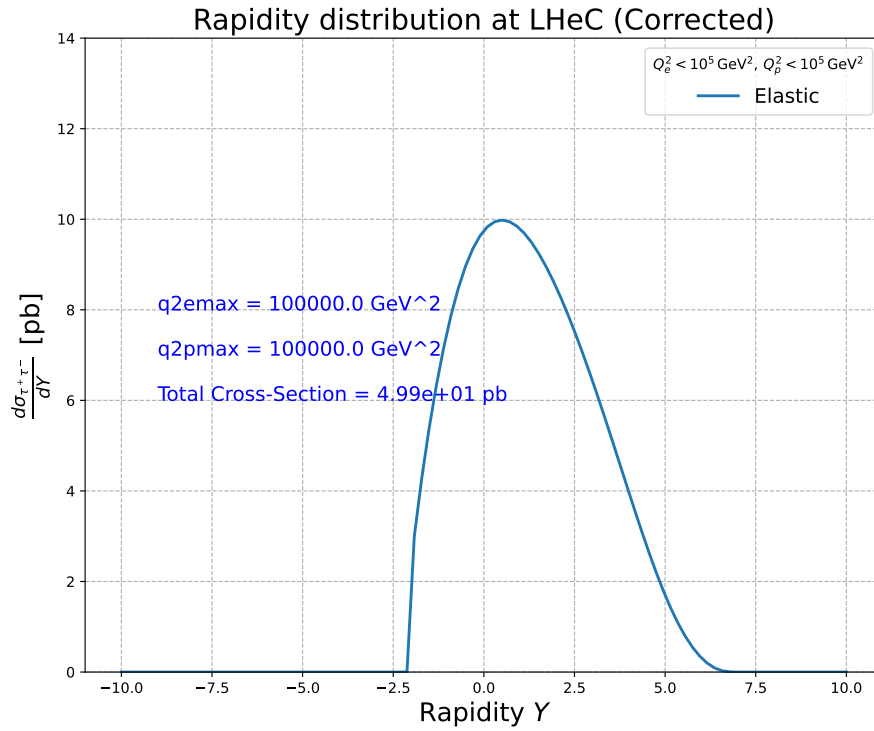
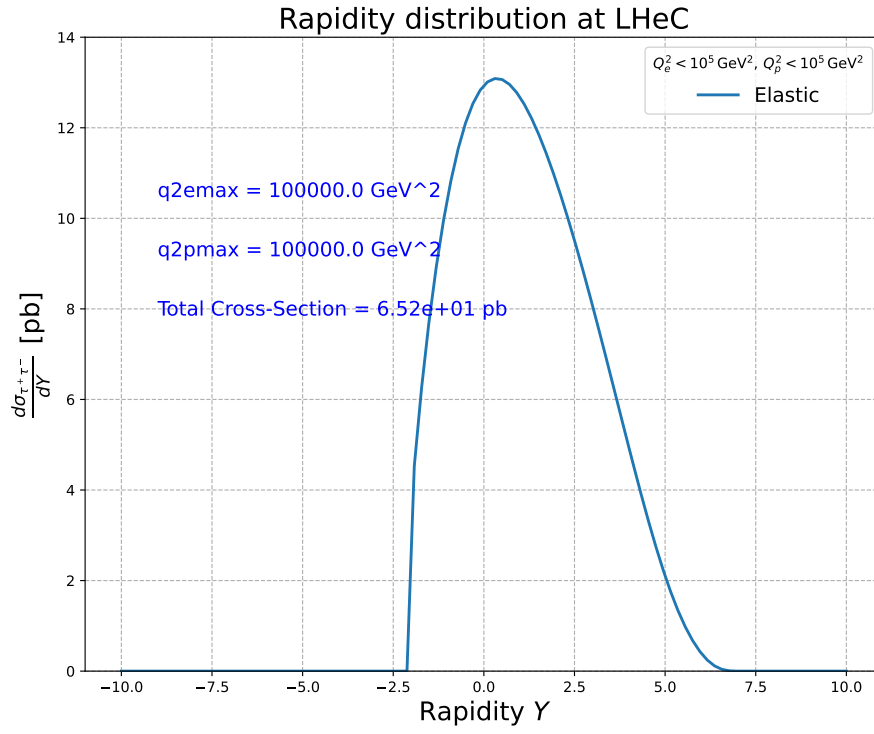


Figure 51. The differential cross section $d\sigma/dY_{\tau^+\tau^-}$ in pb as a function of the tau pairs rapidity, through the elastic process, assuming $Q_{\text{max}}^2 = 10^5 \text{ GeV}^2$ for both electron and proton vertex. The result with the correction factor is also shown for comparison.